

Performance Tools Manual

BASE24[®]



© 2007 by ACI Worldwide, Inc. All rights reserved.

All information contained in this documentation, as well as the software described in it, is confidential and proprietary to ACI Worldwide, Inc., is subject to a license agreement, and may be used or copied only in accordance with the terms of such license. Except as permitted by such license, no part of this documentation may be reproduced, stored in a retrieval system, or transmitted in any form or by electronic, mechanical, recording, or any other means, without the prior written permission of ACI Worldwide, Inc.

ACI, ACI Worldwide, and the ACI product names used in this documentation are trademarks or registered trademarks of ACI Worldwide, Inc. or its subsidiaries.

Other companies' trademarks, service marks, or registered trademarks and service marks are trademarks, service marks, or registered trademarks and service marks of their respective companies.

Contents

Preface	ix
Conventions Used in This Manual	xiii
1: Introduction	1-1
BASE24 Performance Tools	1-2
ATM Transaction Analysis Utility	1-3
POS Transaction Analysis Utility	1-4
Remote Banking Transaction Analysis Utility	1-5
Teller Transaction Analysis Utility	1-6
System and File Analysis Utilities	1-7
BLOCKR2 Utility	1-7
MACKIE Draw Utilities	1-7
DSCRPT Utility	1-8
Purchasing Options	1-9
Event Message Generation	1-10
2: Transaction Analysis Utilities	2-1
Overview	2-2
Transaction Log Files	2-3
Transaction Analysis Utility Transaction History File	2-4
Accessing Report Data	2-4
Configuring Report Dates	2-5
Setting the Release Parameter	2-5
Transaction Analysis Utilities Installation Procedures	2-6
ATM Transaction Analysis Utility Installation Procedures	2-7
POS Transaction Analysis Utility Installation Procedures	2-7
Remote Banking Transaction Analysis Utility Installation Procedures	2-7
Teller Transaction Analysis Utility Installation Procedures	2-8

Transaction Analysis Utilities *continued*

LCONF Configuration	2-9
ATM Transaction Analysis Utility	2-9
POS Transaction Analysis Utility	2-9
Teller Transaction Analysis Utility	2-10
Remote Banking Transaction Analysis Utility	2-10
Reporting Options	2-11
CATEGORY	2-12
CLOCKDATE	2-12
DIVIDEBY	2-12
END	2-13
ENTRIES	2-13
FILE	2-14
HELP	2-14
HISTFILE	2-14
HISTORY	2-15
INTERVAL	2-15
MAXTPS	2-15
NEF	2-16
NOSPOOL	2-16
OPTIONS	2-18
START	2-20
TOMORROW	2-20
Running the Transaction Analysis Utilities	2-21
Command Entry for Running the Transaction Analysis Utilities	2-21
ATM Command Examples	2-23
POS Command Examples	2-27
Remote Banking Command Examples	2-33
Teller Command Examples	2-37
Transaction Analysis Utility Reports	2-41
Transaction Log File Report	2-43
Transaction Log File Report—Summary and Totals	2-54
Response Time Summary Report	2-72
Transaction Rate vs. Response Time Report	2-78
Transaction Volume Summary Report	2-86

Transaction Analysis Utilities *continued*

Approval Amount Report	2-90
Transaction Denial Report	2-101
Reversal Report	2-109
Station Reports	2-114
Stations Sorted by Name, Detail Report	2-114
Stations Sorted by Activity Report	2-121
Stations Sorted by Line Report	2-124
Station Activity Summary Report	2-127
Device Type Report	2-131
Device Handler Report	2-135
Transaction Originator Reports	2-138
Sorted by Interface Instance Name	2-138
Sorted by Activity	2-141
Activity Summary Report	2-144
Transaction Originator Type Report	2-147
Authorization Report	2-150
Institution Reports	2-154
By Card Owner	2-155
By Issuer	2-160
By Terminal Owner FIID	2-164
Category Report	2-169
3: BLOCKR2 Utility	3-1
Overview	3-2
BLOCKR2 Utility Installation Procedures	3-3
Running the BLOCKR2 Utility	3-4
Command Entry for Running the BLOCKR2 Utility	3-4
Command Examples	3-5
BLOCKR2 Utility Options Summary	3-8
Exiting the BLOCKR2 Utility	3-9
Running the BLOCKR2 Utility Interactively	3-10
Running the BLOCKR2 Utility Noninteractively	3-13
Setting Up Input Data Files	3-14

BLOCKR2 Utility *continued*

Input Data File Templates	3-15
Key-Sequenced Template	3-15
Entry-Sequenced and Relative Template	3-16
BLOCKR2 Utility Output	3-17
Key-Sequenced DP2 File	3-18
Entry-Sequenced DP2 Example	3-20
Relative DP2 Example	3-21
 4: MACKIE Draw Utilities	 4-1
Overview	4-2
MACKIE Draw Installation Procedures	4-3
Running the MACKIE Draw Utilities	4-4
Running the MACPUP Utility	4-5
MACPUP Utility RUN Command Examples	4-6
Running the MACPUP Utility Without a Specified Output File	4-6
Running the MACPUP Utility With an Output File Specified	4-6
Output Produced by the MACPUP Utility	4-7
Running the MACPRT Utility	4-9
MACPRT Utility RUN Command Example	4-10
Output Produced by the MACPRT Utility	4-11
System Information	4-11
CPU Configuration Information	4-12
Device Configuration Information	4-13
Example Report Generated by the MACPRT Utility	4-14
MACPRT Diagram Symbols	4-15
Example System Configuration Diagram	4-18
 5: The DISCRPT Utility	 5-1
Overview	5-2
DISCRPT Utility Installation Procedures	5-3
Setting Up the DISCRPT Utility	5-4
Specifying File Locations	5-5
Running the DISCRPT Utility	5-6

The DISCRPT Utility *continued*

Output Produced by the DISCRPT Utility	5-7
Disc Report Detail	5-7
Disc Report Summary	5-12
A: Error Messages	A-1
Transaction Analysis Utility Error Messages	A-2
BLOCKR2 Error Messages	A-46
DSCRPT Error Messages	A-48
MACPUP Error Messages	A-52
MACPRT Error Messages	A-56
Index	Index-1

ACI Worldwide, Inc.

Preface

The ***BASE24 Performance Tools Manual*** documents and describes how to use the following BASE24 performance tools:

- ATM Transaction Analysis utility
- POS Transaction Analysis utility
- Remote Banking Transaction Analysis utility
- Teller Transaction Analysis utility
- System and file analysis utilities

The BASE24 performance tools are packaged to allow several purchasing options. This gives you the flexibility to determine the tools best suited for your specific requirements. The ATM, POS, Remote Banking, and Teller Transaction Analysis utilities are used for specific products and, therefore, are purchased separately. Since all customers do not support all four products (i.e., BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller), you can purchase any combination of the transaction analysis utilities you require.

The system and file analysis utilities (i.e., BLOCKR2, MACKIE Draw, and DSCRPT) can be purchased only as a set because of their relationships with one another. The documentation in this manual further explains these relationships.

Audience

This manual is intended for use by personnel working with the BASE24 performance tools for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller.

Prerequisites

This manual discusses how the BASE24 performance tools are used with BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller. Therefore, readers should have a solid understanding of BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller before using this manual.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24-atm Transaction Processing Manual*** discusses in detail the Standard Internal Message (STM) format used with BASE24-atm.
- The ***BASE24 IEP Configuration Manual*** and ***BASE24 IEP Objects Reference Manual*** describe statistics that include the transaction response times used by the Integrated Endpoint process.
- The ***BASE24 Event Message Reference Manual*** provides a complete explanation of even message documentation.
- The ***BASE24 Logical Network Configuration File Manual*** describes the assigns and params used by the transaction analysis utilities.
- The ***BASE24-pos Transaction Processing Manual*** describes in detail the POS Standard Internal Message (PSTM) format used with BASE24-pos.
- The ***BASE24 Remote Banking Transaction Processing Manual*** describes in detail the Internal Transaction Data (ITD) format used with BASE24 Remote Banking.
- The ***BASE24-teller Files Maintenance Manual*** contains a list of reversal reason codes that can appear on the Reversal Report.
- The ***BASE24-teller Transaction Processing Manual*** describes in detail the Teller Standard Internal Message (TSTM) format used with BASE24-teller.

This manual contains references to the *ACI Make Release 5 User Guide* for detailed information on the ACI Make utilities.

This manual contains references to the International Organization for Standardization (ISO) 8583-1993 standard, *Financial Transaction Card Originated Messages—Interchange Message Specification*, for reversal reason codes that can appear on the Reversal Report for BASE24 Remote Banking.

This manual references the following HP NonStop publications:

- The *ENSCRIBE Programmer's Guide* provides additional information about alternate key files.
- The *Peripheral Utility Program (PUP) Reference Manual* provides a list of valid device types that can be used in the RUN command for the MACPRT utility.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an introduction to the BASE24 performance tools. This section provides an overview of each transaction analysis, system analysis, and file analysis utility.

Section 2, “Transaction Analysis Utilities,” describes how to install, configure, and run the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities. This section also describes the reports that are generated by these utilities, including examples.

Section 3, “BLOCKR2 Utility,” describes the BLOCKR2 file analysis utility. This section includes an overview of the BLOCKR2 utility; the procedures for installing, configuring, and running the utility; and a description of the output produced by the utility.

Section 4, “MACKIE Draw Utilities,” describes the MACKIE Draw system analysis utilities. This section includes an overview of the MACKIE Draw utilities, the procedures for installing and running these utilities, and a description of the output produced by the utilities.

Section 5, “The DSCRPT Utility,” describes the DSCRPT system analysis utility. This section includes an overview of the DSCRPT utility; the procedures for installing, configuring, and running the utility; and a description of the output produced by DISCRPT.

Appendix A, “Error Messages,” provides a list of the error messages that can occur when running the BASE24 performance tools. The error messages contained in this appendix are organized alphabetically by tool.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-UT020-01 for the ***BASE24 Performance Tools Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Nov-2007 for November, 2007) indicates the issue of the manual. The software release information (for example, R6.0v8 for release 6.0, version 8) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how syntax notation is documented in this manual. It also explains BASE24-atm and BASE24-pos Terminal Data files terminology.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Terminal Data Files Terminology

Data entered on BASE24-atm Terminal Data files (ATD) screens is stored in different files, depending on the type of terminal being defined. For Diebold 10XX/478X and NCR 5XXX Device Handler terminals, data is stored in the BASE24-atm Terminal Data Dynamic File—general data (ATDD1) and the BASE24-atm Terminal Data Static File—general data (ATDS1) files. Dynamic data is stored in the ATDD1 file and static data is stored in the ATDS1 file. For all other types of ATM terminals, data is stored in the Terminal Data File (TDF).

Data entered on BASE24-pos Terminal Data files (PTD) screens is actually stored in different files, depending on the type of terminal being defined. For Hypercom and Standard POS Device Handler terminals, dynamic data is stored in the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) and static data is stored in the BASE24-pos Terminal Data Static File—general data (PTDS1). For all other types of POS terminals, all data from the PTD screens is stored in the POS Terminal Data File (PTDF).

For documentation purposes, the terms BASE24-atm Terminal Data files and BASE24-pos Terminal Data files are used throughout this manual instead of the individual file names and acronyms listed above.

1: Introduction

ACI has created a set of BASE24 performance tools designed to assist you in generating statistical data about their system. Each BASE24 performance tool is used to produce a different set of data that you can use to analyze system activity. This section provides overviews for each BASE24 performance tool and describes the functions and benefits of each tool. In addition, it describes the packaging options you can select when purchasing the BASE24 performance tools.

BASE24 Performance Tools

BASE24 performance tools allow you to accumulate, organize, and store system data that otherwise must be obtained manually, if at all. Through the use of the BASE24 performance tools, you can monitor the transaction activity in the system, size files used in the system, generate system configuration data, and produce reports concerning disk use. The BASE24 performance tools consist of the following utilities:

- ATM Transaction Analysis utility
- POS Transaction Analysis utility
- Remote Banking Transaction Analysis utility
- Teller Transaction Analysis utility
- System and file analysis utilities

ATM Transaction Analysis Utility

The ATM Transaction Analysis utility is a BASE24 performance tool that generates reports and report summaries used to analyze transaction activity for BASE24-atm. Information in these reports is obtained from the Transaction Log File (TLF). The report data includes transaction rates, transaction response times, and the number of denials and reversals. Information regarding terminal, host, and switch activity is also included.

The information generated by the ATM Transaction Analysis utility is used to analyze transaction activity occurring throughout the BASE24-atm system on a particular day. You can use these statistics to perform a trend analysis for items such as the number of transactions processed per day, the peak times for transaction processing, the average response time per transaction, and the maximum number of transactions processed per second.

POS Transaction Analysis Utility

The POS Transaction Analysis utility is a BASE24 performance tool that generates reports and report summaries used to analyze transaction activity for BASE24-pos. Information contained in these reports is obtained from the POS Transaction Log File (PTLF). The report data includes transaction rates, transaction response times, and the number of denials and reversals. Information regarding terminal, host, and switch activity is also included.

The information generated by the POS Transaction Analysis utility is used to analyze transaction activity occurring throughout the BASE24-pos system on a particular day. You can use these statistics to perform a trend analysis for items such as the number of approved transactions processed per day, the number of transactions authorized by a particular authorizing entity, the peak times for transaction processing, the average response time per transaction, and the maximum number of transactions processed per second.

Remote Banking Transaction Analysis Utility

The Remote Banking Transaction Analysis utility is a BASE24 performance tool that generates reports and report summaries used to analyze transaction activity for BASE24-telebanking and BASE24-billpay. Information contained in these reports is obtained from the ITS Transaction Log File (ITLF). The report data includes transaction rates, transaction response times, and the number of denials and reversals. Information regarding interface instance, host, and switch activity is also included.

The information generated by the Remote Banking Transaction Analysis utility is used to analyze transaction activity occurring throughout the BASE24 Remote Banking system on a particular day. You can use these statistics to perform a trend analysis for items such as the number of approved transactions processed per day, the number of transactions authorized by a particular authorizing entity, the peak times for transaction processing, the average response time per transaction, and the maximum number of transactions processed per second.

Teller Transaction Analysis Utility

The Teller Transaction Analysis utility is a BASE24 performance tool that generates reports and report summaries used to analyze transaction activity for BASE24-teller. Information contained in these reports is obtained from the Teller Transaction Log File (TTLF). The report data includes transaction rates, transaction response times, and the number of denials and reversals. Information regarding terminal, host, and switch activity is also included.

The information generated by the Teller Transaction Analysis utility is used to analyze transaction activity occurring throughout the BASE24-teller system on a particular day. You can use these statistics to perform a trend analysis for items such as the number of approved transactions processed per day, the number of transactions authorized by a particular authorizing entity, the peak times for transaction processing, the average response time per transaction, and the maximum number of transactions processed per second.

System and File Analysis Utilities

The system and file analysis utilities are a set of BASE24 performance tools used in file sizing, gathering system configuration information, and providing disk usage statistics. The system and file analysis utilities consist of the following:

- BLOCKR2 utility
- MACKIE Draw utilities
- DSCRPT utility

BLOCKR2 Utility

The BLOCKR2 utility calculates ENSCRIBE file sizes for structured files. The BLOCKR2 utility not only calculates file sizing information, but also generates statistics that you can use in analyzing file space. The statistics include the number of records in the file, the record size, and the total amount of space that is unused (i.e., available for use), in a particular file.

MACKIE Draw Utilities

The MACKIE Draw utilities gather system configuration information and generate a physical system diagram. The diagram provides system, CPU, and device configuration information that you can use to determine where changes can be made to your system to enhance performance. You can also use the system configuration diagram to assist in pinpointing where problems are occurring in the system. The MACKIE Draw utilities provide a visual picture of how a particular system is configured. Without the MACKIE Draw utilities, creating a system configuration diagram can be a time consuming process that must be repeated and monitored each time changes are made to the system. Using the MACKIE Draw utilities, each time changes are made to a system configuration, you can run the utilities to generate a new system configuration diagram.

DSCRPT Utility

The DSCRPT utility gathers and organizes statistical information concerning disk usage and configuration. The DSCRPT utility consolidates information generated by HP NonStop utilities and generates two reports based on the information it gathers. The first report is the Disc Report Detail. This report contains an analysis of memory, space allocation, and file usage for a particular disk. The second report is the Disc Report Summary. This report contains a graph that shows the percentage of each disk volume in a system that is full. The information displayed on these reports is organized according to disk volume. You can use the information on these reports to determine if more disk space is required and to provide an overall analysis of the disks used in the system.

Purchasing Options

The BASE24 performance tools are packaged to allow several purchasing options. This gives you the flexibility to determine the tools best suited to meet your specific requirements. The BASE24 performance tools consist of the following utilities, each of which is purchased separately:

- ATM Transaction Analysis utility
- POS Transaction Analysis utility
- Remote Banking Transaction Analysis utility
- Teller Transaction Analysis utility
- System and file analysis utilities

The ATM Transaction Analysis utility, POS Transaction Analysis utility, Remote Banking Transaction Analysis utility, and Teller Transaction Analysis utility must all be purchased separately because each is used with a different product, as follows:

- The ATM Transaction Analysis utility is used only with BASE24-atm.
- The POS Transaction Analysis utility is used only with BASE24-pos.
- The Remote Banking Transaction Analysis utility is used only with BASE24-telebanking and BASE24-billpay.
- The Teller Transaction Analysis utility is used only with BASE24-teller.

The system and file analysis utilities must be purchased separately from the transaction analysis utilities listed above. The system and file analysis utilities consist of BLOCKR2, MACKIE Draw, and DSCRPT. These utilities are available only as a set and cannot be purchased individually.

Event Message Generation

The Transaction Analysis utilities and system and file analysis utilities generate event messages to assist network operators in performing their daily tasks. All system and file analysis utility messages are displayed on the terminal from which the utility was run. For the Transaction Analysis utilities, some messages are displayed on the terminal from which the utility was run and some are sent to the system log. Messages that can be displayed on the terminal from which a utility was run are described in appendix A.

Event messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access event message documentation electronically. Event messages are contained in files on a designated HP NonStop volume and subvolume. Each file contains event messages for a specific BASE24 process. Files on the designated HP NonStop volume and subvolume are organized according to product module ID (PMID). One PMID exists for every process in BASE24. PMIDs identify a specific process in the BASE24 system (e.g., BATAR identifies the Transaction Analysis utilities). Providing event message documentation in files on the HP NonStop system gives customers greater flexibility by allowing them to print out event message documentation as needed at their location or to electronically manipulate the data to suit their needs.

Documentation is available in file form for the event messages generated by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities. It is located on an HP NonStop volume specified by the customer and a subvolume called BAxxLOGM, where xx is the number of the current release, in a file named BATAR. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

2: Transaction Analysis Utilities

This section describes how to run and interpret the output from the transaction analysis utilities for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller. It documents the command syntax used to run the utilities and the reports generated by them. It also provides an overview of items to consider before running the utilities.

Overview

There are separate transaction analysis utilities for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller. The ATM, POS, Remote Banking, and Teller Transaction Analysis utilities generate statistical reports containing transaction volumes, authorization sources, and transaction response times for these products, respectively. Reports are produced based on the product analyzed by each utility. Therefore, separate reports are generated for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller.

Note: These utilities, while written for separate products, work virtually the same and are run using the same syntax.

Each transaction analysis utility reads a different transaction log file to obtain transaction activity data as shown in the following table.

Utility	Transaction Log File
ATM Transaction Analysis utility	Transaction Log File (TLF)
POS Transaction Analysis utility	POS Transaction Log File (PTLF)
Remote Banking Transaction Analysis utility	ITS Transaction Log File (ITLF)
Teller Transaction Analysis utility	Teller Transaction Log File (TTLF)

The following list shows the reports generated by all of the transaction analysis utilities:

- Transaction Log File Report
- Transaction Log File Report—Summary and Totals
- Response Time Summary Report
- Transaction Rate Vs. Response Time Report
- Transaction Volume Summary Report
- Approval Amount Report
- Transaction Denial Report
- Reversal Report

- Authorization Report
- Institution Report

The following reports are generated by the ATM, POS, and Teller Transaction Analysis utilities only:

- Stations Sorted by Name, Detail Report
- Stations Sorted by Activity Report
- Stations Sorted by Line Report
- Stations Activity Summary Report
- Device Type Report
- Device Handler Report
- Category Report (generated by the POS Transaction Analysis utility only)

The following reports are generated by the Remote Banking Transaction Analysis utility only:

- Transaction Originator Report (Sorted by Interface Instance Name)
This report is similar in format to the Stations Sorted by Name, Detail Report.
- Transaction Originator Report (Sorted by Activity)
This report is similar in format to the Stations Sorted by Activity Report.
- Transaction Originator Report (Activity Summary)
This report is similar in format to the Stations Activity Summary Report.
- Transaction Originator Type Report
This report is similar in format to the Device Type Report.

Transaction Log Files

Transaction Log Files (TLFs) contain records of every financial transaction, approved or denied, processed by BASE24-atm during a single processing day. PTLFs, ITLFs, and TTLFs serve the same purpose as TLFs, but are used with BASE24-pos, BASE24 Remote Banking, and BASE24-teller, respectively. PTLFs, ITLFs, and TTLFs can also contain records of certain nonfinancial or administrative transactions. TLFs, PTLFs, and TTLFs also contain settlement records for every terminal in the system. All of these files contain audit records at

the transaction level of system processing, and are extracted daily to provide detailed transaction data for processing by a host. One record is created for each transaction.

The ATM, POS, Remote Banking, and Teller Transaction Analysis utilities use information contained in the TLF, PTLF, ITLF, and TTLF, respectively, to generate a diverse set of statistical reports that provide information about transaction activity.

Transaction Analysis Utility Transaction History File

The Transaction Analysis Utility Transaction History File (THF) is an optional file that provides a central location where ATM, POS, Remote Banking, or Teller transaction analysis reports can be stored in unstructured form for access at a later date. This makes it possible for you to print out reports as needed for transaction activity analysis.

Two parameters are required in the ATM, POS, Remote Banking, or Teller Transaction Analysis utility RUN command if the THF is used—the HISTORY and OPTIONS parameters. The HISTORY parameter causes reports to be written to the THF and the OPTIONS parameter identifies which reports, by report acronym, are written to the THF. Multiple reports can be sent to the same THF if the reports are specified in the RUN command.

The location of the THF is identified in the ATM-TRN-THF, POS-TRN-THF, RB-TRN-THF, and TLR-TRN-THF assigns in the Logical Network Configuration File (LCONF). The HISTORY parameter uses these assigns to locate the appropriate THF.

Note: The Transaction Analysis Utility Transaction History File (THF) should not be confused with the BASE24-telebanking Transaction History File (THF). The BASE24-telebanking THF contains account history information that can be obtained using check clearance and transaction history inquiry transactions. If BASE24-telebanking is installed on your system, the above assigns should use a file name other than THF to avoid confusion.

Accessing Report Data

The reports generated by running the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities are available in 132 column hard copy form or can be kept in unstructured format on disk. Running the report each day and writing the most important data to a disk file allows you to perform trend analysis on the

statistics. You can use the HISTFILE parameter to run reports from an existing THF. The HISTFILE parameter can use any reports specified in the OPTIONS parameter when the THF was created to create new reports. See the topic “Reporting Options” for more information about the HISTFILE and OPTIONS parameters.

Configuring Report Dates

The date format displayed on the transaction analysis reports is specified in the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF). One of two date formats can be selected: either DDMM or MMDD. The date format specified in the LCONF is used on all reports.

Setting the Release Parameter

If you are running the BASE24 performance tools in a release 6.0 logical network, you must set the BASE24-REL param in the LCONF to 60. If you are running the BASE24 performance tools in a release 5.x logical network, you must set the BASE24-rel param to a value of 50. This param determines the release of BASE24 code that the logical network is running.

Transaction Analysis Utilities Installation Procedures

Before the Transaction Analysis utilities can be used, they must be properly installed. To install the utilities, restore the BAxxBTRN subvolume (where xx represents the current release number), and one of the following subvolumes for the transaction analysis utility you purchased, from tape to the volume where the BASE24 performance tools reside:

- BAxxATRN subvolume for BASE24-atm
- BAxxPTRN subvolume for BASE24-pos
- BAxxRTRN subvolume for BASE24 Remote Banking
- BAxxTTRN subvolume for BASE24-teller

The BAxxBTRN subvolume contains base procedures used by all of the transaction analysis utilities. The other subvolumes contain code specific to a particular transaction analysis utility. These subvolumes contain all of the files necessary to run the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities.

After these subvolumes are restored, the following steps are required to complete installation. For detailed information on the ACI Make utilities, refer to the ***ACI Make Release 5 User Guide***.

1. Modify the LCONF as needed to contain the assigns and params used by the transaction analysis utilities purchased. Refer to the topic “LCONF Configuration” for a list of the assigns and params used by each utility.
2. Modify the BTRNFM and BTRNMM files as needed for your site. The BTRNFM file contains configuration options and the BTRNMM file contains file defines.
3. Run the BTRNM MAKE file to compile the BASEO object file.
4. Execute the product-specific installation procedures for the transaction analysis utilities purchased as described on the following pages.

ATM Transaction Analysis Utility Installation Procedures

The following steps are required to complete installation of the ATM Transaction Analysis utility:

1. Modify the ATRNFM and ATRNMM files as needed for your site. The ATRNFM file contains configuration options and the ATRNMM file contains file defines.
2. Run the ATRNM MAKE file to create the TLF executable file.

POS Transaction Analysis Utility Installation Procedures

The following steps are required to complete installation of the POS Transaction Analysis utility:

1. Modify the PTRNFM and PTRNMM files as needed for your site. The PTRNFM file contains configuration options and the PTRNMM file contains file defines.
2. Run the PTRNM MAKE file to create the PTLF executable file.

Remote Banking Transaction Analysis Utility Installation Procedures

The following steps are required to complete installation of the Remote Banking Transaction Analysis utility:

1. Modify the RTRNFM, ITLFTM, RTRNMM, and RTRNNAMS files as needed for your site. The RTRNFM and ITLFTM files contain configuration options, the RTRNMM file contains file defines, and the RTRNNAMS file specifies text that appears on Transaction Originator Reports (some of this text can be modified).
2. Run the RTRNM MAKE file to create the ITLFLO and ITLFTO object files. The ITLFLO file contains utilities used by the ITLFTO file. The Remote Banking Transaction Analysis utility is run by executing the ITLFTO file.

Teller Transaction Analysis Utility Installation Procedures

The following steps are required to complete installation of the Teller Transaction Analysis utility:

1. Modify the TTRNFM and TTRNMM files as needed for your site. The TTRNFM file contains configuration options and the TTRNMM file contains file defines.
2. Run the TTRNM MAKE file to create the TTLF executable file.

LCONF Configuration

This subsection lists the Logical Network Configuration File (LCONF) assigns and params used by each of the transaction analysis utilities. For detailed information on each of these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

ATM Transaction Analysis Utility

The ATM Transaction Analysis utility uses the following LCONF assigns and params:

Assigns	Params
ATM-ATD-DYN-GNRL	BASE24-REL
ATM-ATD-STATIC-GNRL	DATE-DISPLAY-TYPE
ATM-ATD-EXTMEM-SWAPVOL	LOGICAL-NET
ATM-TRN-THF	
ICF	
ICFE	
IDF	
NETLIST-PROG	
TDF	
TLF	

POS Transaction Analysis Utility

The POS Transaction Analysis utility uses the following LCONF assigns and params:

Assigns	Params
ICF	BASE24-REL
ICFE	DATE-DISPLAY-TYPE
IDF	LOGICAL-NET
NETLIST-PROG	
POS-PTD-DYN-GNRL	

Assigns

POS-PTD-STATIC-GNRL
POS-PTD-EXTMEM-SWAPVOL
POS-PTDF
POS-PTLF
POS-TRN-THF

Params

Teller Transaction Analysis Utility

The Teller Transaction Analysis utility uses the following LCONF assigns and params:

Assigns

IDF
NETLIST-PROG
TLR-TRN-THF
TLF-TTDF
TLF-TTLF

Params

DATE-DISPLAY-TYPE
LOGICAL-NET

Remote Banking Transaction Analysis Utility

The Remote Banking Transaction Analysis utility uses the following LCONF assigns and params:

Assigns

IDF
IFCF
LCF
NETLIST-PROG
RB-TRN-THF
TLF-DEFAULT
TLF-group

Params

CURRENCY-CODE
DATE-DISPLAY-TYPE
LOGICAL-NET

Reporting Options

The ATM, POS, Remote Banking, and Teller Transaction Analysis utilities provide a set of reporting parameters that can be specified in the RUN command. These parameters allow you to tailor the output of the statistical reports produced by each program and determine the locations where these reports are sent. For more information on the ATM, POS, Remote Banking, and Teller Transaction Analysis utility RUN commands, refer to the topic “Running the Transaction Analysis Utilities” later in this section. The remainder of this subsection defines the following parameters and how they are used.

- CATEGORY
- CLOCKDATE
- DIVIDEBY
- END
- ENTRIES
- FILE
- HELP
- HISTFILE
- HISTORY
- INTERVAL
- MAXTPS
- NEF
- NOSPOOL
- OPTIONS
- START
- TOMORROW

CATEGORY

A size is entered with this parameter that specifies the maximum number of unique category records expected on the Category Report (range is 1 to 5000). The Category Report cannot be printed in hard copy form by the POS Transaction Analysis utility and can only be accessed through the THF.

The Category Report is only produced for BASE24-pos. Therefore, the CATEGORY parameter can only be used when running the POS Transaction Analysis utility. The CATEGORY parameter is required in the RUN command only if the Category Report is to be produced.

In addition, the HISTORY and OPTIONS parameters must also be specified in the RUN command if the CATEGORY parameter is used. The acronym CR (Category Report) must be included following the OPTIONS parameter in the RUN command. The HISTORY and OPTIONS parameters are discussed later.

CLOCKDATE

The CLOCKDATE parameter instructs the ATM, POS, Remote Banking and Teller Transaction Analysis utilities to produce a 24 hour midnight-to-midnight report. The CLOCKDATE parameter specifies that the report start time is midnight and the report spans 24 hours, including the next day's TLF, PTLF, ITLF, or TTLF. When the CLOCKDATE parameter is specified in the RUN command, the transaction analysis utilities open the TLF, PTLF, ITLF or TTLF with the date specified in the FILE parameter and the next day's TLF, PTLF, ITLF, or TTLF and merge the data according to transaction time. The CLOCKDATE parameter cannot be used with the TOMORROW parameter. The CLOCKDATE parameter is optional in the RUN command.

Note: The CLOCKDATE parameter requires that both day's TLF, PTLF, ITLF, or TTLF contain financial transaction records. If one of the files is empty (e.g., over a holiday when no transactions are processed), the utility will not function correctly and should be stopped.

DIVIDEBY

The DIVIDEBY parameter causes all monetary amounts reported from the TLF, PTLF, ITLF, or TTLF to first be divided by a number specified with the parameter (range is 1 to 32767). For example, you can use this parameter to determine the average amount of transactions processed at a BASE24-atm, BASE24-pos,

BASE24 Remote Banking, or BASE24-teller device. In this case, the total number of devices in the system for the product would be specified with the parameter. The DIVIDEBY parameter is optional in the RUN command.

END

The END parameter specifies the time, in hours and minutes, within the specified TLF, PTLF, ITLF, or TTLF at which transactions stop being included in the statistical reports produced by the ATM, POS, Remote Banking and Teller Transaction Analysis utilities. The default is to report on the entire TLF, PTLF, ITLF, or TTLF. This parameter is specified in 24-hour time (hh:mm), where the hours can range from 00 through 23 and the minutes can range from 00 through 59. The END parameter is used in conjunction with the START parameter and is optional in the RUN command, unless the START parameter has been specified.

ENTRIES

The ENTRIES parameter specifies the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Response Time Report. The ENTRIES parameter is used in conjunction with the MAXTPS parameter.

The ENTRIES parameter controls the size of the Transaction Rate Vs. Response Time Report. It correlates directly to the number of rows in the System TPS field of the report. System transactions per second (TPS) categories are determined using the value of the MAXTPS parameter specified at runtime. The MAXTPS parameter identifies the maximum number of transactions per second (MAXTPS) that is expected. You should select the next highest whole number over your expected peak average transaction rate for the specified time interval (specified in the INTERVAL parameter). The utility then takes the number specified in the MAXTPS parameter and divides it by the number specified in the ENTRIES parameter to determine the different ranges of throughput rates that are displayed in the System TPS field on the report. If the number of throughput rate ranges on the report exceeds 40, the report will automatically span multiple pages. The ENTRIES parameter is optional in the RUN command. Valid values are 1 through 99.

FILE

The FILE parameter identifies the date of the TLF, PTLF, ITLF, or TTLF from which the reports are to be generated. The FILE parameter keyword is followed by the date (YYMMDD) of a transaction log file to be analyzed. While the FILE parameter allows a new transaction log file to be analyzed, the HISTFILE parameter allows reports to be created from an existing Transaction Analysis Utility Transaction History File (THF). Either the FILE or HISTFILE parameter is required in the RUN command.

If either the CLOCKDATE or TOMORROW parameter is also specified in the RUN command, data is extracted from two transaction log files—the transaction log file for the date specified in the FILE parameter and the transaction log file for the next day. If either the CLOCKDATE or TOMORROW parameter is not also specified in the RUN command, data is extracted only from the transaction log file for the date specified.

HELP

The HELP parameter displays RUN command syntax help text for the transaction analysis utility without creating a report.

HISTFILE

The HISTFILE parameter allows reports to be created from an existing Transaction Analysis Utility Transaction History File (THF). Reports can be created from up to 10 different THFs by specifying the dates (YYMMDD) of the files separated by spaces. Successive dates can be specified by entering the first date followed by a hyphen (-) and the last date (e.g. 020601-020604 specifies THFs from June 1–4, 2002).

When running reports using the HISTFILE parameter, the options specified when the THF was created override any options specified in the current RUN command, except for the NOSPOOL option. The NOSPOOL option in the current RUN command is the only option that is not overridden.

While the HISTFILE parameter allows reports to be created from an existing THF, the FILE parameter allows a new transaction log file to be analyzed. Either the HISTFILE or the FILE parameter is required in the RUN command.

HISTORY

The HISTORY parameter causes report data to be written in unstructured format to the Transaction Analysis Utility Transaction History File (THF) specified in the ATM-TRN-THF, POS-TRN-THF, RB-TRN-THF, or TLR-TRN-THF assign in the LCONF. Reports can be stored in the THF and analyzed at a later date.

Each record written to the THF includes a record header that specifies the date of the transaction log file from which the information was obtained, a two-digit code indicating the type of information contained in the record, and the length of the record in bytes.

The HISTORY parameter is used in conjunction with the OPTIONS parameter. The HISTORY parameter specifies that reports are written to the THF and the OPTIONS parameter specifies which reports are written to the THF. The HISTORY parameter is optional in the RUN command.

Note: There is no file retention param in the LCONF for purging old THFs. Therefore, it is your responsibility to purge your old THFs when necessary. For more information on the THF assigns, refer to the *BASE24 Logical Network Configuration File Manual*.

INTERVAL

The INTERVAL parameter specifies the interval, in minutes, at which statistics are accumulated for use on the statistical reports generated by the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities. The valid range is 1 through 60 minutes with a default of one minute.

This parameter is used in conjunction with the MAXTPS parameter to determine the time ranges for which statistics are accumulated and displayed on several reports. The INTERVAL parameter is optional in the RUN command.

MAXTPS

The MAXTPS parameter causes the Transaction Rate Vs. Response Time Report to be produced using the value specified in this field as the largest transaction rate per second. A rate is entered with this parameter that specifies a whole number representing the highest expected transaction rate, in transactions per second,

averaged over the value specified in the INTERVAL parameter, and is used to set the time ranges for the report. The valid range for the MAXTPS parameter is 0 through 302.

The MAXTPS parameter is used in conjunction with the ENTRIES parameter. The MAXTPS parameter identifies the maximum number of transactions per second that is expected. The ENTRIES parameter is used to specify the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Response Time Report. The range for the ENTRIES parameter is 0 through 100. The default for the ENTRIES parameter is 40 (i.e., 40 graph lines are printed per page).

The MAXTPS parameter is optional in the RUN command. You should keep in mind that, generally, the accuracy of the graph produced increases with a lower number of MAXTPS, an INTERVAL of 1, and a higher number of ENTRIES as long as no occurrences are above the maximum line specified.

NEF

The NEF parameter causes a line activity report for terminals to be produced based on the specified Network Environment File (NEF). A file name is entered with this parameter that specifies the name of the NEF for the XPNET node reported on. This parameter can be repeated for up to 16 XPNET nodes. The NEF parameter is optional in the RUN command.

NOSPOOL

The NOSPOOL parameter stops specific reports from being sent to the spooler location specified in the RUN command. By default, all reports generated by the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities are sent to the spooler location specified in the RUN command. By using the NOSPOOL parameter, you can control which reports are sent to the spooler location.

There are two options available when using the NOSPOOL parameter. The first option prevents all statistical reports generated by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities from being sent to the spooler. In this case, the NOSPOOL parameter is specified in the RUN command by itself.

The second option is to eliminate a particular report or group of reports from being sent to the spooler. All reports generated by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities are associated with a corresponding report acronym in the NOSPOOL parameter. These acronyms are used to identify which

reports are not sent to the spooler. In this case, the NOSPOOL parameter keyword is followed by one or more of the report acronyms listed below in the RUN command. If more than one acronym is specified, the subsequent acronyms must be separated by a blank.

The valid report acronyms for the ATM, POS, and Teller Transaction Analysis utilities are as follows:

DIR = Transaction Log File Report
TTR = Transaction Log File Report—Summary and Totals
RTS = Response Time Summary Report
TVR = Transaction Rate Vs. Response Time Summary
TVS = Transaction Volume Summary Report
AAR = Approval Amount Report
TDR = Transaction Denial Report
RR = Reversal Report
SA = Stations Sorted by Activity Report
SR = Stations Sorted by Name, Detail Report
SAS = Station Activity Summary Report
LAS = Stations Sorted by Line Report
DTR = Device Type Report
DHR = Device Handler Report
AR = Authorization Report
IR = Institution Report
CR = Category Report (BASE24-pos only)

The valid report acronyms for the Remote Banking Transaction Analysis utility are as follows:

RBDIR = Transaction Log File Report
RBTTT = Transaction Log File Report—Summary and Totals
RBRTS = Response Time Summary Report
RBTVR = Transaction Rate Vs. Response Time Summary
RBTVS = Transaction Volume Summary Report
RBAAR = Approval Amount Report
RBTDTR = Transaction Denial Report
RBRR = Reversal Report
RBTOA = Transaction Originator Report (Sorted by Activity)
RBTOD = Transaction Originator Report (Sorted by Interface Instance Name)
RBTOS = Transaction Originator Report (Activity Summary)
RBTOT = Transaction Originator Type Report
RBAR = Authorization Report
RBIR = Institution Report

Valid acronyms for this parameter are the same as those listed under the **OPTIONS** parameter, except that acronym value of **B** cannot be used for the **NOSPOOL** parameter. In addition, the **NOSPOOL** parameter and the **OPTIONS** parameter with an acronym value of **B** cannot be used together. The **NOSPOOL** parameter is optional in the **RUN** command.

OPTIONS

The **OPTIONS** parameter specifies which reports are written to the Transaction Analysis Utility Transaction History File (THF). It is used in conjunction with the **HISTORY** parameter. The **HISTORY** parameter specifies that reports are written to the THF.

There are two options available when using the **OPTIONS** parameter. First, you can have all statistical reports generated by the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities written to the THF. In this case, the **OPTIONS** parameter is specified in the **RUN** command by itself.

The second option is to select a particular report or group of reports to be written to the THF. All reports generated by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities are associated with a corresponding report acronym in the **OPTIONS** parameter. These acronyms are used to identify which reports are written to the THF. In this case, the **OPTIONS** parameter is specified in the **RUN** command followed by one or more of the report acronyms listed below. If more than one acronym is specified, the subsequent acronyms must be separated by a blank.

The valid report acronyms for the ATM, POS, and Teller Transaction Analysis utilities are as follows:

B	= Send reports not specified to the spooler location (see below)
DIR	= Transaction Log File Report
TTR	= Transaction Log File Report—Summary and Totals
RTS	= Response Time Summary Report
TVR	= Transaction Rate Vs. Response Time Summary
TVS	= Transaction Volume Summary Report
AAR	= Approval Amount Report
TDR	= Transaction Denial Report
RR	= Reversal Report
SR	= Stations Sorted by Name, Detail Report
SAS	= Station Activity Summary Report
LAS	= Stations Sorted by Line Report
DTR	= Device Type Report

DHR = Device Handler Report
AR = Authorization Report
IR = Institution Report
CR = Category Report (BASE24-pos only)

The valid report acronyms for the Remote Banking Transaction Analysis utility are as follows:

B = Send reports not specified to the spooler location (see below)
RBDIR = Transaction Log File Report
RBTTT = Transaction Log File Report—Summary and Totals
RBRTS = Response Time Summary Report
RBTVR = Transaction Rate Vs. Response Time Summary
RBTVS = Transaction Volume Summary Report
RBAAR = Approval Amount Report
RBTDR = Transaction Denial Report
RBRR = Reversal Report
RBTOA = Transaction Originator Report (Sorted by Activity)
RBTOD = Transaction Originator Report (Sorted by Interface Instance Name)
RBTOS = Transaction Originator Report (Activity Summary)
RBTOT = Transaction Originator Type Report
RBAR = Authorization Report
RBIR = Institution Report

By default, all reports generated by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities are sent to the spooler location specified in the RUN command. Using the value B causes the reports **not** specified in the OPTIONS parameter to be sent to the spooler location. Reports specified in the OPTIONS parameter are sent to the Transaction Analysis Utility Transaction History File (THF). The value B must be listed first, if it is used.

When using the TTR or RBTTT options, the AAR or RBAAR option must also be specified so that the information needed to generate the summary portion of the Transaction Log File Report—Summary and Totals is available. When using the SR, LAS, or DTR options, the SAS option must also be specified so that all station summary information is available. When using the RBTOD or RBTOT options, the RBTOA option must also be specified so that all interface instance summary information is available.

Reports to be written to the Transaction Analysis Utility Transaction History File (THF) are identified by acronyms specified following the OPTIONS parameter in the Transaction Analysis utility RUN command. Each report produced by the ATM, POS, Remote Banking, or Teller Transaction Analysis utility has a corresponding acronym in the OPTIONS parameter. If the acronym for a

particular report is specified following the **OPTIONS** parameter in the **RUN** command, that report is written to the THF. If the acronym is not specified, the report is not written to the THF.

The **OPTIONS** parameter is optional in the **RUN** command.

START

The **START** parameter specifies the time, in hours and minutes, within the specified TLF, PTLF, ITLF, or TTLF at which transactions start being included in the statistical reports produced by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities (the default is to report the entire TLF, PTLF, ITLF, or TTLF). This parameter is specified in 24-hour time (hh:mm), where the hours can range from 00 through 23 and the minutes can range from 00 through 59. The **START** parameter is used in conjunction with the **END** parameter and is optional in the **RUN** command, unless the **END** parameter has been specified.

TOMORROW

The **TOMORROW** parameter specifies that the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities produce a report including transactions processed between midnight and financial institution cutover (i.e., transactions from yesterday's and today's TLF, PTLF, ITLF, or TTLF). When the **TOMORROW** parameter is specified in the **RUN** command, the transaction analysis utilities open the log file with the date specified in the **FILE** parameter and the next day's log file. Data from the log file specified in the **FILE** parameter that does not apply is discarded. The **TOMORROW** parameter cannot be used with the **CLOCKDATE** parameter. The **TOMORROW** parameter is optional in the **RUN** command.

Running the Transaction Analysis Utilities

Once the ATM, POS, Remote Banking, or Teller Transaction Analysis utility has been installed, it can be run at any time from a TACL prompt. This subsection provides instructions for running the utilities and includes examples of the parameters you can enter when using each utility.

Command Entry for Running the Transaction Analysis Utilities

The command syntax for running any of the transaction analysis utilities is described below, followed by an explanation of each field that can be entered. To run a transaction analysis utility, move to the volume where the BASE24 performance tools reside and enter the following command.

Syntax

RUN *pgm* / IN *lconf*, [OUT *rptfile*] / *parameter* [,*parameter* [...]]

pgm — The object file name of the report program. For the BASE24-atm utility, the object file name is TLFO. For the BASE24-pos utility, the object file name is PTLFO. For the BASE24 Remote Banking utility, the object file name is ITLFTO. For the BASE24-teller utility, the object file name is TTLFO.

lconf — The name of the current BASE24 Logical Network Configuration File (LCONF).

rptfile — The destination where the report is sent. The destination can be a spooler location or a file. If this field contains a file name, the file must be created before the transaction analysis utility is run. To create the file, enter the following command from a TACL prompt:

```
FUP CREATE file-name, TYPE E, REC 132, BLOCK 4096, EXT 40,
BUFFERED
```

The *file-name* consists of eight characters. The first character is user-defined. The second character is usually an A to identify the file as a BASE24-atm file, a P to identify the file as a BASE24-pos file, an R to identify the file as a BASE24 Remote Banking file, or a T to identify the file as a BASE24-teller file. The final six characters of the file name provide the date of the TLF, PTLF, ITLF, or TTLF in YYMMDD format. The BUFFERED parameter should be used only if the destination for the report specified by the *rptfile* variable resides on a DP2 volume.

parameter — The parameters that can be specified in the RUN command. If a string of parameters is specified, each parameter must be separated by a comma in the RUN command.

In most cases, it is not necessary to specify the complete name of the parameter in the RUN command. Most keywords can be abbreviated to their first three characters. Characters that must be entered for a particular parameter are identified in the list below by uppercase letters. Optional characters or variables that are not required for a particular parameter are enclosed within brackets. Any variables required with the parameter are identified in the list by lowercase italicized letters (e.g. *rate*). For an explanation of each parameter and how it is used, refer to the topic “Reporting Options” provided earlier in this section. The following list identifies the available parameters:

- CAT[EGORY] *size* (Valid for the POS Transaction Analysis utility only)
- CLO[CKDATE]
- DIV[IDEBY] *number*
- END *hh:mm*
- ENT[RIES] *number*
- FIL[E] *yymmdd*
- HEL[P]
- [HIST]FIL[E] *yymmdd*
- HIS[TORY]
- INT[ERVAL] *mm*
- MAX[TPS] *rate*
- NEF *r lane f*
- NOS[POOL] *acronym* [*acronym* [...]]
- OPT[IONS] *acronym* [*acronym* [...]]
- STA[RT] *hh:mm*
- TOM[ORROW]

ATM Command Examples

The command for running the ATM Transaction Analysis utility can be modified to achieve certain results. The following examples describe two of the options available when running the ATM Transaction Analysis utility.

Sending Reports to a Spooler Location

By default, all reports produced by the ATM Transaction Analysis utility are sent to the spooler location specified in the RUN command. However, it is possible to eliminate reports from being sent to the spooler location by also specifying the NOSPOOL parameter in the RUN command. By using NOSPOOL, you can eliminate all or a specified group of reports from being sent to the spooler location. An example of the ATM Transaction Analysis utility RUN command to eliminate a specific group of reports from being sent to the spooler location is shown below. This example is followed by a description of each element entered in the RUN command.

Example

```
RUN TLFO / IN L1CONF, OUT $S.#TLFRPT / FILE 020625, CLO, INT 1,  
MAXTPS 4, ENTRIES 40, NOSPOOL TTR AAR TDR
```

In this example, the ATM Transaction Analysis utility being run is TLFO. This is the object name of the BASE24-atm utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the Transaction Log Files (TLFs) and BASE24-atm Terminal Data files used by the ATM Transaction Analysis utility.

The spooler location in the example is \$S.#TLFRPT. This is the location where the reports produced by the ATM Transaction Analysis utility are sent.

The FILE parameter specifies the date (in this case June 25, 2002) of the TLF against which the ATM Transaction Analysis utility is run. A date must be specified in the RUN command.

The CLOCKDATE parameter (CLO) in the example specifies that the ATM Transaction Analysis utility opens the TLFs for 06/25/02 (specified in the FILE parameter) and 06/26/02 and produces a 24-hour report beginning at midnight on 06/25/02.

The INTERVAL parameter (INT) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In this example, statistics are accumulated in one minute intervals.

In the example, the maximum transactions per second rate (MAXTPS) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the INTERVAL parameter. For more information on the MAXTPS parameter, refer to the “Reporting Options” topic. From the reports produced by the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities, the number of transactions per second at which the BASE24 system peaked over the busiest minute during a processing day can be determined. In this example, the maximum transactions per second (MAXTPS) was set to 4 and the interval (INT) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The ENTRIES parameter is used in conjunction with the MAXTPS parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In this example, the number of entries to be tracked on the report is 40.

The final parameter specified in the example is the NOSPOOL parameter. This parameter specifies the reports to exclude from being sent to the specified spooler location. To exclude all reports, you can specify the NOSPOOL parameter by itself in the RUN command. To exclude a specific group of reports, you can specify the NOSPOOL parameter followed by the acronyms associated with the reports you want to eliminate. Refer to the “Reporting Options” topic for a complete list of report names and their corresponding acronyms. In this example, the NOSPOOL parameter is followed by the TTR, AAR, and TDR report acronyms. These acronyms correspond to the Transaction Log File Totals Report, the Approval Amount Report, and the Transaction Denial Report, respectively. Therefore, these three reports are **not** sent to the spooler location specified in the RUN command.

Sending Reports to a Transaction Analysis Utility Transaction History File (THF)

You can also specify to send reports to a Transaction Analysis Utility Transaction History File (THF) where they can be stored for future evaluation and analysis. An example of the ATM Transaction Analysis utility RUN command used to send reports to a THF is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN TLFO / IN L1CONF, OUT $$.#TLFRPT / FILE 020625, CLO, INT 1,  
MAXTPS 4, ENTRIES 40, HISTORY, OPTIONS B TTR RTS TVS IR
```

In this example, the ATM Transaction Analysis utility being run is TLFO. This is the object name of the BASE24-atm utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the Transaction Log Files (TLFs) and BASE24-atm Terminal Data files used by the ATM Transaction Analysis utility.

The spooler location in the example is \$\$.#TLFRPT. This is the location where the reports produced by the ATM Transaction Analysis utility are sent.

The RUN command parameters specified in this example are the same as that specified when sending reports to a spooler location, except that the HISTORY and OPTIONS parameters are included and the NOSPOOL parameter is not needed.

The HISTORY parameter in the example causes report data to be written in unstructured format to the THF specified in the ATM-TRN-THF assign in the LCONF. The HISTORY parameter is used in conjunction with the OPTIONS parameter.

The OPTIONS parameter specifies which reports are written to the THF. In the example, the OPTIONS parameter is followed by TTR, RTS, TVS, and IR. These acronyms correspond to the Transaction Log File Totals Report, the Response Time Summary Report, the Transaction Volume Summary Report, and the Institution Report, respectively. The data for these four reports is written to the THF specified in the RUN command.

In addition, the letter B is specified as the first entry following the OPTIONS parameter. By default, all reports generated by the ATM Transaction Analysis utility are sent to the spooler location specified in the RUN command. Using the letter B option causes only the reports **not** specified in the OPTIONS parameter to be sent to the spooler location. Reports specified in the OPTIONS parameter are written to the THF. Therefore, in this example, all reports except for the Transaction Log File Totals Report, Response Time Summary Report, Transaction Volume Summary Report, and the Institution Report are sent to the specified spooler location. The data for these four reports is written to the specified THF.

Running Reports from Existing Transaction Analysis Utility Transaction History Files

You can also use the ATM Transaction Analysis utility to create a report using stored information from one or more existing Transaction Analysis Utility Transaction History Files (THFs). An example of the ATM Transaction Analysis utility RUN command used to generate reports from existing THFs is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN TLFO / IN L1CONF, OUT $$.#TLFRPT / HISTFILE 020625 020627-  
020630 020703, NOSPOOL TVR
```

In this example, the ATM Transaction Analysis utility being run is TLFO. This is the object name of the BASE24-atm utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the Transaction Log Files (TLFs) and BASE24-atm Terminal Data files used by the ATM Transaction Analysis utility.

The spooler location in the example is \$\$.#TLFRPT. This is the location where the reports produced by the ATM Transaction Analysis utility are sent.

The HISTFILE parameter specifies the dates (in this case, June 25, June 27–30, and July 3, 2002) of the THFs against which the ATM Transaction Analysis utility is run. A date must be specified in the RUN command.

The final parameter specified in the example is the NOSPOOL parameter. This parameter specifies the reports to exclude from being sent to the specified spooler location. In this example, the NOSPOOL parameter is followed by TVR. This acronym corresponds to the Transaction Rate Vs. Response Time Summary Report. Therefore, this report is **not** sent to the spooler location specified in the RUN command.

POS Command Examples

The command for running the POS Transaction Analysis utility can be modified to achieve specific results. The following examples describe two of the options available when running the POS Transaction Analysis utility.

Specifying Transaction Start and End Times

You can specify the time within the PTLF at which transactions start being included in the statistical reports produced by the POS Transaction Analysis utility. An example of the POS Transaction Analysis utility RUN command used when specifying the transaction start and end times is shown below. This example is followed by a description of each element entered in the RUN command.

Example

```
RUN PTLFO / IN L1CONF, OUT $$.#PTLFRPT / FILE 020625, START  
09:00, END 23:59, INT 1, MAXTPS 4, ENTRIES 40
```

In the example, the POS Transaction Analysis utility being run is PTLFO. This is the object name of the BASE24-pos utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the POS Transaction Log Files (PTLFs) and BASE24-pos Terminal Data files used by the POS Transaction Analysis utility.

The spooler location in the example is \$\$.#PTLFRPT. This is the location where the reports produced by the POS Transaction Analysis utility are sent.

The FILE parameter specifies the date (in this case June 25, 2002) of the PTLF against which the POS Transaction Analysis utility is run. A date must be specified in the RUN command.

The START parameter specifies the time, in hours and minutes, within the PTLF at which transactions start being included in the statistical reports produced by the POS Transaction Analysis utility. The START time in this example is 9:00 a.m.

The END parameter specifies the time, in hours and minutes, within the PTLF at which transactions stop being included in the statistical reports produced by the POS Transaction Analysis utility. The END time in the example is 23:59 p.m.

The INTERVAL parameter (INT) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In the example, statistics are accumulated in one minute intervals.

In the example, the maximum transactions per second rate (MAXTPS) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the INTERVAL parameter. For more information on the MAXTPS parameter, refer to the “Reporting Options” topic. From the reports produced by the POS Transaction Analysis utility, the number of transactions per second at which the BASE24 system peaked over the busiest minute during a processing day can be determined. In the command string shown in the example, the maximum transactions per second rate (MAXTPS) was set to 4 and the interval (INT) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The ENTRIES parameter is used in conjunction with the MAXTPS parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In this example, the number of entries to be tracked on the report is 40.

Sending Reports to a Transaction Analysis Utility Transaction History File (THF)

You can also specify to send reports to a Transaction Analysis Utility Transaction History File (THF) where they can be stored for future evaluation and analysis. An example of the POS Transaction Analysis utility RUN command used to send reports to a THF is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN PTLFO / IN L1CONF, OUT $$.#PTLFRPT / FILE 020625, CLO, INT  
1, MAXTPS 4, ENTRIES 40, HISTORY, OPTIONS B TTR RTS TVS IR
```

In the example, the POS Transaction Analysis utility being run is PTLFO. This is the object name of the BASE24-pos utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the POS Transaction Log Files (PTLFs) and BASE24-pos Terminal Data files used by the POS Transaction Analysis utility.

The spooler location in the example is `$$.#PTLFRPT`. This is the location where the reports produced by the POS Transaction Analysis utility are sent.

The `FILE` parameter specifies the date (in this case June 25, 2002) of the PTLF against which the POS Transaction Analysis utility is run. A date must be specified in the `RUN` command.

The `CLOCKDATE` parameter (`CLO`) in the example specifies that the POS Transaction Analysis utility opens the PTLFs for 06/25/02 (specified in the `FILE` parameter) and 06/26/02 and produces a 24-hour report beginning at midnight on 06/25/02.

The `INTERVAL` parameter (`INT`) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In the example, statistics are accumulated in one minute intervals.

In the example, the maximum transactions per second rate (`MAXTPS`) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the `INTERVAL` parameter. For more information on the `MAXTPS` parameter, refer to the “Reporting Options” topic. From the reports produced by the POS Transaction Analysis utility, the number of transactions per second at which the `BASE24` system peaked over the busiest minute during a processing day can be determined. In the command string shown in the example, the maximum transactions per second rate (`MAXTPS`) was set to 4 and the interval (`INT`) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The `ENTRIES` parameter is used in conjunction with the `MAXTPS` parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In this example, the number of entries to be tracked on the report is 40.

The `HISTORY` parameter in the example causes report data to be written in unstructured format to the THF specified in the `POS-TRN-THF` assign in the `LCONF`. The `HISTORY` parameter is used in conjunction with the `OPTIONS` parameter.

The `OPTIONS` parameter specifies which reports are written to the THF. In the example, the `OPTIONS` parameter is followed by `TTR`, `RTS`, `TVS`, and `IR`. These acronyms correspond to the Transaction Log File Totals Report, the Response Time Summary Report, the Transaction Volume Summary Report, and the Institution Report, respectively. The data for these four reports is written to the THF specified in the `RUN` command.

In addition, the letter B is specified as the first entry following the OPTIONS parameter. By default, all reports generated by the POS Transaction Analysis utility are sent to the spooler location specified in the RUN command. Using the value B causes only the reports **not** specified in the OPTIONS parameter to be sent to the spooler location. Reports specified in the OPTIONS parameter are written to the THF. If used, B must be the first entry following the OPTIONS parameter in the RUN command. Therefore, in this example, all reports except for the Transaction Log File Totals Report, Response Time Summary Report, Transaction Volume Summary Report, and Institution Report are sent to the specified spooler location. The data for these four reports is written to the specified THF.

Creating the Category Report

Although much of the information produced by the ATM, POS, and Teller Transaction Analysis utilities is similar in nature, the POS Transaction Analysis utility can create an additional report produced exclusively for BASE24-pos. This report is the Category Report. An example of the POS Transaction Analysis utility RUN command used to create the Category Report is shown below. This example is followed by a description of each element entered in the RUN command.

Example

```
RUN PTLFO / IN L1CONF, OUT $$.#PTLFRPT / FILE 020625, CLO, INT  
1, MAXTPS 4, ENTRIES 40, CAT 5000, HISTORY, OPTIONS B CR  
TTR RTS
```

In the example, the POS Transaction Analysis program being run is PTLFO. This is the object name of the BASE24-pos report program.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the POS Transaction Log Files (PTLFs) and BASE24-pos Terminal Data files used by the POS Transaction Analysis utility.

The spooler location in the example is \$\$.#PTLFRPT. This is the location where the reports produced by the POS Transaction Analysis utility are sent.

The FILE parameter specifies the date (in this case June 25, 2002) of the PTLF for which the POS Transaction Analysis utility is run. A date must be specified in the RUN command.

The **CLOCKDATE** parameter (**CLO**) in the example specifies that the POS Transaction Analysis utility opens the PTLFs for 06/25/02 (specified in the **FILE** parameter) and 06/26/02 and produces a 24-hour report beginning at midnight on 06/25/02.

The **INTERVAL** parameter (**INT**) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In the example, statistics are accumulated in one minute intervals.

In the example, the maximum transactions per second rate (**MAXTPS**) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the **INTERVAL** parameter. For more information on the **MAXTPS** parameter, refer to the “Reporting Options” topic. From the reports produced by the POS Transaction Analysis utility, the number of transactions per second at which the **BASE24** system peaked over the busiest minute during a processing day can be determined. In the command string shown in the example, the maximum transactions per second rate (**MAXTPS**) was set to 4 and the interval (**INT**) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The **ENTRIES** parameter is used in conjunction with the **MAXTPS** parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In this example, the number of entries to be tracked on the report is 40.

The **CATEGORY** parameter (**CAT**) specifies that a maximum of 5000 unique category records are expected in the Category Report.

The **HISTORY** parameter in the example causes report data to be written in unstructured format to the THF specified in the **POS-TRN-THF** assign in the **LCONF**. The **HISTORY** parameter is used in conjunction with the **OPTIONS** parameter.

The **OPTIONS** parameter specifies which reports are written to the THF. In the example, the **OPTIONS** parameter is followed by **CR**, **TTR**, and **RTS**. These acronyms correspond to the Category Report, the Transaction Log File Totals Report, and the Response Time Summary Report. The data for these four reports is written to the THF specified in the **RUN** command.

In addition, the letter **B** is specified as the first entry following the **OPTIONS** parameter. By default, all reports generated by the POS Transaction Analysis utility are sent to the spooler location specified in the **RUN** command. Using the value **B** causes only the reports **not** specified in the **OPTIONS** parameter to be sent

to the spooler location. Reports specified in the **OPTIONS** parameter are written to the THF. If used, the value **B** must be the first entry following the **OPTIONS** parameter in the **RUN** command. Therefore, in this example, all reports except for the Category Report, Transaction Log File Totals Report, and Response Time Summary Report are sent to the specified spooler location. The data for these four reports is written to the specified THF.

Note: The Category Report is never sent to the spooler location. If the Category Report is produced, the **HISTORY** and **OPTIONS** parameters must be included in the **RUN** command, and the acronym **CR** must be listed following the **OPTIONS** parameter.

Remote Banking Command Examples

The command for running the Remote Banking Transaction Analysis utility can be modified to achieve certain results. The following examples describe two of the options available when running the Remote Banking Transaction Analysis utility.

Sending Reports to a Spooler Location

By default, all reports produced by the Remote Banking Transaction Analysis utility are sent to the spooler location specified in the RUN command. However, it is possible to eliminate reports from being sent to the spooler location by also specifying the NOSPOOL parameter in the RUN command. By using the NOSPOOL parameter, you can eliminate all or a specified group of reports from being sent to the spooler location. An example of the Remote Banking Transaction Analysis utility RUN command to eliminate a specific group of reports from being sent to the spooler location is shown below. This example is followed by a description of each element entered in the RUN command.

Example

```
RUN ITLFTO / IN L1CONF, OUT $$.#ITLFRPT / FILE 020625, CLO, INT 1,  
MAXTPS 4, ENTRIES 40, NOSPOOL RBTTR RBAAR RBTDR
```

In this example, the Remote Banking Transaction Analysis utility being run is ITLFTO. This is the object name of the BASE24 Remote Banking utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the ITS Transaction Log Files (ITLFs) used by the Remote Banking Transaction Analysis utility.

The spooler location in the example is \$\$.#ITLFRPT. This is the location where the reports produced by the Remote Banking Transaction Analysis utility are sent.

The FILE parameter specifies the date (in this case June 25, 2002) of the ITLF against which the Remote Banking Transaction Analysis utility is run. A date must be specified in the RUN command.

The CLOCKDATE parameter (CLO) in the example specifies that the Remote Banking Transaction Analysis utility opens the ITLFs for 06/25/02 (specified in the FILE parameter) and 06/26/02 and produces a 24-hour report beginning at midnight on 06/25/02.

The INTERVAL parameter (INT) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In this example, statistics are accumulated in one minute intervals.

In the example, the maximum transactions per second rate (MAXTPS) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the INTERVAL parameter. For more information on the MAXTPS parameter, refer to the “Reporting Options” topic. From the reports produced by the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities, the number of transactions per second at which the BASE24 system peaked over the busiest minute during a processing day can be determined. In this example, the maximum transactions per second (MAXTPS) was set to 4 and the interval (INT) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The ENTRIES parameter is used in conjunction with the MAXTPS parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In this example, the number of entries to be tracked on the report is 40.

The final parameter specified in the example is the NOSPOOL parameter. This parameter specifies the reports to exclude from being sent to the specified spooler location. To exclude all reports, you can specify the NOSPOOL parameter by itself in the RUN command. To exclude a specific group of reports, you can specify the NOSPOOL parameter followed by the acronyms associated with the reports you want to eliminate. Refer to the “Reporting Options” topic for a complete list of report names and their corresponding acronyms. In this example, the NOSPOOL parameter is followed by the RBTTR, RBAAR, and RBTDR report acronyms. These acronyms correspond to the Transaction Log File Totals Report, the Approval Amount Report, and the Transaction Denial Report, respectively. Therefore, these three reports are **not** sent to the spooler location specified in the RUN command.

Sending Reports to a Transaction Analysis Utility Transaction History File (THF)

You can also specify to send reports to a Transaction Analysis Utility Transaction History File (THF) where they can be stored for future evaluation and analysis. An example of the Remote Banking Transaction Analysis utility RUN command used to send reports to a THF is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN ITLFTO / IN L1CONF, OUT $$.#ITLFRPT / FILE 020625, CLO, INT 1,  
MAXTPS 4, ENTRIES 40, HISTORY, OPTIONS B RBTTR RBRTS  
RBTVS RBIR
```

In this example, the Remote Banking Transaction Analysis utility being run is ITLFTO. This is the object name of the BASE24 Remote Banking utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the ITS Transaction Log Files (ITLFs) used by the Remote Banking Transaction Analysis utility.

The spooler location in the example is \$\$.#ITLFRPT. This is the location where the reports produced by the Remote Banking Transaction Analysis utility are sent.

The RUN command parameters specified in this example are the same as that specified when sending reports to a spooler location, except that the HISTORY and OPTIONS parameters are included and the NOSPOOL parameter is not needed.

The HISTORY parameter in the example causes report data to be written in unstructured format to the THF specified in the RB-TRN-THF assign in the LCONF. The HISTORY parameter is used in conjunction with the OPTIONS parameter.

The OPTIONS parameter specifies which reports are written to the THF. In the example, the OPTIONS parameter is followed by RBTTR, RBRTS, RBTVS, and RBIR. These acronyms correspond to the Transaction Log File Totals Report, the Response Time Summary Report, the Transaction Volume Summary Report, and the Institution Report, respectively. The data for these four reports is written to the THF specified in the RUN command.

In addition, the letter B is specified as the first entry following the OPTIONS parameter. By default, all reports generated by the Remote Banking Transaction Analysis utility are sent to the spooler location specified in the RUN command. Using the letter B option causes only the reports **not** specified in the OPTIONS parameter to be sent to the spooler location. Reports specified in the OPTIONS parameter are written to the THF. Therefore, in this example, all reports except for the Transaction Log File Totals Report, Response Time Summary Report, Transaction Volume Summary Report, and Institution Report are sent to the specified spooler location. The data for these four reports is written to the specified THF.

Running Reports from Existing Transaction Analysis Utility Transaction History Files

You can also use the Transaction Analysis utility to create a report using stored information from one or more existing Transaction Analysis Utility Transaction History Files (THFs). An example of the Remote Banking Transaction Analysis utility RUN command used to generate reports from existing THFs is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN ITLFTO / IN L1CONF, OUT $$.#ITLFRPT / HISTFILE 020625  
020627-020630 020703, NOSPOOL RBTVR
```

In this example, the Remote Banking Transaction Analysis utility being run is ITLFTO. This is the object name of the BASE24 Remote Banking utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the ITS Transaction Log Files (ITLFs) used by the Remote Banking Transaction Analysis utility.

The spooler location in the example is \$\$.#ITLFRPT. This is the location where the reports produced by the Remote Banking Transaction Analysis utility are sent.

The HISTFILE parameter specifies the dates (in this case, June 25, June 27–30, and July 3, 2002) of the THFs against which the Remote Banking Transaction Analysis utility is run. A date must be specified in the RUN command.

The final parameter specified in the example is the NOSPOOL parameter. This parameter specifies the reports to exclude from being sent to the specified spooler location. In this example, the NOSPOOL parameter is followed by RBTVR. This acronym corresponds to the Transaction Rate Vs. Response Time Summary Report. Therefore, this report is **not** sent to the spooler location specified in the RUN command.

Teller Command Examples

The command for running the Teller Transaction Analysis utility can be modified to achieve specific results. The following examples describe two of the options available when running the Teller Transaction Analysis utility.

Specifying Transaction Start and End Times

You can specify the time within the TTLF at which transactions start being included in the statistical reports produced by the Teller Transaction Analysis utility. An example of the Teller Transaction Analysis utility RUN command used when specifying the transaction start and end times is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN TTLFO / IN L1CONF / FILE 020625, START 09:00, END 23:59, INT 1,  
MAXTPS 4, ENTRIES 40
```

In the example, the Teller Transaction Analysis program being run is TTLFO. This is the object name of the BASE24-teller utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the Teller Transaction Log Files (TTLFs) and Teller Terminal Data Files (TTDFs) used by the Teller Transaction Analysis utility.

The FILE parameter specifies the date (in this case June 25, 2002) of the TTLF for which the Teller Transaction Analysis utility is run. A date must be specified in the RUN command.

The START parameter specifies the time, in hours and minutes, within the TTLF at which transactions start being included in the statistical reports produced by the Teller Transaction Analysis utility. The START time in the example is 9:00 a.m.

The END parameter specifies the time, in hours and minutes, within the TTLF at which transactions stop being included in the statistical reports produced by the Teller Transaction Analysis utility. The END time in the example is 23:59 p.m.

The INTERVAL parameter (INT) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In the example, statistics are accumulated in one minute intervals.

In the example, the maximum transaction per second rate (MAXTPS) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the INTERVAL parameter. For more information on the MAXTPS parameter, refer to the “Reporting Options” topic. From the reports produced by the Teller Transaction Analysis utility, the number of transactions per second at which the BASE24 system peaked over the busiest minute during a processing day can be determined. In the command string shown in the example, the Maximum Transactions Per Second (MAXTPS) was set to 4 and the Interval (INT) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The ENTRIES parameter is used in conjunction with the MAXTPS parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In the example, the number of entries to be tracked on the report is 40.

Sending Reports to a Transaction Analysis Utility Transaction History File (THF)

You can also specify to send reports to a Transaction Analysis Utility Transaction History File (THF) where they can be stored for future evaluation and analysis. An example of the Teller Transaction Analysis utility RUN command used to send reports to a THF is shown below. The example is followed by a description of each element entered in the RUN command.

Example

```
RUN TTLFO / IN L1CONF, OUT $$.#TTLFRPT / FILE 020625, CLO, INT 1,  
MAXTPS 4, ENTRIES 40, HISTORY, OPTIONS B TTR RTS TVS IR
```

In this example, the Teller Transaction Analysis program being run is TTLFO. This is the object name of the BASE24-teller utility.

L1CONF is the name of the current Logical Network Configuration File (LCONF). If only the file name is specified in the RUN command, TACL uses the current system, volume, and subvolume. The LCONF contains the names and locations of the Teller Transaction Log Files (TTLFs) and Teller Terminal Data Files (TTDFs) used by the Teller Transaction Analysis utility.

The spooler location in the example is `$$.#TTLFRPT`. This is the location where the reports produced by the Teller Transaction Analysis utility are sent.

The `FILE` parameter specifies the date (in this case June 25, 2002) of the TTLF against which the Teller Transaction Analysis utility is run. A date must be specified in the `RUN` command.

The `CLOCKDATE` parameter (`CLO`) in the example specifies that the Teller Transaction Analysis utility opens the TTLFs for 06/25/02 (specified in the `FILE` parameter) and 06/26/02 and produces a 24-hour report beginning at midnight on 06/25/02.

The `INTERVAL` parameter (`INT`) specifies the interval, in minutes, at which statistics are accumulated for use on statistical reports. In the example, statistics are accumulated in one minute intervals.

In the example, the maximum transaction per second rate (`MAXTPS`) is calculated as four. This calculation is performed by averaging the highest expected transaction rate, in transactions per second, over the value specified in the `INTERVAL` parameter. For more information on the `MAXTPS` parameter, refer to the “Reporting Options” topic. From the reports produced by the Teller Transaction Analysis utilities, the number of transactions per second at which the BASE24 system peaked over the busiest minute during a processing day can be determined. In the command string shown in the example, the Maximum Transactions Per Second (`MAXTPS`) was set to 4 and the Interval (`INT`) was set to 1 because the user estimated a peak transaction rate of 3 to 4 transactions per second, averaged over the busiest minute.

The `ENTRIES` parameter is used in conjunction with the `MAXTPS` parameter to determine the number of transaction throughput rate ranges that are displayed in the System TPS field on the Transaction Rate Vs. Transaction Response Report. In the example, the number of entries to be tracked on the report is 40.

The `HISTORY` parameter in the example causes report data to be written in unstructured format to the THF specified in the `TLR-TRN-THF` assign in the `LCONF`. The `HISTORY` parameter is used in conjunction with the `OPTIONS` parameter.

The final parameter specified in the example is the `OPTIONS` parameter. This parameter is used to determine which reports are written to the THF. In the example, the `OPTIONS` parameter is followed by `TTR`, `RTS`, `TVS`, and `IR`. These acronyms correspond to the Transaction Log File Totals Report, Response Time

Summary Report, Transaction Volume Summary Report, and Institution Report, respectively. These four reports are written to the THF specified in the RUN command.

In addition, the letter B is specified as the first entry following the OPTIONS parameter. By default, all reports generated by the Teller Transaction Analysis utility are sent to the spooler location specified in the RUN command. Using the value B causes only the reports **not** specified in the OPTIONS parameter to be sent to the spooler location. Reports specified in the OPTIONS parameter are written to the THF. If used, the value B must be the first entry following the OPTIONS parameter in the RUN command. Therefore, in the example, all reports except for the Transaction Log File Totals Report, Response Time Summary Report, Transaction Volume Summary Report, and Institution Report are sent to the specified spooler location. These four reports are sent to the specified THF.

Transaction Analysis Utility Reports

The following reports are produced by the ATM, POS, Remote Banking, and Teller Transaction Analysis utilities. The reports are similar, if not identical, for all four products. Therefore, the report documentation has been combined for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller, and the differences between the reports for each product are identified in the documentation.

- Transaction Log File Report
- Transaction Log File Report — Summary and Totals
- Response Time Summary Report
- Transaction Rate Vs. Response Time Summary
- Transaction Volume Summary Report
- Approval Amount Report
- Transaction Denial Report
- Reversal Report
- Authorization Report
- Institution Report

The next set of reports listed below are produced for BASE24-atm, BASE24-pos, and BASE24-teller, except for the Category Report. The Category Report is produced for BASE24-pos only. The reports are similar, if not identical, for all three products. Therefore, the report documentation has been combined for BASE24-atm, BASE24-pos, and BASE24-teller, and the differences between the reports for each product are identified in the documentation.

- Stations Sorted by Activity Report
- Stations Sorted by Name, Detail Report
- Station Activity Summary Report
- Stations Sorted by Line Report
- Device Type Report
- Device Handler Report
- Category Report

The following reports are generated by the Remote Banking Transaction Analysis utility only:

- Transaction Originator Report (Sorted by Interface Instance Name)
- Transaction Originator Report (Sorted by Activity)
- Transaction Originator Report (Activity Summary)
- Transaction Originator Type Report

Each report can be written to a Transaction Analysis Utility Transaction History File (THF) if the HISTORY and OPTIONS parameters are specified in the RUN command.

Transaction Log File Report

The Transaction Log File Report provides a representation of transaction activity that occurred during each specified time interval taken from the Transaction Log Files (TLFs), POS Transaction Log Files (PTLFs), ITS Transaction Log Files (ITLFs), or Teller Transaction Log Files (TTLFs).

For BASE24-atm, BASE24-pos, and BASE24-teller, this report contains financial transactions only (including approved or denied transactions, reversals, and force posts). For BASE24 Remote Banking, this report also includes non-financial transactions. The report does not include such records as administrative or settlement records.

Two sample Transaction Log File Reports are shown on the following pages—one for BASE24-atm and one for BASE24 Remote Banking. The format of the report for BASE24 Remote Banking is slightly different from the format for BASE24-atm, BASE24-pos, and BASE24-teller. The format of the report for BASE24-pos and BASE24-teller is identical to the format for BASE24-atm. The sample reports are followed by descriptions of their fields.

Page 10	System \ACI	ATM	TRANSACTION LOG			FILE	REPORT			Logical network LLLL			MM/DD/YY	
!-----PERFORMANCE-----!-----ORIGIN-----!-----TRANSACTION PROCESSING-----!-----Switch-----!														
Time	Interval	Max Total	Trans	Rate	Foreign	Local	Aprvd	Denied	N-A	Resp time	Aprvd	Denied	N-A	Resp time
17:06-17:06	3	39	0.65	100%	69%	5%	3.07	3%	5%	0.75	13%	5%	3.71	
17:07-17:07	3	43	0.72	100%	60%	2%	3.26	2%	14%	0.82	21%		4.88	
17:08-17:08	2	39	0.65	100%	67%	3%	3.42		15%	0.79	15%		3.66	
17:09-17:09	2	39	0.65	100%	72%		3.29	3%	5%	0.83	21%		3.75	
17:10-17:10	3	43	0.72	100%	70%	2%	3.58	2%	2%	0.75	23%		4.10	
17:11-17:11	4	55	0.92	100%	69%		3.57	2%	9%	0.79	20%		3.90	
17:12-17:12	3	47	0.78	100%	62%	2%	3.50		4%	0.75	30%	2%	4.66	
17:13-17:13	2	31	0.52	100%	71%	6%	3.20	3%	6%	0.83	13%		5.00	
17:14-17:14	3	49	0.82	100%	57%	4%	3.60		16%	0.78	18%	4%	7.36	
17:15-17:15	3	55	0.92	100%	69%	5%	3.90		4%	0.87	22%		3.83	
17:16-17:16	3	49	0.82	100%	63%	8%	3.97		2%	0.75	24%	2%	4.38	
17:17-17:17	3	43	0.72	100%	65%	12%	3.54	2%	5%	7.58	14%	2%	4.57	
17:18-17:18	4	50	0.83	100%	78%	10%	3.54		4%	0.75	8%		3.25	
17:19-17:19	3	48	0.80	100%	75%	4%	3.52	2%	4%	7.58	13%	2%	5.00	
17:20-17:20	4	42	0.70	100%	74%	7%	3.88		5%	0.87	12%	2%	7.16	
17:21-17:21	2	43	0.72	100%	67%	9%	3.48	2%	5%	0.83	14%	2%	8.00	
17:22-17:22	3	32	0.53	100%	66%	9%	3.62		6%	0.87	16%	3%	5.66	
17:23-17:23	4	47	0.78	100%	64%	11%	3.62		11%	0.80	15%		3.42	
17:24-17:24	2	41	0.68	100%	63%	7%	3.72		10%	0.81	20%		4.37	
17:25-17:25	3	38	0.63	100%	79%		3.63		8%	0.83	13%		3.40	
17:26-17:26	4	42	0.70	100%	71%	2%	3.25	2%	5%	0.75	19%		5.87	
17:27-17:27	3	37	0.62	100%	78%	3%	3.46	3%	8%	0.75	8%		3.33	
17:28-17:28	3	39	0.65	100%	77%		3.33		3%	0.75	18%	3%	2.75	
17:29-17:29	3	33	0.55	100%	76%	3%	3.26	3%	3%	0.75	15%		3.20	
17:30-17:30	3	43	0.72	100%	63%		3.48		7%	0.91	28%	2%	4.00	
17:31-17:31	3	43	0.72	100%	65%	2%	3.37		16%	0.78	12%	5%	5.42	
17:32-17:32	4	53	0.88	100%	77%		3.36		2%	0.75	15%	6%	4.10	
17:33-17:33	4	55	0.92	100%	60%		3.09		7%	0.75	27%	5%	4.33	
17:34-17:34	2	54	0.90	100%	67%		3.36	2%	2%	0.75	26%	4%	3.68	
17:35-17:35	4	52	0.87	100%	79%	2%	3.73				17%	2%	5.00	
17:36-17:36	3	54	0.90	100%	57%	2%	3.58		4%	0.75	30%	7%	4.55	
17:37-17:37	3	39	0.65	100%	64%		3.29	3%	8%	0.81	18%	8%	5.20	
17:38-17:38	3	30	0.50	100%	77%		3.30		7%	0.87	13%	3%	6.40	
17:39-17:39	3	39	0.65	100%	77%		3.10		5%	0.87	15%	3%	5.00	
17:40-17:40	3	54	0.90	100%	70%		3.24		6%	0.75	24%		5.07	
17:41-17:41	4	50	0.83	100%	58%		3.60		4%	0.87	26%	12%	4.66	
17:42-17:42	3	49	0.82	100%	73%	2%	3.33	4%	2%	0.83	18%	2%	5.90	
17:43-17:43	3	57	0.95	100%	68%	2%	3.46	5%	4%	0.87	18%	5%	5.15	
17:44-17:44	4	51	0.85	100%	71%		3.80	2%	8%	0.75	16%	4%	4.00	
17:45-17:45	4	44	0.73	100%	64%		3.53	2%	11%	0.83	16%	7%	5.80	
17:46-17:46	3	49	0.82	100%	69%		3.55	2%	6%	0.75	18%	4%	4.18	
17:47-17:47	3	44	0.73	100%	68%		4.06	2%	5%	0.75	25%		4.81	
17:48-17:48	4	41	0.68	100%	73%		4.06		10%	0.87	15%	2%	6.00	
17:49-17:49	2	37	0.62	100%	76%	8%	3.22	3%		0.75	14%		3.60	

Page	10	System \ACI	RB	TRANSACTION	LOG	FILE	REPORT	Logical network	LLLL	MM/DD/YY
!---PERFORMANCE---!-----ORIGIN-----!-----TRANSACTION PROCESSING-----! !-----Host-----!-----BASE24-telebanking-----! !-----BASE24-billpay-----!-----Switch-----! Time !Max Total Tran ! Interval !TPS Trans Rate ! Foreign Local BASE24-billpay ! Aprvd Denied N-A Resp time ! Aprvd Denied N-A Resp time !										
	0:00- 0:59									
	1:00- 1:59									
	2:00- 2:59									
	3:00- 3:59									
	4:00- 4:59									
	5:00- 5:59									
	6:00- 6:59									
	7:00- 7:59	2	11	9%	81%		54%	10.28		
	8:00- 8:59	2	25		45%	12.18	48%	16.52		
	9:00- 9:59	2	8		52%	19.85	50%	107.82		
	10:00-10:59	1	2		50%	19.00	50%	18.05		
	11:00-11:59	1	2		50%	26.82	100%	161.83		
	12:00-12:59	1	1		100%					
	13:00-13:59	1	2		100%	21.05	100%	172.89		
	14:00-14:59									
	15:00-15:59									
	16:00-16:59									
	17:00-17:59									
	18:00-18:59									
	19:00-19:59									
	20:00-20:59									
	21:00-21:59									
	22:00-22:59									

Time Interval

The time interval for which transaction activity is monitored. Each line in this report shows the transaction activity that occurred during a specified time interval. The length of this reporting time interval can be specified at runtime using the INTERVAL parameter. The interval defaults to one minute.

For the BASE24 Remote Banking report, there are two lines of data for each time interval in the Transaction Processing section. The first line of data provides statistics for transactions authorized by a host or BASE24-telebanking. The second line of data provides statistics for transactions authorized by BASE24-billpay or that were sent to an interchange or Interchange Interface process for authorization.

PERFORMANCE

The Performance section includes information about the maximum number of transactions logged per second during the specified time interval, the total number of transactions logged during the specified time interval, and the average transaction rate at which transactions were logged during a specified time interval.

Max TPS — The maximum number of transactions logged during a single second within the specified time interval.

Total Trans — The total number of transactions logged within the specified time interval.

Tran Rate — The average rate, in transactions per second, at which transactions were logged during the specified time interval.

ORIGIN

The Origin section indicates where transactions originated. Two fields are displayed under the Origin heading for BASE24-atm, BASE24-pos, and BASE24-teller—Foreign and Local. An additional field is displayed under the Origin heading for BASE24 Remote Banking—BASE24-billpay.

The percentages displayed in the fields in this section should total 100 percent.

Foreign — The percentage of transactions (including reversals and adjustments) logged during the specified time interval that originated from a host, switch, or third party acquiring interface.

Local — The percentage of transactions (including reversals) logged during the specified time interval that originated from a terminal or device controlled by the BASE24 system or from the BASE24 system itself (e.g., an Authorization process or module).

BASE24-billpay — The percentage of transactions (including reversals) logged during the specified time interval that originated from a BASE24-billpay process (i.e., Billpay Server process, Scheduled Transaction Initiator process, or Bank Advice process). This field is only displayed on reports generated by the Remote Banking Transaction Analysis utility.

TRANSACTION PROCESSING

The Transaction Processing section describes in detail where transactions that were processed during a specified time interval were authorized. For BASE24-atm, BASE24-pos, and BASE24-teller, three possibilities exist: host, BASE24, and interchange (switch). For BASE24 Remote Banking, four possibilities exist: host, BASE24-telebanking, BASE24-billpay, and interchange (switch).

Transactions that fall into the Host N-A (the host was not available) and Switch N-A (the interchange was not available) categories are also counted in either the approved or denied columns for the BASE24 system, depending on whether the BASE24 system performed stand-in authorization. Stand-in authorization takes place because the BASE24 system authorizes a transaction when communications lines to the authorizer are unavailable. If the host or interchange is not available and either stand-in authorization is not performed or is denied as a result of stand-in authorization, the transaction is counted in the denied column. If the host or interchange is not available and the transaction is approved as a result of stand-in authorization, the transaction is counted in the approved column. Therefore, the sum of the percentages under the Transaction Processing heading should equal 100 percent, excluding the Host N-A and Switch N-A fields.

Note: Due to rounding, the sum total of the values in the three approval and three denial fields may not equal exactly 100 percent.

Host — The fields in this section provide information about transactions that were sent to the host for authorization and includes transactions authorized by the host or a Host Interface process.

Statistics in this section include the percentage of the total transactions processed during the specified time interval that were approved or denied by the host. These percentages are shown in the Aprvd and Denied fields on the report. When this report is generated for BASE24-teller, an additional field, the Denied w/Override field, is also included.

The report includes the percentage of the total transactions processed during the specified time interval that were intended to be authorized by a host, but the host was not available. This information is shown in the N-A field on the report.

The average response time, in seconds, is shown in the Resp time field on this report. The response time is the approximate amount of time it took to respond to a transaction. When a transaction authorization request enters the BASE24 system, the first application process (i.e., Device Handler, Integrated Authorization Server, Host Interface, or Interchange Interface) that receives it places the current timestamp into the Standard Internal Message (STM, PSTM, or TSTM) or into the Internal Transaction Data (for BASE24 Remote Banking only). The transaction is then routed to the host for authorization. When the response from the host is ready to be logged to the TLF, PTLF, ITLF, or TTLF, the Authorization process or module enters the current time in the Standard Internal Message or Internal Transaction Data before it is returned to the transaction source. The response time is the time difference between these two timestamps.

The response time includes most of the BASE24 transaction processing time and all of the external time. It does not include the time required to receive the request and route the response over the communications line to the transaction source (e.g., for BASE24 Remote Banking, it does not include Integrated Endpoint processing time). It also does not include the time period starting with logging of the response by the Authorization process or module through handling of the response by the Host Interface process. However, it does provide a figure that can be used to get the majority of transaction response times and provide a representative approximation of actual user response times.

Response times for transactions that fall into the host not available (N-A) category are counted as BASE24 response times. When timeouts occur waiting on a host response, the BASE24 response time increases. In order to discard any possible invalid data in determining response times, the transaction analysis utilities do not allow the response time to exceed 60 seconds. If a response time in the Resp time field is 60 seconds, it is possible that an application process is not filling in the entry timestamp properly, and this condition should be investigated.

BASE24 — The fields in this section provide information about transactions that were sent to a BASE24 Authorization process or module for authorization.

Statistics in this section include the percentage of total transactions processed during the specified time interval that were approved or denied by a BASE24 Authorization process or module. These percentages are shown in the Aprvd and Denied fields on the report. When this report is generated for BASE24-teller, another field, the Denied w/Override field, is also included.

In addition, the report includes the percent of total transactions during the specified time interval that were intended to be authorized by a BASE24 Authorization process or module, but BASE24 authorization was not available. This information is shown in the N-A field on the report. These are considered force post transactions. Force post transactions are transactions that were approved or denied by a terminal or remote database while communications lines were down. These transactions are reported to the BASE24 system in advice messages.

The average response time is shown in the Resp time field on this report. The response time is the approximate amount of time it took to respond to the transaction. When a transaction request enters the BASE24 system, the first application process (i.e., Device Handler, Host Interface, or Interchange Interface) that receives it places the current timestamp into the Standard Internal Message (STM, PSTM, or TSTM). The transaction is then authorized. When the response from the host is ready to be logged to the TLF, PTLF, or TTLF, the Authorization process or module enters the current time in the Standard Internal Message before it is returned to the transaction source. The response time is the time difference between these two timestamps.

The response time includes most of the BASE24 transaction processing time. It does not include the time required to receive the request and route the response over the communications line to the transaction source. It also does not include the time required to log the response by the Authorization process or module. However, it does provide a figure that can be used to get the majority of transaction response times and provide a representative approximation of actual user response times.

Response times are not calculated for force post transactions. Force post transactions are transactions that were authorized by a terminal or host when the communications lines to the authorizer were unavailable.

In order to discard any possible invalid data in these two timestamp fields, the transaction analysis utilities do not allow the response time to exceed 60 seconds. If a response time in the Resp time field is 60 seconds, it is possible that an application process is not filling in the entry timestamp properly, and this condition should be investigated.

BASE24-telebanking — The fields in this section provide information about transactions that were sent to a BASE24-telebanking Integrated Authorization Server process or module for authorization.

Statistics in this section include the percentage of total transactions processed during the specified time interval that were approved or denied by an Integrated Authorization Server process. These percentages are shown in the Aprvd and Denied fields on the report.

In addition, the report includes the percent of total transactions during the specified time interval that were intended to be authorized by an Integrated Authorization Server process, but a process was not available. This information is shown in the N-A field on the report. These are considered force post transactions. Force post transactions are transactions that were approved or denied by a remote database while communications lines were down with a host. These transactions are reported to the BASE24 system from a host. Force post transactions are not currently allowed from a remote banking device.

The average response time is shown in the Resp time field on this report. The response time is the approximate amount of time it took to respond to the transaction. When an Integrated Authorization Server process receives a transaction request, it enters the current timestamp into the Internal Transaction Data (ITD). The transaction is then authorized. When the response is ready to be logged to the ITLF, the Integrated Authorization Server process enters the current time in the ITD before it is returned to the transaction source. The response time is the time difference between these two timestamps.

The response time includes most of the BASE24 transaction processing time. It does not include the time required for the BASE24 system to receive the request, route it to an Integrated Authorization Server process, and route the response back to the transaction source (i.e., it does not include the processing time required by the Integrated Endpoint process to route the request or the time needed to route the response over the communications line to the transaction source). It also does not include the time required to log the response by the Integrated Authorization Server process.

Response times that include the time used by the Integrated Endpoint process can be obtained from the statistics maintained by the Integrated Endpoint process. For more information on these statistics, refer to the ***BASE24 IEP Configuration Manual*** and the ***BASE24 IEP Objects Reference Manual***.

Response times are not calculated for force post transactions. Force post transactions are transactions that were authorized by a remote banking device or host when the communications lines to the authorizer were unavailable.

In order to discard any possible invalid data in these two timestamp fields, the transaction analysis utilities do not allow the response time to exceed 60 seconds. If a response time in the Resp time field is 60 seconds, it is possible that an application process is not filling in the entry timestamp properly, and this condition should be investigated.

BASE24-billpay — The fields in this section provide information about transactions that were sent to BASE24-billpay Billpay Server process for authorization.

Statistics in this section include the percentage of the total number of transactions processed during the specified time interval that were approved or denied by a Billpay Server process. These percentages are shown in the Aprvd and Denied fields on the report.

In addition, the report includes the percent of total transactions during the specified time interval that were intended to be authorized by a Billpay Server process, but a process was not available. This information is shown in the N-A field on the report. These are considered force post transactions. Force post transactions are transactions that were approved or denied by a remote database while communications lines were down with a host. These transactions are reported to the BASE24 system from a host. Force post transactions are not currently allowed from a remote banking device.

The average response time is shown in the Resp time field on this report. The response time is the approximate amount of time it took to respond to the transaction. When an Integrated Authorization Server process receives a transaction request, it enters the current timestamp into the Internal Transaction Data (ITD). The transaction is then sent to and authorized by a Billpay Server process. When the response is ready to be logged to the ITLF, the Integrated Authorization Server process enters the current time in the ITD before it is returned to the transaction source. The response time is the time difference between these two timestamps.

The response time includes most of the BASE24 transaction processing time. It does not include the time required for the BASE24 system to receive the request, route it to an Integrated Authorization Server process, and route the response back to the transaction source (i.e., it does not include the processing time required by the Integrated Endpoint process to route the request or the time needed to route the response over the communications line to the transaction source). It also does not include the time required to log the response by the Integrated Authorization Server process.

Response times that include the time used by the Integrated Endpoint process can be obtained from the statistics maintained by the Integrated Endpoint process. For more information on these statistics, refer to the *BASE24 IEP Configuration Manual* and the *BASE24 IEP Objects Reference Manual*.

Response times are not calculated for force post transactions. Force post transactions are transactions that were authorized by a remote banking device or host when the communications lines to the authorizer were unavailable.

In order to discard any possible invalid data in these two timestamp fields, the transaction analysis utilities do not allow the response time to exceed 60 seconds. If a response time in the Resp time field is 60 seconds, it is possible that an application process is not filling in the entry timestamp properly, and this condition should be investigated.

Switch — The fields in this section provide information about transactions that were sent to an interchange or Interchange Interface process for authorization. This field is not present on the BASE24-teller version of this report.

Statistics in this section include the percentage of the total number of transactions processed during the specified time interval that were approved or denied by an interchange or Interchange Interface process. These percentages are shown in the Aprvd and Denied fields on the report. When this report is generated for BASE24-teller, an additional field, the Denied W/Override field, is also included.

In addition, the report includes the percent of total transactions during the specified time interval that were intended to be authorized by an interchange, but the interchange was not available. These transactions are authorized by the Interchange Interface process rather than by the interchange itself. Therefore, this field can include transactions rejected by the Interchange Interface process, even though the interchange is available. For example, transactions routed to the Interchange Interface process with a transaction type that is not allowed in the Interchange Configuration File (ICF) or that is not allowed for the transaction profile configured in the Enhanced Interchange Configuration File (ICFE), are counted in this column.

The average response time is also shown on this report. The response time is the approximate amount of time it took to respond to the transaction. When a transaction request enters the BASE24 system, the first application process (Device Handler, Integrated Authorization Server, Host Interface, or Interchange Interface) that receives it places the current timestamp into the Standard Internal Message (STM, PSTM, or TSTM) or Internal Transaction Data (for BASE24 Remote Banking only). The transaction is then routed to the interchange or Interchange Interface process for authorization. When the response is ready to be logged to the TLF, PTLF, ITLF, or TTLF, the Authorization process or module enters the current time in the Standard Internal Message or Internal Transaction Data before it is returned to the transaction source. The response time is the time difference between these two timestamps.

The response time includes most of the BASE24 transaction processing time and all of the external time. It does not include the time required to receive the request and route the response over the communications line to the transaction source (e.g., for BASE24 Remote Banking, it does not include Integrated Endpoint processing time). It also does not include the time period starting with logging of the response by the Authorization process or module through handling of the response by the interchange or Interchange Interface process. However, it does provide a figure that can be used to get the majority of transaction response times and provide a representative approximation of actual user response times.

Response times for transactions that fall into the Switch N-A category are counted as BASE24 response times. When timeouts occur waiting on an interchange response, the BASE24 response time increases. In order to discard any possible invalid data in determining response times, the transaction analysis utilities do not allow the response time to exceed 60 seconds. If a response time in the Resp time field is 60 seconds, it is possible that an application process is not filling in the entry timestamp properly, and this condition should be investigated.

Transaction Log File Report—Summary and Totals

This report provides a one-page summary of the detailed data found in the Transaction Log File Report. It also provides additional summary information concerning the number of reversal transactions and transaction types other than administrative or settlement transactions. Four sample reports are included in this subsection, one for BASE24-atm, one for BASE24-pos, one for BASE24-teller, and one for BASE24 Remote Banking. The sample reports are followed by descriptions of their fields.

Page	42	System \ACI	ATM	TRANSACTION	LOG	FILE	REPORT	Logical network	LLLL	MM/DD/YY
Summary and Totals										
Financial transactions read										
Administrative transactions read										
Exceptions financial transaction read										
Key maintenance records read										
Unclassifiable records read										
Total TLF records read										
Maximum Transactions Per Second										
Maximum Transaction Rate										
BASE24 Average Response Time										
Host Average Response Time										
Switch Average Response Time										
Transaction Source:										
Transaction Processing:										
N-A										
Approved Denied										
Host										
BASE24										
Switch										
TOTAL										
Transaction Type:										
Withdrawal										
Deposit										
Deposit w/Cash Back										
Split Deposit										
Balance Inquiry										
Transfer										
Payment										
Summary										

Nov-2007 R6.0v8 BA-UT020-01
ACI Worldwide, Inc.

Page 43	System \ACI	R B	T R A N S A C T I O N	L O G	F I L E	R E P O R T	Logical network	LLLL	MM/DD/YY
Approved									
Transaction Type:									

Customer Initiated									

CSR Initiated									

Page 42	System \ACI	TELLER	TRANSACTION LOG				FILE	REPORT				Logical network LLLL				MM/DD/YY
Summary and Totals																
Financial transactions read 132																
Administrative transactions read 48																
Exceptions financial transaction read 0																
Key maintenance records read 1																
Unclassifiable records read 0																

Total TTTLF records read 181																
=====																
Maximum Transactions Per Second 6																
Maximum Transaction Rate 2.43																
BASE24 Average Response Time .14																
Transaction Source:																
132 Local (0 0% reversed)																
0 Foreign (0 0% reversed)																
When Normal Destination N-A:																
Approved Denied Overrid																
Host	0 !	0 !	0 !	Host N-A !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0,	
0%																
BASE24	120 !	70 !	!	190,100% BASE24 N-A!	12 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	12,	
100%																
Switch	0 !	0 !	!	0, 0% Switch N-A!	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0,	
0%																
TOTAL	120 !	70 !	0 !	190	TOTAL N-A !	12 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	0 !	12	
12																
Transaction Type:																
Withdrawal	52	39%	!	360.00	52	74%	!	104	51%	!	!	!	!	!	!	
Deposit	24	18%	!	160.00	4	6%	!	28	14%	!	!	!	!	!	!	
Payment	48	36%	!		8	11%	!	56	28%	!	!	!	!	!	!	
Other	8	6%	!		2	3%	!	10	5%	!	!	!	!	!	!	

Summary																
132 65%																
70 35%																
202 100%																

Summary and Totals

This section provides a breakdown of the combined total number of transactions processed during the specified processing date by the host, the BASE24 system, and the interchange or Interchange Interface process. It also includes totals for transaction rates and response times for each of the three authorizing entities.

Financial transactions read — The number of records processed by the host, the BASE24 system, and the interchange or Interchange Interface process from the TLF, PTLF, ITLF, or TTLF that were financial transactions.

Non-Financial transactions read — The number of records processed by the host, the BASE24 system, and the interchange or Interchange Interface process from the ITLF that were non-financial transactions. This field is displayed for BASE24 Remote Banking only.

Administrative transactions read — The number of records processed by the host, the BASE24 system, and the interchange or Interchange Interface process from the TLF, PTLF, ITLF, or TTLF that were administrative transactions. No administrative transactions are currently logged to the ITLF.

Exceptions financial transactions read — The number of financial records in the TLF, PTLF, ITLF, or TTLF that could not be processed by the host, the BASE24 system, or an interchange or Interchange Interface process because the record contained invalid data. On the BASE24 Remote Banking report, this field is called Exceptions transactions read, and includes non-financial as well as financial transaction exceptions.

Key maintenance records read — The number of records processed by the host, the BASE24 system, and the interchange or Interchange Interface process from the TLF, PTLF, or TTLF that were used to maintain the alternate key files. Usually only one of these records exists—the first control record in the TLF, PTLF, or TTLF. It is used to control updating the alternate key files, which are used to index a file. This allows you to access a file randomly by a field value. Additional information about alternate key files can be found in the HP NonStop *ENSCRIBE Programmer's Guide*. Key maintenance records are not currently used by BASE24 Remote Banking.

Unclassifiable records read — The number of records that do not fall into any of the previously described categories.

Total TLF records read — The combined total number of TLF, PTLF, ITLF, or TTLF records that were processed by the host, the BASE24 system, and the interchange or Interchange Interface process for a specific processing day. On the BASE24-pos report, this field is called Total PTLF records read. On the BASE24-teller report, this field is called Total TTLF records read. On the BASE24 Remote Banking report, this field is called Total ITLF records read.

Maximum Transactions Per Second — The highest number of transactions logged in a single second. This is the highest value from the Max TPS field on the Transaction Log File Report.

Maximum Transaction Rate — The highest transaction rate found and averaged over the specified time interval in transactions per second. If a one minute interval was used, then this is the highest average one minute transaction rate. This is the highest value from the Tran Rate field on the Transaction Log File Report.

BASE24 Average Response Time — The average response time, in seconds, required for the BASE24 system to authorize a transaction. This is an average for the entire time period for which the report was run. If the BASE24 system did not authorize any transactions, this field does not appear on the report. This field is not displayed on the report for BASE24 Remote Banking.

BASE24-telebanking Average Response Time — The average response time, in seconds, required for BASE24-telebanking Integrated Authorization Server processes to authorize a transaction. This is an average for the entire time period for which the report was run. If BASE24-telebanking did not authorize any transactions, this field does not appear on the report. This field is displayed on the report for BASE24 Remote Banking only.

BASE24-billpay Average Response Time — The average response time, in seconds, required for BASE24-billpay Billpay Server processes to authorize a transaction. This is an average for the entire time period for which the report was run. If BASE24-billpay did not authorize any transactions, this field does not appear on the report. This field is displayed on the report for BASE24 Remote Banking only.

Host Average Response Time — The average response time, in seconds, required for the host to authorize a transaction. This is an average for the entire time period for which the report was run. If the host did not authorize any transactions, this field does not appear on the report.

Switch Average Response Time — The average response time, in seconds, required for an interchange or Interchange Interface process to authorize a transaction. This is an average for the entire time period for which the report was run. If an interchange or Interchange Interface process did not authorize any transactions, this field does not appear on the report.

Transaction Source

Totals identifying where the transaction requests originated. This section is divided into two totals for BASE24-atm, BASE24-pos, and BASE24-teller—local and foreign. An additional total field is displayed for BASE24 Remote Banking—BASE24-billpay. These totals are taken from the ORIGIN fields on the Transaction Log File Report.

Local — The number of transactions (including reversals) that originated from terminals or devices controlled by the BASE24 system or from the BASE24 system itself. This line also indicates the total number of local transactions that were reversed and indicates the percentage of local transactions that were reversed.

Foreign — The number of transactions that originated at a system outside of BASE24 (i.e., a host, switch, or third party acquiring interface). This line also indicates the total number of foreign transactions that were reversed and indicates the percentage of foreign transactions that were reversed.

BASE24-billpay — The number of transactions that originated from a BASE24-billpay process (i.e., Billpay Server process, Scheduled Transaction Initiator process, or Bank Advice process). This line also indicates the total number of BASE24-billpay transactions that were reversed and indicates the percentage of BASE24-billpay transactions that were reversed. This field is displayed on the report for BASE24 Remote Banking only.

Transaction Processing

The table on the left side of the report provides totals for financial transactions that were approved, denied, or the authorizer was not available (as defined previously in the detail time period report) for the host, the BASE24 system, or an interchange or Interchange Interface process. For BASE24 Remote Banking, this table includes non-financial as well as financial transactions and totals for the BASE24 system are divided between BASE24-telebanking and BASE24-billpay.

Host — These two rows provide a breakdown of the transactions that were approved or denied by a host. It also shows the total number of transactions that were intended to be authorized by the host, but the host was not available for processing. The first row provides the number of host transactions and the second row provides the percentage of host transactions that fall in each category.

The fourth value in the first row provides the total number of transactions authorized by the host during the specified processing date. The fifth value in the first row provides the percentage that the total number of transactions authorized by the host represents in relationship to all transactions authorized by all authorizers (i.e., the host, the BASE24 system, and an interchange or Interchange Interface process) during the specified processing date.

BASE24 — These two rows provide a breakdown of the transactions that were approved or denied by the BASE24 system. It also shows the total number of transactions that were intended to be authorized by the BASE24 system, but the BASE24 system was not available for processing. The first row provides the number transactions authorized by the BASE24 system and the second row provides the percentage of transactions authorized by the BASE24 system that fall in each category.

The fourth value in the first row provides the total number of transactions authorized by the BASE24 system during the specified processing date. The fifth value in the first row provides the percentage that the total number of transactions authorized by the BASE24 system represents in relationship to all transactions authorized by all authorizers (i.e., the host, the BASE24 system, and an interchange or Interchange Interface process) during the specified processing date. These rows are not displayed on the report for BASE24 Remote Banking.

BASE24-tb — These two rows provide a breakdown of the transactions that were approved or denied by BASE24-telebanking. It also shows the total number of transactions that were intended to be authorized by BASE24-telebanking, but BASE24-telebanking was not available for processing. The first row provides the number of BASE24-telebanking transactions and the second row provides the percentage of BASE24-telebanking transactions that fall in each category.

The fourth value in the first row provides the total number of transactions authorized by BASE24-telebanking during the specified processing date. The fifth value in the first row provides the percentage that the total number of transactions authorized by BASE24-telebanking represents in relationship to all transactions authorized by all authorizers (i.e., the host, BASE24-telebanking, BASE24-billpay,

and an interchange or Interchange Interface process) during the specified processing date. These rows are displayed on the report for BASE24 Remote Banking only.

BASE24-bp — These two rows provide a breakdown of the transactions that were approved or denied by BASE24-billpay. It also shows the total number of transactions that were intended to be authorized by BASE24-billpay, but BASE24-billpay was not available for processing. The first row provides the number of BASE24-billpay transactions and the second row provides the percentage of BASE24-billpay transactions that fall in each category.

The fourth value in the first row provides the total number of transactions authorized by BASE24-billpay during the specified processing date. The fifth value in the first row provides the percentage that the total number of transactions authorized by BASE24-billpay represents in relationship to all transactions authorized by all authorizers (i.e., the host, BASE24-telebanking, BASE24-billpay, and an interchange or Interchange Interface process) during the specified processing date. These rows are displayed on the report for BASE24 Remote Banking only.

Switch — These two rows provide a breakdown of the transactions that were approved or denied by an interchange or Interchange Interface process. It also shows the total number of transactions that were intended to be authorized by an interchange, but the interchange or Interchange Interface process was not available for processing. The first row provides the number of interchange transactions and the second row provides the percentage of interchange transactions that fall in each category.

The fourth value in the first row provides the total number of transactions authorized by the interchange or Interchange Interface process during the specified processing date. The fifth value in the first row provides the percentage that the total number of transactions authorized by the interchange or Interchange Interface process represents in relationship to all transactions authorized by all authorizers (i.e., the host, the BASE24 system, and an interchange or Interchange Interface process) during the specified processing date.

TOTAL — These two rows provide a breakdown of the transactions that were approved or denied by all authorizers. It also shows the total number of transactions that were intended to be authorized, but the host, the BASE24 system, or the interchange or Interchange Interface process was not available for processing. The first row provides the number of host, BASE24, and interchange transactions and the second row provides the percentage of host, BASE24, and interchange transactions that fall in each category.

The fourth value in the first row provides the combined total number of transactions authorized by the host, the BASE24 system, or an interchange or Interchange Interface process during the specified processing date.

When Normal Destination N-A

This table shows the number of approvals and denials when the intended transaction destination was not available. It provides a breakdown of the number contained in the N-A column in the left table into transactions that were approved or denied for the host, the BASE24 system, or an interchange or Interchange Interface process.

Host N-A — This section contains the total number and percentage of transactions that were approved or denied by the BASE24 Authorization process or module because the host was not available for processing.

A transaction is counted in this category when the intended authorizer is not available for processing. As a result, the BASE24 Authorization process or module either approves or denies the transaction using stand-in authorization, or denies the transaction if stand-in authorization is not used.

BASE24 N-A — This section contains the total number and percentage of transactions that were logged as force post transactions because the BASE24 system was not available for processing. This section is not displayed on the report for BASE24 Remote Banking.

BASE24-tb N-A — These rows are displayed for BASE24 Remote Banking only and are reserved for future use. Force post transactions are not currently supported by BASE24 Remote Banking.

BASE24-bp N-A — These rows are displayed for BASE24 Remote Banking only and are reserved for future use. Force post transactions are not currently supported by BASE24 Remote Banking.

Switch N-A — This section contains the total number and percentage of transactions that were approved or denied by the BASE24 Authorization process or module because the interchange or Interchange Interface process was not available for processing.

A transaction is counted in this category when the intended authorizer is not available for processing. As a result, the BASE24 Authorization process or module either approves or denies the transaction using stand-in authorization, or denies the transaction if stand-in authorization is not used.

BASE24 Remote Banking does not currently support stand-in processing for any switch interfaces.

TOTAL N-A — This row provides a breakdown of the total number of transactions that were approved or denied by the BASE24 Authorization process or module, or were logged as force post transactions, because the intended authorizer was unavailable. It also includes the percentage of the total number of N-A transactions that were approved or denied.

This row includes the combined total number of N-A transactions authorized by the host, the BASE24 system, or an interchange or Interchange Interface process during the specified processing date.

Transaction Type

This section provides a breakdown, by transaction type, of the number of approved, denied, and combined total of approved and denied transactions that occurred on the specified processing date. In addition, the percentage of approved, denied, and combined total of approved and denied transactions for each transaction type in relationship to the total number of transactions processed on the specified processing date is also included.

This section also displays the total amount, by transaction type, of approved transactions that occurred during the specified processing date. Approved monetary amounts are computed for any supported transactions.

For BASE24 Remote Banking, the report is further broken down by the number of approved, denied, reversed, and total transactions by transaction type, according to whether they were initiated by a customer or a customer service representative. The BASE24 Remote Banking report also provides a breakdown of reversed transactions by transaction type. For BASE24 Remote Banking transactions that originated from the Scheduled Transaction Initiator process, the transactions are counted according to the original transaction originator, rather than according to the Scheduled Transaction Initiator process itself.

The transaction types listed in this section of the report differ based on product. This report lists totals for any transaction set you support. Transaction types are displayed on the report only if a transaction has been performed on the specified processing date for the particular transaction type. Lists of the valid transaction types that can be displayed for each product appear below and on the following page.

ATM	POS	Teller
Withdrawal	Normal Purchase	Withdrawal
Deposit	Preauthorization purchase	Deposit
Deposit with cash back	Preauthorization purchase completed	Payment
Split deposit	Mail/phone order	Purchase
Balance inquiry	Merchandise return	Verify PIN
Transfer	Cash advance	Other
PIN change	Card verification	
Payment other	Balance inquiry	
EMV PIN unblock	Purchase with cash back	
	Check verification	
	Check guarantee	
	Purchase adjustment	
	Merchandise return with adjustment	
	Purchase with cash back adjustment	
	Other	

The valid transaction types for BASE24 Remote Banking are as follows:

Code	Transaction Description
30	Available funds inquiry
3A	Check clearance inquiry
3B	Last count of debit/credit transactions inquiry
3C	Last count of account by source transactions inquiry
3D	Last count of check transactions inquiry
3E	Last count of debit transactions inquiry
3F	Last count of credit transactions inquiry
3G	Last count of account transfer transactions inquiry
3H	Customer vendor list inquiry
3J	Scheduled payments list inquiry
3K	Scheduled transfers list inquiry
3L	History inquiry – payments and transfers
3M	Scheduled transactions list inquiry
3N	Customer account list inquiry
3P	Multiple account balance inquiry/Account list inquiry with balances
3Q	Statement closing download
3R	Statement download
40	Transfer – immediate
4A	Transfer – future
4B	Transfer – recurring
50	Payment – immediate
5A	Payment – future

Code	Transaction Description
5B	Payment – recurring
90	PIN change
91	PIN verify
9A	Schedule payment – immediate
9B	Schedule payment – future
9C	Schedule payment – recurring
9D	Schedule transfer – future
9E	Schedule transfer – recurring
9F	Scheduled payment delete
9G	Scheduled transfer delete
9H	Scheduled payment update
9J	Scheduled transfer update
9K	Customer add
9L	Customer inquiry
9M	Customer update
9N	Customer vendor add
9P	Customer vendor deactivate
9Q	Customer vendor update
9R	Master vendor list inquiry
9S	Master vendor add
9T	Customer vendor request mailing
9V	Loan application
9W	Miscellaneous request

Code	Transaction Description
9X	History inquiry – all transactions
A0	Check stop payment add
B0	Customer account list inquiry
B1	Bank list inquiry
B2	Bank branch list inquiry
B3	Cancel immediate payment
B4	Miscellaneous requests inquiry
B5	Customer account add
B6	Customer account activate
B7	Customer account deactivate
B8	Customer account delete
B9	Customer account update
BB	Customer activate
BC	Customer deactivate
BD	Customer vendor activate
BE	Customer vendor mask verify inquiry
BF	Customer verify
BG	Customer select

Summary

This section provides separate grand totals for approved and denied transactions. The grand total is calculated for approved transactions by adding together the number of approved transactions for each transaction type listed. The grand total is calculated for denied transactions by adding together the number of denied transactions for each transaction type listed.

For BASE24 Remote Banking, the report also provides grand totals for reversed transactions and further breaks down the grand totals of approved, denied, and reversed transactions according to whether they were initiated by a customer or a customer service representative. For BASE24 Remote Banking transactions that originated from the Scheduled Transaction Initiator process, the transactions are counted according to the original transaction originator, rather than according to the Scheduled Transaction Initiator process itself.

This section also includes the percentage of all transactions that occurred on the specified processing date that were approved, denied, or reversed (BASE24 Remote Banking only), and provides a grand total of all transactions that occurred on the specified transaction date.

Approved — The number, percentage, and amount of transactions approved for a particular transaction type. The number of reversals are not included in these counts.

Denied — The number and percentage of transactions denied for a particular transaction type. The number of reversals are not included in these counts.

Reversed — The number and percentage of transactions reversed for a particular transaction type. These fields are displayed for BASE24 Remote Banking only.

Total — The total number and percentage of all transaction activity. The number of reversals are included in these counts for BASE24 Remote Banking only. On the BASE24 Remote Banking report, this field is called Total Customers and CSR.

Response Time Summary Report

This report provides a detailed breakdown of transaction response times. Response times are split into time intervals and transactions are logged according to the time interval in which a response was received. The report is also separated by the entity that authorized the transaction (i.e., host, BASE24 system, interchange, or Interchange Interface). Response times for reversals are not included on this report.

Two sample Response Time Summary Reports are shown on the following pages—one for BASE24-atm and one for BASE24 Remote Banking. The format of the report for BASE24 Remote Banking is slightly different from the format for BASE24-atm, BASE24-pos, and BASE24-teller. The format of the report for BASE24-pos and BASE24-teller is identical to the format for BASE24-atm. The sample reports are followed by descriptions of their fields.

Page 43 System \ATM		A T M		R E S P O N S E T I M E S U M M A R Y		Logical network LLLL		MM/DD/YY	
(Reversals and foreign originated transactions not included)									
Response Time		Host		BASE24		Switch (Not On Us)			
-----		-----		-----		-----		-----	
Trans,		Trans,		Trans,		Trans,		Percent	
-----		-----		-----		-----		-----	
> 20	36	56	2%(100%)	27					
20	3	4		5					
19	4	2		3					
18	0	2		6					
17	1	2		14					
16	2	7		99				1% (99%)	
15	6	3		75				1% (98%)	
14	2	4		34					
13	4	0		59				1% (97%)	
12	0	1		65				1% (96%)	
11	12	1		117				1% (95%)	
10	18	1		111				1% (94%)	
9	24	1		228				3% (92%)	
8	62	5		334				4% (89%)	
7	150	3	1% (99%)	559				7% (85%)	
6	546	0	3% (98%)	766				10% (78%)	
5	1556	4	8% (96%)	988				13% (68%)	
4	6110	3	31% (88%)	1214				15% (56%)	
3	10034	4	50% (57%)	1633				21% (40%)	
2	1311	16	7% (7%)	1196		1% (96%)		15% (19%)	
1	1	529		257		21% (95%)		3% (4%)	
< 1	0	1878		61		74% (74%)		1% (1%)	
-----		-----		-----		-----			
19882		2526		7851					
=====		=====		=====					

Page	5	System \ACI	R B	R E S P O N S E		T I M E	S U M M A R Y		Logical network LLLL		MM/DD/YY
(Reversals and foreign originated transactions not included)											
Response Time	Host	Trans,	Percent	BASE24-telebanking	Trans,	Percent	BASE24-billpay	Trans,	Percent	Switch (Not On Us)	
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	
> 20	0	0		7	21%(100%)		5	50%(100%)		0	
20	0	0		0			0			0	
19	0	0		0			0			0	
18	0	0		1	3% (78%)		0			0	
17	0	0		1	3% (75%)		1	10% (50%)		0	
16	0	0		0			1	10% (40%)		0	
15	0	0		1	3% (71%)		0			0	
14	0	0		0			0			0	
13	0	0		0			0			0	
12	0	0		0			0			0	
11	0	0		0			0			0	
10	0	0		0			0			0	
9	0	0		0			0			0	
8	0	0		0			0			0	
7	0	0		0			0			0	
6	0	0		0			0			0	
5	0	0		1	3% (68%)		0			0	
4	0	0		0			0			0	
3	0	0		0			0			0	
2	0	0		0			0			0	
1	0	0		1	3% (65%)		2	20% (30%)		0	
< 1	0	0		20	62% (62%)		1	10% (10%)		0	
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	
	0			32			10			0	
=====	=====	=====	=====	=====	=====	=====	=====	=====	=====	=====	

Response Time

This column shows the response time ranges for which transaction response times are counted. The response time categories range from over 20 seconds to less than one second, at intervals of one second. Fractional response times are truncated. For example, the category of 3 seconds includes response times greater than or equal to 3 seconds and less than 4 seconds.

Host

The following fields show the number and percent of transactions authorized by the host according to transaction response time.

Trans — The number of transactions authorized by a host or Host Interface process for the specified response time interval. The last figure in this column shows the total number of transactions authorized by the host or Host Interface process.

Percent — The percentage of total transactions authorized by the host or Host Interface process that were authorized within the specified response time interval. A cumulative percentage, starting from the minimum response time interval, is listed in parentheses to the right.

BASE24

The following fields show the number and percent of transactions authorized by the BASE24 system according to transaction response time. These fields are not displayed on the report for BASE24 Remote Banking.

Trans — The number of transactions authorized by the BASE24 system for the specified response time interval. The last figure in this column shows the total number of transactions authorized by the BASE24 system.

Percent — The percentage of total transactions authorized by the BASE24 system that were authorized within the specified response time interval. A cumulative percentage, starting from the minimum response time interval, is listed in parentheses to the right.

BASE24-telebanking

The following fields show the number and percent of transactions authorized by BASE24-telebanking according to transaction response time. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of transactions authorized by BASE24-telebanking for the specified response time interval. The last figure in this column shows the total number of transactions authorized by BASE24-telebanking.

Percent — The percentage of total transactions authorized by BASE24-telebanking that were authorized within the specified response time interval. A cumulative percentage, starting from the minimum response time interval, is listed in parentheses to the right.

BASE24-billpay

The following fields show the number and percent of transactions authorized by BASE24-billpay according to transaction response time. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of transactions authorized by BASE24-billpay for the specified response time interval. The last figure in this column shows the total number of transactions authorized by BASE24-billpay.

Percent — The percentage of total transactions authorized by BASE24-billpay that were authorized within the specified response time interval. A cumulative percentage, starting from the minimum response time interval, is listed in parentheses to the right.

Switch (Not On Us)

The following fields show the number and percent of transactions authorized by an interchange according to transaction response time.

Trans — The number of transactions authorized by an interchange or Interchange Interface process for the specified response time interval. The last figure in this column shows the total number of transactions authorized by an interchange or Interchange Interface process.

Percent — The percentage of total transactions authorized by an interchange or Interchange Interface process that were authorized within the specified response time interval. A cumulative percentage, starting from the minimum response times, is listed in parentheses to the right.

Transaction Rate vs. Response Time Report

This report provides a comparison of transaction response times at various transaction throughput rates. Using this report, you can determine if there is a decline in response times at high transaction processing rates.

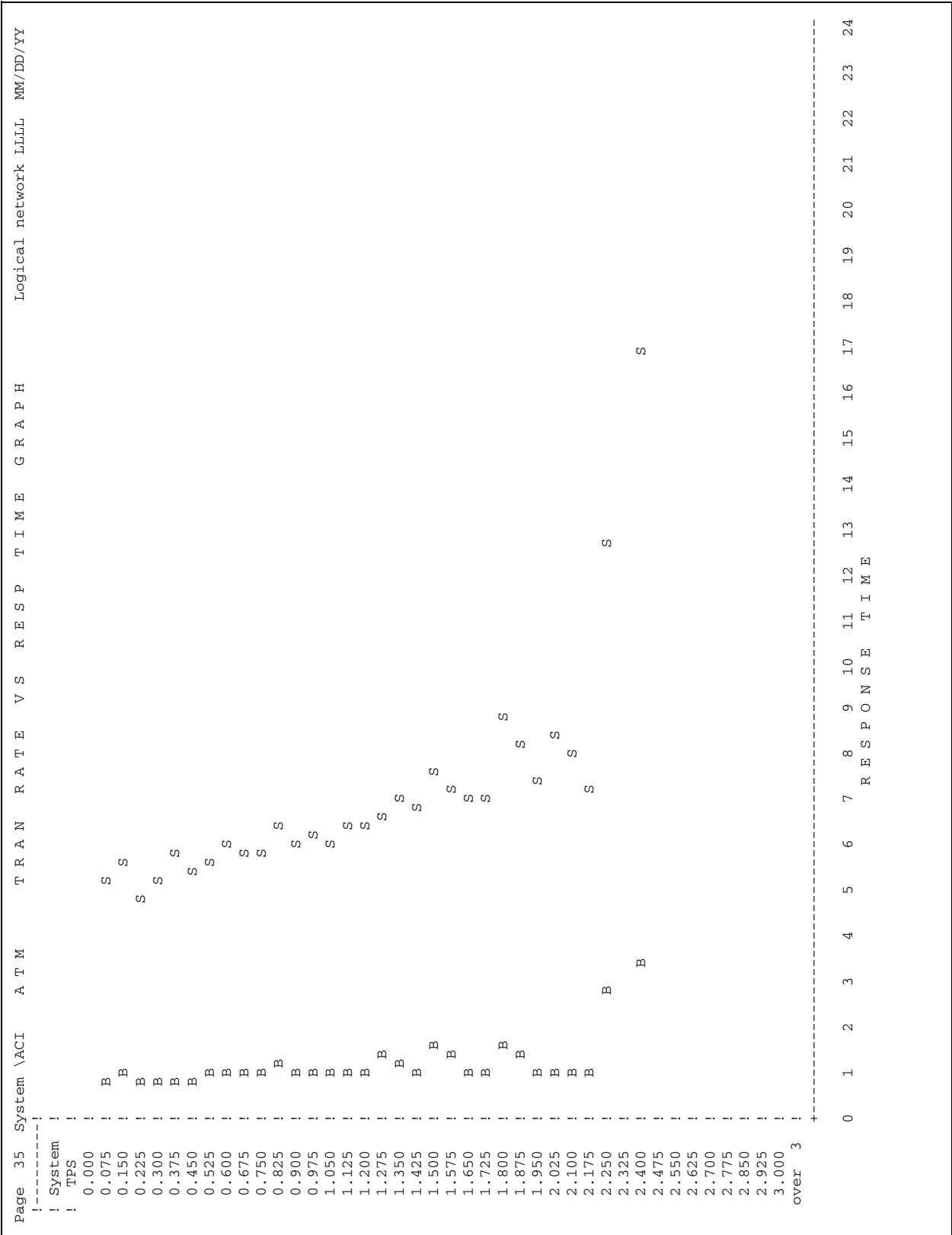
Transaction rates are averaged from the time intervals in which the program was run. They are not categorized by single second transaction rates. If you select a 60 minute time interval, and the TLF, PTLF, ITLF, or TTLF contains 24 hours of data, only 24 data points exist. A time interval that is less than 60 minutes is recommended to get the most from this report. If this report is used in great detail, a time interval of one minute is suggested. The interval to use is specified using the INTERVAL parameter.

There are two representations of this data. The first representation lists the raw data. The second representation plots response times versus transaction throughput, providing a graphical representation of response times. Response time categories range from 0 to 24 seconds.

This report is generated only if you specify the MAXTPS runtime parameter. MAXTPS is documented in the “Reporting Options” discussion provided earlier in this section.

Two sample Transaction Rate vs. Response Time Reports are shown on the following pages—one for BASE24-atm and one for BASE24 Remote Banking. The format of the report for BASE24 Remote Banking is slightly different from the format for BASE24-atm, BASE24-pos, and BASE24-teller. The format of the report for BASE24-pos and BASE24-teller is identical to the format for BASE24-atm. These sample reports are followed by descriptions of their fields.

Page	34	System \ACI	A T M	T R A N	R A T E	V S	R E S P	T I M E	S T A T S	Logical network	LLLL	MM/DD/YY
!-----THROUGHPUT-----!-----TRANSACTION DESTINATION-----!												
! System !			Number of	Percent	!-----Host-----!			!-----BASE24-----!			!-----Switch-----!	
! TPS !			Transactions	of Total	! N-A Resp Time !			! N-A Resp Time !			! N-A Resp Time !	
0.075	526	.9%							0.84	5.0%	5.25	
0.150	721	1.2%							0.99	1.9%	5.50	
0.225	828	1.3%							0.87	1.7%	4.89	
0.300	737	1.2%							0.87	2.2%	5.26	
0.375	838	1.4%							0.88	1.6%	5.78	
0.450	859	1.4%							0.88	3.1%	5.46	
0.525	1293	2.1%							0.90	1.6%	5.61	
0.600	2043	3.3%							0.91	3.6%	6.04	
0.675	3586	5.8%							0.91	2.2%	5.82	
0.750	2030	3.3%							0.92	2.1%	5.89	
0.825	2342	3.8%							1.19	1.6%	6.35	
0.900	1596	2.6%							0.93	2.1%	5.94	
0.975	2024	3.3%							0.93	.8%	6.18	
1.050	2854	4.6%							0.92	.9%	6.03	
1.125	3906	6.4%							0.93	1.2%	6.37	
1.200	2575	4.2%							0.92	1.7%	6.38	
1.275	3478	5.7%							1.40	.8%	6.64	
1.350	2269	3.7%							1.20	1.0%	7.06	
1.425	4378	7.1%							0.95	.9%	6.71	
1.500	3682	6.0%							1.54	1.7%	7.51	
1.575	4604	7.5%							1.39	3.0%	7.21	
1.650	2888	4.7%							0.96	.9%	7.06	
1.725	2421	3.9%							0.98	1.9%	6.90	
1.800	1687	2.7%							1.56	.9%	8.71	
1.875	2101	3.4%							1.37	.7%	8.13	
1.950	2062	3.4%							1.04	8.0%	7.31	
2.025	1301	2.1%							1.03	1.9%	8.45	
2.100	860	1.4%							1.06	2.0%	8.05	
2.175	383	.6%							1.02	3.1%	7.12	
2.250	395	.6%							2.88		12.57	
2.325	0											
2.400	143	.2%							3.47	3.6%	16.71	
2.475	0											
2.550	0											
2.625	0											
2.700	0											
2.775	0											
2.850	0											
2.925	0											
3.000	0											
over	3											



THROUGHPUT

This section provides statistics about the throughput rates of transactions on the specified processing date. Throughput rate means the same as transactions per second (TPS). This section provides time intervals that represent the TPS. For each time interval, the number of transactions that occurred in that time interval is displayed. In addition, a separate field lists the percentage of the total number of transactions for the specified processing date represented by the number in each particular time interval.

For each time interval, counts are kept by authorizer (i.e., host, BASE24 system, and interchange or Interchange Interface process). For the BASE24 Remote Banking report only, the BASE24 authorization statistics are divided between BASE24-telebanking and BASE24-billpay. The average response times for each range and the percent of total transactions that fell into the N-A category are listed. This provides an availability rating of the host and interchange at high transaction rates. For example, if the authorizer goes offline during peaks of high activity, this is probably a result of a performance or throughput problem.

System TPS — A list of time values that represent the number of TPS. On the sample reports, 41 transaction throughput categories are listed.

System TPS categories are determined using the value of the MAXTPS parameter specified at runtime. The MAXTPS parameter identifies the maximum number of transactions per second (MAXTPS) that is expected. You should select the next highest whole number over your expected peak average transaction rate for the specified time interval. The MAXTPS parameter is used in conjunction with the ENTRIES parameter. The utility takes the number specified in the MAXTPS parameter and divides it by the value in the ENTRIES parameter. This determines the ranges of different throughput rates. The MAXTPS parameter is described in the “Reporting Options” discussion provided earlier in this section.

A MAXTPS number that is too small causes a large amount of data from the important high transaction rates to be grouped into a single range, the *over* range. A number that is too large reduces the amount of data available to plot and makes each range unnecessarily large.

Number of Transactions — Specifies the number of transactions that were processed within the rate indicated on this line of the report for the processing date.

Percent of Total — Specifies the percentage of the total number of transactions processed for the specified processing date that were processed within the transaction rate indicated on this line of the report.

TRANSACTION DESTINATION

This section provides statistics for the entities that authorized the transactions (host, BASE24 system, and interchange or Interchange Interface process). For the BASE24 Remote Banking report, BASE24 statistics are divided between BASE24-telebanking and BASE24-billpay.

This section shows the average amount of time each authorizing entity took to authorize a transaction during each System TPS time interval. It also shows the percentage of transactions at the System TPS time interval that were intended for authorization, but the authorizing entity was unavailable.

Host

The following two fields provide response time statistics for transactions authorized by a host.

N-A — The percentage of transactions at a particular System TPS time interval that were intended for host authorization, but the host was unavailable.

Resp Time — The average amount of time, in seconds, it took for the host to authorize a transaction for this System TPS time interval.

BASE24

The following two fields provide response time statistics for transactions authorized by the BASE24 system. These fields are not displayed on the report for BASE24 Remote Banking.

N-A — The percentage of transactions at a particular System TPS time interval that are force post records, rather than authorization requests.

Resp Time — The average amount of time, in seconds, it took the BASE24 system to authorize a transaction.

BASE24-tb

The following two fields provide response time statistics for transactions authorized by BASE24-telebanking. These fields are displayed on the report for BASE24 Remote Banking only.

N-A — The percentage of transactions at a particular System TPS time interval that are force post records, rather than authorization requests. Force post transactions are not currently supported by BASE24 Remote Banking.

Resp Time — The average amount of time, in seconds, it took BASE24-telebanking to authorize a transaction.

BASE24-bp

The following two fields provide response time statistics for transactions authorized by BASE24-billpay. These fields are displayed on the report for BASE24 Remote Banking only.

N-A — The percentage of transactions at a particular System TPS time interval that are force post records, rather than authorization requests. Force post transactions are not currently supported by BASE24 Remote Banking.

Resp Time — The average amount of time, in seconds, it took BASE24-billpay to authorize a transaction.

Switch

The following two fields provide response time statistics for transactions authorized by an interchange or Interchange Interface process.

N-A — The percentage of transactions at a particular System TPS time interval that were intended for authorization by an interchange or Interchange Interface process, but the interchange was unavailable or the Interchange Interface process rejected the transaction. The Interchange Interface process passes the transaction to the interchange.

Resp Time — The average amount of time, in seconds, it took the interchange or Interchange Interface process to authorize the transaction.

Graph

This section of the report plots the response times versus the transaction rates.

For the BASE24-atm, BASE24-pos, and BASE24-teller reports, the letter B is used to indicate the BASE24 system, S is used to indicate interchange (switch), and H is used to indicate host.

For the BASE24 Remote Banking report, the letter B is used to indicate BASE24-billpay, T is used to indicate BASE24-telebanking, S is used to indicate interchange (switch), and H is used to indicate host.

System TPS — This column lists the transaction throughput categories, as determined by the MAXTPS and ENTRIES parameters, available as shown in the report. TPS stands for transactions per second.

RESPONSE TIME — The average amount of time, in seconds, it took the host, the BASE24 system, or the interchange or Interchange Interface process to authorize a transaction.

Transaction Volume Summary Report

This report summarizes transaction activity for each hour of the day, listing average and peak rates within each hour. This report is most helpful when the transaction analysis utility is run using the optional **CLOCKDATE** parameter. The **CLOCKDATE** parameter is discussed in the “Reporting Options” topic provided earlier in this section. The figures are a total of transaction rates generated by the system on the specified date. Because the system can be logging to two different TLFs, PTLFs, ITLFs, or TTLFs at any given time, the **CLOCKDATE** parameter merges the data from the TLFs, PTLFs, ITLFs, or TTLFs for two days, based on transaction time and date.

When the **CLOCKDATE** parameter is not used, the statistics on this report require greater clarification. The following example best illustrates what can happen. In this example, logical network cutover occurs at 17:00. Some terminals, devices, acquirer hosts, or acquirer interchanges begin posting to the next day’s TLF, PTLF, ITLF, or TTLF at 14:00. The Transaction Volume Summary Report shows activity from 14:00 of day 1 to 17:00 of day 2. Transactions processed on both days between these hours are summed into the respective time intervals on the report. The average transaction rate on this report for these hours shows the sum of both days. However, the peak interval and peak second shows the highest rates that occurred from either day 1 or day 2, whichever is higher. Therefore, the peaks can actually be less than the averages.

Even when the **CLOCKDATE** parameter is used, the statistics on this report need clarification during the period when Daylight Savings Time ends. The figures for the hour before the time change appearing on the report represent two hours of activity. For example, if Daylight Savings Time ends at 02:00 and the time is reset to 01:00, the transactions that would have appeared on the report between 02:00 and 03:00 without the time change instead appear on the report between 01:00 and 02:00. As a result, the report combines transactions for two hours in the 01:00 to 02:00 time period.

When this report is properly used, it can be a valuable asset for capacity planning. Usually, marketing projections for transaction volume are provided for expected daily or monthly volume. The system analyst can apply past ratios of peak hour to peak day and peak minute to peak hour to get probable future peak one minute transaction rates. These rates are required for accurate capacity planning.

Examples of the Transaction Volume Summary Report are shown on the following pages. Although the heading format of the report is slightly different for BASE24 Remote Banking, the same fields are displayed on the report for all BASE24 products. The sample reports are followed by descriptions of their fields.

Page 46 System \ACI A T M T R A N S A C T I O N V O L U M E S U M M A R Y Logical network LLLL MM/DD/YY									
Time		Transactions		Average		Transaction Rates		Graph	
-----		-----		-----		-----		-----	
		Peak Interval		Peak		Second			
		-----		-----		-----			
00:00-00:59	357,	1%	0.10	0.27	2	***			
01:00-01:59	196,		0.05	0.45	5	**			
02:00-02:59	114,		0.03	0.10	2	*			
03:00-03:59	60,		0.02	0.08	2				
04:00-04:59	80,		0.02	0.13	2				
05:00-05:59	203,		0.06	0.18	2	**			
06:00-06:59	595,	1%	0.17	0.40	3	*****			
07:00-07:59	1338,	3%	0.37	0.65	4	*****			
08:00-08:59	1575,	3%	0.44	0.72	4	*****			
09:00-09:59	2038,	5%	0.57	0.82	4	*****			
10:00-10:59	2338,	5%	0.65	0.87	5	*****			
11:00-11:59	3888,	9%	1.08	1.63	5	*****			
12:00-12:59	4166,	9%	1.16	1.88	6	*****			
13:00-13:59	3179,	7%	0.88	1.32	4	*****			
14:00-14:59	2791,	6%	0.78	0.87	4	*****			
15:00-15:59	2779,	6%	0.77	0.58	4	*****			
16:00-16:59	3321,	7%	0.92	0.73	4	*****			
17:00-17:59	3987,	9%	1.11	0.95	6	*****			
18:00-18:59	3988,	9%	1.11	2.18	6	*****			
19:00-19:59	3137,	7%	0.87	1.27	5	*****			
20:00-20:59	2112,	5%	0.59	0.98	5	*****			
21:00-21:59	1360,	3%	0.38	0.60	4	*****			
22:00-22:59	841,	2%	0.23	0.48	3	*****			
23:00-23:59	617,	1%	0.17	0.30	3	*****			
total		45060	=====						

Page	8	System \ACI	R B	T R A N S A C T I O N	L O G	F I L E	R E P O R T	Logical network	L I L L	M M / D D / Y Y
Transaction Volume Summary										
Time		Transactions	Average	Transaction Rates		Peak Interval	Peak Second	Graph		
-----		-----	-----	-----		-----	-----	----		
0:00- 0:59		0,	0.00			0.00	0			
1:00- 1:59		0,	0.00			0.00	0			
2:00- 2:59		0,	0.00			0.00	0			
3:00- 3:59		0,	0.00			0.00	0			
4:00- 4:59		0,	0.00			0.00	0			
5:00- 5:59		0,	0.00			0.00	0			
6:00- 6:59		0,	0.00			0.00	0			
7:00- 7:59		11, 21%	0.00			0.03	2	*****		
8:00- 8:59		25, 49%	0.00			0.03	2	*****		
9:00- 9:59		8, 15%	0.00			0.03	2	*****		
10:00-10:59		2, 3%	0.00			0.01	1	**		
11:00-11:59		2, 3%	0.00			0.01	1	**		
12:00-12:59		1, 1%	0.00			0.01	1	*		
13:00-13:59		2, 3%	0.00			0.01	1	**		
14:00-14:59		0,	0.00			0.00	0			
15:00-15:59		0,	0.00			0.00	0			
16:00-16:59		0,	0.00			0.00	0			
17:00-17:59		0,	0.00			0.00	0			
18:00-18:59		0,	0.00			0.00	0			
19:00-19:59		0,	0.00			0.00	0			
20:00-20:59		0,	0.00			0.00	0			
21:00-21:59		0,	0.00			0.00	0			
22:00-22:59		0,	0.00			0.00	0			
23:00-23:59		0,	0.00			0.00	0			
total		51		=====						

Time — A range of time intervals during the day in which transactions occurred. The time of day is expressed in one hour intervals using 24-hour time (hh:mm).

Transactions — The number of transactions that occurred during a specific hour of the day, and the percentage that this number of transactions represents in relationship to the total number of transactions that occurred during the entire processing day.

Transaction Rates

The following fields record the rate of transaction activity during each hour of the day.

Average — The average number of transactions processed per second. This average is calculated by dividing the total number of transactions processed during this hour by 3600, the number of seconds in an hour.

Peak Interval — The highest average number of transactions processed per second within a specific hour.

Peak Second — The highest number of transactions processed within a specific hour that were logged in a single second.

Graph — A graphical representation of the percent of transaction activity during each hour of the specified day. This graph provides an overall depiction of transaction distribution throughout the day.

Total — The total number of transactions processed for a specific processing day.

Approval Amount Report

This report itemizes the number of approved financial transactions according to their transaction amounts. It is divided into two sections: approved withdrawal and approved deposit transactions for BASE24-atm and BASE24-teller, approved purchase and preauthorization purchase transactions for BASE24-pos, and approved transfer and payment transactions for BASE24 Remote Banking. Reversal transactions have not been subtracted from the totals shown on this report.

You can use the statistical data provided in this report to analyze activity for approved financial transactions processed in the system and to help determine how your system is performing. For example, you can use the Approval Amount Report in conjunction with the Transaction Denial Report to provide a complete picture of the transaction activity that occurred on a particular day.

Transactions are grouped on the report into monetary categories that cover a specific range of transaction amounts. For example, transactions performed for amounts between 0.00 and 10.00 monetary units (e.g., U.S. dollars) are grouped into one category on the report, transactions performed for amounts between 10.01–20.00 monetary units are grouped into another category, and so on.

This report also provides the percentage of the grand total that the number and amount of transactions shown in a particular monetary category represents. The following example describes how these percentages are calculated. For example purposes, 2821 transactions are performed in the monetary range of 0–10.00 dollars. These transactions total \$27,540. Furthermore, the total number of transactions performed by combining all monetary categories displayed on the report was 30,735, totaling \$1,309,372.60. Therefore, 9.2 percent of the total number of transactions performed on the report date were in the 0.00–10.00 dollar range. In addition, 2.1 percent of the total amount of money resulting from transactions performed on the report date were in the 0.00–10.00 dollar range.

In addition, the report includes separate grand totals sections for each type of approved transactions. The grand totals are calculated for both the number of transactions and the amount of transactions. The grand total for the number of transactions is calculated by adding together the number of transactions contained in each monetary category. The grand totals for the amount of transactions is calculated by adding together the amounts contained in each monetary category. Included as separate statistics in the grand totals sections are the average amount of each type of transaction for the report date.

Examples of the Approval Amount Report for BASE24-atm, BASE24-pos, BASE24 Remote Banking, and BASE24-teller are shown on the following pages. The sample reports are followed by descriptions of their fields.

Page	47	System \ACI	A T M	A P P R O V A L	A M O U N T	R E P O R T	Logical network	LLLL	MM/DD/YY
(Reversals not impacted)									
		Withdrawals		Deposits					
		Trans,	Percent	Amount,	Percent	Trans,	Percent	Amount,	Percent
		-----	-----	-----	-----	-----	-----	-----	-----
0.00 -	10.00	6728	21.9%	56,971.21	4.4%	121	3.8%	720.50	.1%
10.01 -	20.00	8375	27.3%	158,577.44	12.1%	125	4.0%	2,202.45	.2%
20.01 -	30.00	4157	13.5%	114,105.98	8.7%	153	4.8%	4,049.60	.3%
30.01 -	40.00	2899	9.4%	11,601.65	8.7%	110	3.5%	4,132.26	.3%
40.01 -	50.00	2971	9.7%	147,594.18	11.3%	137	4.3%	6,551.45	.5%
50.01 -	60.00	977	3.2%	58,038.55	4.4%	78	2.5%	4,469.75	.3%
60.01 -	70.00	216	.7%	14,768.49	1.1%	54	1.7%	3,584.43	.3%
70.01 -	80.00	579	1.9%	45,021.13	3.4%	67	2.1%	5,134.47	.4%
80.01 -	90.00	69	.2%	6,046.49	.5%	61	1.9%	5,305.86	.4%
90.01 -	100.00	1932	6.3%	193,078.84	14.7%	145	4.6%	14,359.58	1.0%
100.01 -	125.00	247	.8%	29,282.22	2.2%	147	4.7%	16,865.92	1.2%
125.01 -	150.00	379	1.2%	55,550.56	4.2%	131	4.2%	18,589.80	1.3%
150.01 -	175.00	78	.3%	12,780.00	1.0%	101	3.2%	16,526.45	1.2%
175.01 -	200.00	692	2.3%	137,772.28	10.5%	126	4.0%	24,345.16	1.7%
200.01 -	225.00	39	.1%	8,429.42	.6%	88	2.8%	18,918.09	1.4%
225.01 -	250.00	90	.3%	22,316.07	1.7%	86	2.7%	20,713.27	1.5%
250.01 -	275.00	18	.1%	4,757.07	.4%	71	2.3%	18,794.72	1.3%
275.01 -	300.00	155	.5%	46,339.45	3.5%	94	3.0%	27,536.03	2.0%
300.01 -	325.00	9		2,858.63	.2%	62	2.0%	19,408.66	1.4%
325.01 -	350.00	16	.1%	5,501.94	.4%	71	2.3%	24,121.71	1.7%
350.01 -	375.00	8		2,906.34	.2%	46	1.5%	16,694.48	1.2%
375.01 -	400.00	31	.1%	12,371.28	.9%	63	2.0%	24,752.50	1.8%
400.01 -	425.00	2		843.97	.1%	57	1.8%	23,572.65	1.7%
425.01 -	450.00	4		1,737.26	.1%	52	1.6%	22,813.63	1.6%
450.01 -	475.00	2		929.24	.1%	38	1.2%	17,499.73	1.3%
475.01 -	500.00	18	.1%	9,000.00	.7%	61	1.9%	30,017.07	2.1%
500.01 -	600.00	8		4,481.82	.3%	187	5.9%	102,676.96	7.3%
600.01 -	700.00	3		1,860.21	.1%	95	3.0%	61,834.08	4.4%
700.01 -	800.00	5		3,664.44	.3%	86	2.7%	65,265.46	4.7%
800.01 -	900.00	3		2,646.29	.2%	74	2.3%	63,256.97	4.5%
900.01 -	1,000.00	2		1,819.18	.1%	72	2.3%	69,745.80	5.0%
1,000.01 -	2,000.00	11		16,206.58	1.2%	207	6.6%	286,374.97	20.5%
2,000.01 -	3,000.00	1		2,225.51	.2%	58	1.8%	138,969.99	9.9%
3,000.01 -	4,000.00					10	.3%	37,452.13	2.7%
4,000.01 -	5,000.00					6	.2%	25,486.99	1.8%
5,000.01 -	6,000.00					5	.2%	26,432.27	1.9%
6,000.01 -	7,000.00					1		6,206.00	.4%
7,000.01 -	8,000.00								
8,000.01 -	9,000.00								
9,000.01 -	10,000.00								
Over	10,000.00								
Total		30,725		15,288.88	1.2%	3,155		1,397,206.08	
Average				42.61				442.85	

Page	47	System \ACI	P O S	A P P R O V A L	A M O U N T	R E P O R T	Logical network LLLL	MM/DD/YY
						(Reversals not impacted)		
		Amount	Purchases		Pre-Auth			
		-----	Trans, Percent	Amount, Percent	Trans, Percent	Amount, Percent		
0.00 -	10.01		6728 21.9%	56,971.21 4.4%	121 3.8%	720.50 .1%		
10.01 -	20.00		8375 27.3%	158,577.44 12.1%	125 4.0%	2,202.45 .2%		
20.01 -	30.00		4157 13.5%	114,105.98 8.7%	153 4.8%	4,049.60 .3%		
30.01 -	40.00		2899 9.4%	113,601.65 8.7%	110 3.5%	4,132.26 .3%		
40.01 -	50.00		2971 9.7%	147,594.18 11.3%	137 4.3%	6,551.45 .5%		
50.01 -	60.00		977 3.2%	58,038.55 4.4%	78 2.5%	4,469.75 .3%		
60.01 -	70.00		216 .7%	14,768.49 1.1%	54 1.7%	3,584.43 .3%		
70.01 -	80.00		579 1.9%	45,021.13 3.4%	67 2.1%	5,134.47 .4%		
80.01 -	90.00		69 .2%	6,046.49 .5%	61 1.9%	5,305.86 .4%		
90.01 -	100.00		1932 6.3%	193,078.84 14.7%	145 4.6%	14,359.58 1.0%		
100.01 -	125.00		247 .8%	29,282.22 2.2%	147 4.7%	16,865.92 1.2%		
125.01 -	150.00		379 1.2%	55,550.56 4.2%	131 4.2%	18,589.80 1.3%		
150.01 -	175.00		78 .3%	12,780.00 1.0%	101 3.2%	16,526.45 1.2%		
175.01 -	200.00		692 2.3%	137,772.28 10.5%	126 4.0%	24,345.16 1.7%		
200.01 -	225.00		39 .1%	8,429.42 .6%	88 2.8%	18,918.09 1.4%		
225.01 -	250.00		90 .3%	22,316.07 1.7%	86 2.7%	20,713.27 1.5%		
250.01 -	275.00		18 .1%	4,757.07 .4%	71 2.3%	18,794.72 1.3%		
275.01 -	300.00		155 .5%	46,339.45 3.5%	94 3.0%	27,536.03 2.0%		
300.01 -	325.00		9	2,858.63 .2%	62 2.0%	19,408.66 1.4%		
325.01 -	350.00		16 .1%	5,501.94 .4%	71 2.3%	24,121.71 1.7%		
350.01 -	375.00		8	2,906.34 .2%	46 1.5%	16,694.48 1.2%		
375.01 -	400.00		31 .1%	12,371.28 .9%	63 2.0%	24,752.50 1.8%		
400.01 -	425.00		2	843.97 .1%	57 1.8%	23,572.65 1.7%		
425.01 -	450.00		4	1,737.26 .1%	52 1.6%	22,813.63 1.6%		
450.01 -	475.00		2	929.24 .1%	38 1.2%	17,499.73 1.3%		
475.01 -	500.00		18 .1%	9,000.00 .7%	61 1.9%	30,017.07 2.1%		
500.01 -	600.00		8	4,481.82 .3%	187 5.9%	102,676.96 7.3%		
600.01 -	700.00		3	1,860.21 .1%	95 3.0%	61,834.08 4.4%		
700.01 -	800.00		5	3,664.44 .3%	86 2.7%	65,265.46 4.7%		
800.01 -	900.00		3	2,646.29 .2%	74 2.3%	63,256.97 4.5%		
900.01 -	1,000.00		2	1,819.18 .1%	72 2.3%	69,745.80 5.0%		
1,000.01 -	2,000.00		11	16,206.58 1.2%	207 6.6%	286,374.97 20.5%		
2,000.01 -	3,000.00		1	2,225.51 .2%	58 1.8%	138,969.99 9.9%		
3,000.01 -	4,000.00				10 .3%	37,452.13 2.7%		
4,000.01 -	5,000.00				6 .2%	25,486.99 1.8%		
5,000.01 -	6,000.00				5 .2%	26,432.27 1.9%		
6,000.01 -	7,000.00				1	6,206.00 .4%		
7,000.01 -	8,000.00							
8,000.01 -	9,000.00							
9,000.01 -	10,000.00							
Over	10,000.00							
Total			30725	15,288.88 1.2%	3155	1,397,206.08		
Average				42.61		442.85		

Page	9	System \ACI	R B	A P P R O V A L		A M O U N T		R E P O R T		Logical network LLLL		MM/DD/YY
		Amount		Transfers		Transfers		Bill Payments				
		-----		Trans, Percent	-----	Amount, Percent	-----	Trans, Percent	-----	Amount, Percent	-----	
		0.00 - 10.00						26	100.0%	3.97	100.0%	
		10.01 - 20.00										
		20.01 - 30.00										
		30.01 - 40.00										
		40.01 - 50.00										
		50.01 - 60.00										
		60.01 - 70.00										
		70.01 - 80.00										
		80.01 - 90.00										
		90.01 - 100.00										
		100.01 - 125.00										
		125.01 - 150.00										
		150.01 - 175.00										
		175.01 - 200.00										
		200.01 - 225.00										
		225.01 - 250.00										
		250.01 - 275.00										
		275.01 - 300.00										
		300.01 - 325.00										
		325.01 - 350.00										
		350.01 - 375.00										
		375.01 - 400.00										
		400.01 - 425.00										
		425.01 - 450.00										
		450.01 - 475.00										
		475.01 - 500.00										
		500.01 - 600.00										
		600.01 - 700.00										
		700.01 - 800.00										
		800.01 - 900.00										
		900.01 - 1,000.00										
		1,000.01 - 2,000.00										
		2,000.01 - 3,000.00										
		3,000.01 - 4,000.00										
		4,000.01 - 5,000.00										
		5,000.01 - 6,000.00										
		6,000.01 - 7,000.00										
		7,000.01 - 8,000.00										
		8,000.01 - 9,000.00										
		9,000.01 -10,000.00										
		Over 10,000.00										
		Total		0				26		3.97		
		Average								0.15		

Page 47	System \ACI	TELLER	A P P R O V A L			A M O U N T			R E P O R T			Logical network LLLL			MM/DD/YY
(Reversals not impacted)															
Amount			Withdrawals			Deposits									
-----			-----			-----			-----			-----			
			Trans,	Percent	Amount,	Trans,	Percent	Amount,	Trans,	Percent	Amount,	Trans,	Percent	Amount,	Percent
			-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
	0.00	-	6728	21.9%	56,971.21		4.4%		121	3.8%	720.50				1%
	10.01	-	8375	27.3%	158,577.44		12.1%		125	4.0%	2,202.45				2%
	20.01	-	4157	13.5%	114,105.98		8.7%		153	4.8%	4,049.60				3%
	30.01	-	2899	9.4%	113,601.65		8.7%		110	3.5%	4,132.26				3%
	40.01	-	2971	9.7%	147,594.18		11.3%		137	4.3%	6,551.45				5%
	50.01	-	977	3.2%	58,038.55		4.4%		58	2.5%	4,469.75				3%
	60.01	-	216	.7%	14,768.49		1.1%		74	1.7%	3,584.43				3%
	70.01	-	579	1.9%	45,021.13		3.4%		67	2.1%	5,134.47				4%
	80.01	-	69	.2%	6,046.49		.5%		61	1.9%	5,305.86				4%
	90.01	-	1932	6.3%	193,078.84		14.7%		145	4.6%	14,359.58				1.0%
	100.01	-	247	.8%	29,282.22		2.2%		147	4.7%	16,865.92				1.2%
	125.01	-	379	1.2%	55,550.56		4.2%		131	4.2%	18,589.80				1.3%
	150.01	-	78	.3%	12,780.00		1.0%		101	3.2%	16,526.45				1.2%
	175.01	-	692	2.3%	137,772.28		10.5%		126	4.0%	24,345.16				1.7%
	200.01	-	39	.1%	8,429.42		.6%		88	2.8%	18,918.09				1.4%
	225.01	-	90	.3%	22,316.07		1.7%		86	2.7%	20,713.27				1.5%
	250.01	-	18	.1%	4,757.07		.4%		71	2.3%	18,794.72				1.3%
	275.01	-	155	.5%	46,339.45		3.5%		94	3.0%	27,536.03				2.0%
	300.01	-	9		2,858.63		.2%		62	2.0%	19,408.66				1.4%
	325.01	-	16	.1%	5,501.94		.4%		71	2.3%	24,121.71				1.7%
	350.01	-	8		2,906.34		.2%		46	1.5%	16,694.48				1.2%
	375.01	-	31	.1%	12,371.28		.9%		63	2.0%	24,752.50				1.8%
	400.01	-	2		843.97		.1%		57	1.8%	23,572.65				1.7%
	425.01	-	4		1,737.26		.1%		52	1.6%	22,813.63				1.6%
	450.01	-	2		929.24		.1%		38	1.2%	17,499.73				1.3%
	475.01	-	18	.1%	9,000.00		.7%		61	1.9%	30,017.07				2.1%
	500.01	-	8		4,481.82		.3%		187	5.9%	102,676.96				7.3%
	600.01	-	3		1,860.21		.1%		95	3.0%	61,834.08				4.4%
	700.01	-	5		3,664.44		.3%		86	2.7%	65,265.46				4.7%
	800.01	-	3		2,646.29		.2%		74	2.3%	63,256.97				4.5%
	900.01	-	2		1,819.18		.1%		72	2.3%	69,745.80				5.0%
	1,000.01	-	11		16,206.58		1.2%		207	6.6%	286,374.97				20.5%
	2,000.01	-	1		2,225.51		.2%		58	1.8%	138,969.99				9.9%
	3,000.01	-							10	.3%	37,452.13				2.7%
	4,000.01	-							6	.2%	25,486.99				1.8%
	5,000.01	-							5	.2%	26,432.27				1.9%
	6,000.01	-									6,206.00				.4%
	7,000.01	-													
	8,000.01	-													
	9,000.01	-													
	Over	10,000.00													
			1		15,288.88		1.2%								
			30725		1,309,372.60				3155		1,397,206.08				
					42.61						442.85				
	Total														
	Average														

Amount

Monetary categories that are broken down into intervals. These categories represent ranges of transaction amounts. Transactions that have occurred in the system are grouped according to their amounts. For example, the category 90.01–100.00 includes monetary transactions valued between, and including, 90.01 and 100.00 monetary units (e.g., U.S. dollars).

Separate transaction number and amount totals are shown for each type of transaction reported (i.e., withdrawals and deposits, purchases and preauthorization purchases, or transfers and payments) for each monetary category.

Withdrawals

The following four fields are displayed on the report for BASE24-atm and BASE24-teller only.

The number and amount of approved withdrawal transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of withdrawal transactions that occurred for a particular monetary category in relationship to the total number and amount of withdrawal transactions that occurred for the specified day.

Trans — The number of withdrawal transactions that occurred for the specified monetary category on the specified day.

Percent — The percentage of withdrawal transactions that occurred for a particular monetary category in relationship to the total number of withdrawal transactions that occurred on the specified day.

Amount — The total amount of the withdrawal transactions that occurred for a particular monetary category on the specified day.

Percent — The percent of withdrawal transaction amounts that occurred for a particular monetary category in relationship to the total amount of withdrawal transactions that occurred on the specified day.

Deposits

The following four fields are displayed on the report for BASE24-atm and BASE24-teller only.

The number and amount of approved deposit transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of deposit transactions that occurred for a particular monetary category in relationship to the total number and amount of deposit transactions that occurred for the specified day.

Trans — The number of deposit transactions that occurred during a specified day for a particular monetary category.

Percent — The percentage of deposit transactions that occurred for a particular monetary category in relationship to the total number of deposit transactions that occurred on the specified day.

Amount — Total amount of the deposit transactions that occurred for a particular monetary category on the specified day.

Percent — The percentage of deposit transactions that occurred for a particular monetary category in relationship to the total amount of deposit transactions that occurred for the specified day.

Purchases

The following four fields are displayed on the report for BASE24-pos only.

The number and amount of approved purchase transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of purchase transactions that occurred for a particular monetary category in relationship to the total number and amount of purchase transactions that occurred for the specified day.

Trans — The number of purchase transactions that occurred for the specified monetary category on the specified day.

Percent — The percentage of purchase transactions that occurred for a particular monetary category in relationship to the total number of purchase transactions that occurred on the specified day.

Amount — The total amount of the purchase transactions that occurred for a particular monetary category on the specified day.

Percent — The percent of purchase transaction amounts that occurred for a particular monetary category in relationship to the total amount of purchase transactions that occurred on the specified day.

Pre-Auth

The following four fields are displayed on the report for BASE24-pos only.

The number and amount of approved preauthorization purchase transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of preauthorization purchase transactions that occurred for a particular monetary category in relationship to the total number and amount of preauthorization purchase transactions that occurred for the specified day.

Trans — The number of preauthorization purchase transactions that occurred during a specified day for a particular monetary category.

Percent — The percentage of preauthorization purchase transactions that occurred for a particular monetary category in relationship to the total number of preauthorization purchase transactions that occurred on the specified day.

Amount — Total amount of the preauthorization purchase transactions that occurred for a particular monetary category on the specified day.

Percent — The percentage of preauthorization purchase transactions that occurred for a particular monetary category in relationship to the total amount of preauthorization purchase transactions that occurred for the specified day.

Transfers

The following four fields are displayed on the report for BASE24 Remote Banking only.

The number and amount of approved transfer transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of transfer transactions that occurred for a particular monetary category in relationship to the total number and amount of transfer transactions that occurred for the specified day. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of transfer transactions that occurred for the specified monetary category on the specified day.

Percent — The percentage of transfer transactions that occurred for a particular monetary category in relationship to the total number of transfer transactions that occurred on the specified day.

Amount — The total amount of the transfer transactions that occurred for a particular monetary category on the specified day.

Percent — The percent of transfer transaction amounts that occurred for a particular monetary category in relationship to the total amount of transfer transactions that occurred on the specified day.

Bill Payments

The following four fields are displayed on the report for BASE24 Remote Banking only.

The number and amount of approved payment transactions that occurred for the specified monetary category. Statistics are also shown for the percentage of payment transactions that occurred for a particular monetary category in relationship to the total number and amount of payment transactions that occurred for the specified day. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of payment transactions that occurred during a specified day for a particular monetary category.

Percent — The percentage of payment transactions that occurred for a particular monetary category in relationship to the total number of payment transactions that occurred on the specified day.

Amount — Total amount of the payment transactions that occurred for a particular monetary category on the specified day.

Percent — The percentage of payment transactions that occurred for a particular monetary category in relationship to the total amount of payment transactions that occurred for the specified day.

Total — The grand totals for withdrawals and deposits, purchases and preauthorization purchases, or transfers and payments. These totals are displayed at the bottom of the report.

Average — The average transaction amount is displayed at the bottom of the report.

Transaction Denial Report

This report summarizes the number of transactions denied authorization by the BASE24 system, a host, or an interchange. For BASE24-atm, BASE24-pos, and BASE24-teller, the report is divided into three sections and transactions are grouped according to the authorizing entity that denied them (i.e., BASE24, host, or interchange). For BASE24 Remote Banking, the BASE24 section is divided between BASE24-telebanking and BASE24-billpay. Denied transactions are broken down further by the specific reason they were denied and the denial code associated with the reason.

This report also shows the percent of the total number of approved, denied, and authorizer not available (N-A) transactions that were denied by the BASE24 system, a host, or an interchange for a specific reason category.

The percentages are calculated with information taken from the Transaction Log File Report—Summary and Totals Report. The number of transactions falling under a particular denial code for the host, the BASE24 system, or the interchange is divided by the total number of approved, denied, and N-A transactions for the host, the BASE24 system, or the interchange. For example, the total number of approved, denied, and N-A transactions for the host on the Transaction Log File Report—Summary and Totals Report described earlier is 32,537. On the Transaction Denial Report, the number of transactions denied by the host with a denial code 58 (Insufficient funds no amount 3) is 746. Therefore, of all the approved, denied, or N-A transactions handled by the host, 2.3 percent of them fall under the denial code 58 (Insufficient funds no amount 3).

Note: Before computing BASE24 percentages, the total number of normal destination N-A transactions are added to the total number of approved, denied, and N-A transactions for the BASE24 system.

Separate grand totals are also shown for transactions denied by each of the authorizing entities.

The percentage for grand totals is calculated much in the same manner as the percentages for each denial code. The percentage is calculated by taking from the Transaction Log File Report—Summary and Totals Report the total number of approved, denied, and N-A transactions for the host, the BASE24 system, or the interchange and dividing it by the total number of transactions denied by the host, the BASE24 system, or the interchange. For example, the total number of approved, denied, and N-A transactions for the host on the Transaction Log File Report—Summary and Totals Report described earlier is 32,537. On the

Transaction Denial Report, the number of transactions denied by the host is 856. Therefore, of all the approved, denied, or N-A transactions handled by the host, 3 percent of them were denied.

Two sample Transaction Denial Reports are shown on the following pages—one for BASE24-atm and one for BASE24 Remote Banking. The format of the report for BASE24 Remote Banking is slightly different from the format for BASE24-atm, BASE24-pos, and BASE24-teller. The format of the report for BASE24-pos and BASE24-teller is identical to the format for BASE24-atm. Field descriptions are provided following the sample reports.

Page 48	System \ACI	ATM	T R A N S A C T I O N	D E N I A L	R E P O R T	Logical network LLLL			MM/DD/YY
	Reason		Denial Code	Host Trans, Percent	BASE24 Trans, Percent	Switch Trans, Percent			
	Unauthorized usage		50	0	24 .6%	5 .1%			
	Expired card		51	0	57 1.4%	0			
	Invalid card		52	0	101 2.4%	23 .2%			
	Incorrect PIN		53	0	809 19.2%	224 2.3%			
	Not able to process transaction		54	67 .2%	153 3.6%	573 5.8%			
	Ineligible transaction		55	35 .1%	64 1.5%	120 1.2%			
	Ineligible account		56	5	189 4.5%	44 .4%			
	ATM owner not support transaction		57	0	55 1.3%	17 .2%			
	Insufficient funds no amount 3		58	746 2.3%	0	154 1.6%			
	Insufficient funds with amount 3		59	0	0	54 .5%			
	No further withdrawal		60	0	0	56 .6%			
	Enter lesser amount		61	3	152 3.6%	20 .2%			
	PIN tries exceeded		62	0	81 1.9%	9 .1%			
	Daily withdrawal limit reached		63	0	71 1.7%	26 .3%			
	Invalid credit card advance amount		64	0	0	4			
	No sharing with card issuer		69	0	631 15.0%	18 .2%			
	System err-retry transaction		70	0	0	2			
	System err-contact bank		71	0	0	6 .1%			
	System err-invalid destination		73	0	0	43 .4%			
	<totals line>			856 3%	2387 57%	1398 14%			

Page 10	System \ACI	R B	T R A N S A C T I O N	D E N I A L	R E P O R T	Logical network LLLL	MM/DD/YY
Reason	Denial Code	Host Trans, Percent	BASE24-tb Trans, Percent	BASE24-bp Trans, Percent	Switch Trans, Percent		
Deny, no acct of type	114	0	1 2.2%	0	0		
Deny, txn invalid	902	0	11 24.4%	0	0		
Deny, issuer unavailable	912	0	1 2.2%	0	0		
<totals line>		0 28%	13 28%	0	0		

Reason

The reason the transactions were denied. For BASE24-atm, the reason is taken from the RQST.RESP field in the BASE24-atm Standard Internal Message (STM). For a complete description of the reasons the transactions were denied, refer to the ***BASE24-atm Transaction Processing Manual***.

For BASE24-pos, the reason is taken from the TRAN.RESP-CDE field in the BASE24-pos Standard Internal Message (PSTM). For a complete description of the reasons the transactions were denied, refer to the ***BASE24-pos Transaction Processing Manual***.

For BASE24 Remote Banking, the reason is taken from the ITD-ACT.CDE and ITD-RSN-CDE-EXCPT.INFO.CDE fields of the Internal Transaction Data (ITD). For a complete description of the reasons the transactions were denied, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.

For BASE24-teller, the reason is taken from the RESP-HDR.RESP field in the BASE24-teller Standard Internal Message Header (TSTMH). For a complete description of the reasons the transactions were denied, refer to the ***BASE24-teller Transaction Processing Manual***.

Denial Code

A code identifying the reason the transaction was denied as specified in the Reason field.

For BASE24-atm, the denial code is determined by the BASE24-atm Authorization process and placed in the RQST.RESP field in the BASE24-atm Standard Internal Message (STM). The code is also logged to the Transaction Log File (TLF). Denial reasons and codes are displayed only if there were transactions containing these codes. For a complete listing of the denial reasons and their associated codes, refer to the ***BASE24-atm Transaction Processing Manual***.

For BASE24-pos, the denial code is determined by the BASE24-pos Authorization module and placed in the TRAN.RESP-CDE field in the BASE24-pos Standard Internal Message (PSTM). The code is also logged to the POS Transaction Log File (PTLF). Denial reasons and codes are displayed only if there were transactions containing these codes. For a complete listing of the denial reasons and their associated codes, refer to the ***BASE24-pos Transaction Processing Manual***.

For BASE24 Remote Banking, the denial code is determined by the BASE24-telebanking Integrated Authorization Server process and placed in the ITD-ACT.CDE and ITD-RSN-CDE-EXCPT.INFO.CDE fields of the Internal Transaction Data (ITD). The code is also logged to the ITS Transaction Log File (ITLF). Denial reasons and codes are displayed only if there were transactions containing these codes. For a complete listing of the denial reasons and their associated codes, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.

For BASE24-teller, the denial code is taken from the RESP-HDR.RESP field in the BASE24-teller Standard Internal Message Header (TSTMH). For a complete description of the reasons the transactions were denied, refer to the ***BASE24-teller Transaction Processing Manual***.

Host

The following fields display the number and percent of transactions denied by the host for the specified reasons.

Trans — The number of transactions denied by the host for the reason indicated.

Percent — The percent of the total number of approved, denied, and authorizer not available (N-A) transactions in relationship to the number of transactions that were denied by the host for the specified reason.

BASE24

The following fields display the number and percent of transactions denied by the BASE24 system for the specified reasons. These fields are displayed on the report for BASE24-atm, BASE24-pos, and BASE24-teller only.

Trans — The number of transactions denied by the BASE24 system for the reason indicated.

Percent — The percent of the total number of approved, denied, and authorizer not available (N-A) transactions in relationship to the number of transactions that were denied by the BASE24 system for the specified reason.

BASE24-tb

The following fields display the number and percent of transactions denied by BASE24-telebanking for the specified reasons. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of transactions denied by BASE24-telebanking for the reason indicated.

Percent — The percent of the total number of approved, denied, and authorizer not available (N-A) transactions in relationship to the number of transactions that were denied by BASE24-telebanking for the specified reason.

BASE24-bp

The following fields display the number and percent of transactions denied by BASE24-billpay for the specified reasons. These fields are displayed on the report for BASE24 Remote Banking only.

Trans — The number of transactions denied by BASE24-billpay for the reason indicated.

Percent — The percent of the total number of approved, denied, and authorizer not available (N-A) transactions in relationship to the number of transactions that were denied by BASE24-billpay for the specified reason.

Switch

The following fields display the number and percent of transactions denied by an interchange or Interchange Interface process for the specified reasons.

Trans — The number of transactions denied by an interchange or an Interchange Interface process for the reason indicated.

Percent — The percent of the total number of approved, denied, and authorizer not available (N-A) transactions in relationship to the number of transactions that were denied by an interchange or Interchange Interface process for the specified reason.

<totals line> — Displays separate grand totals for transactions denied by a host, the BASE24 system, BASE24-telebanking, BASE24-billpay, or an interchange or Interchange Interface process.

Reversal Report

This report provides a breakdown of reversal transactions according to reversal code and reason. The report is divided into two sections for BASE24-atm, BASE24-pos, and BASE24-teller—reversal transactions reported by local authorizers and reversal transactions reported by foreign authorizers. For BASE24 Remote Banking, an additional section is used to differentiate reversals generated by BASE24-telebanking (local) and BASE24-billpay. A local transaction originates from a terminal or device that is connected to the BASE24 system. A foreign transaction originates from a host, switch, or third party acquiring interface. A BASE24-billpay transaction originates from a BASE24-billpay process (i.e., Billpay Server process, Scheduled Transaction Initiator process, or Bank Advice process). Separate totals are also provided for local, foreign, and BASE24-billpay authorizers.

A reversal is a transaction that reverses a previous transaction that did not complete as authorized. Since transactions are logged before they are completed, the incomplete transactions must be backed out of the system. For example, a customer wants to withdraw funds from a terminal. The customer types in the transaction, but the terminal does not give the customer the requested funds. Even though the customer does not possess the funds, the records show that a transaction occurred.

Two sample Reversal Reports are shown on the following pages—one for BASE24-atm and one for BASE24 Remote Banking. The format of the report for BASE24 Remote Banking is slightly different from the format for BASE24-atm, BASE24-pos, and BASE24-teller. The format of the report for BASE24-pos and BASE24-teller is identical to the format for BASE24-atm. Field descriptions are provided following the sample reports.

Page	49	System \ACI	ATM	REVERSAL	REPORT	Logical network	LLLL	MM/DD/YY
		Reason	Reversal Code	Local Count	Foreign Count			
		-----	-----	-----	-----			
		<UNKNOWN reversal code>	0	0	1			
		Time-out	1	88	59			
		Command reject	2	33	0			
		Destination not available	3	7	1			
		Transaction Cancelled	8	12	8			
		Hardware error	10	80	78			
		Suspect transaction	20	25	1			
		<totals line>		----- 245 =====	----- 148 =====			

Page 11	System \ACI	R B	R E V E R S A L	R E P O R T	Logical network	LLLL	MM/DD/YY
		Reason	Reversal Code	Local Count	Foreign Count	BASE24-bp	
		-----	-----	-----	-----	-----	
		Response Received Too Late	4006	0	0	3	
				-----	-----	-----	
		<totals line>		0	0	3	
				=====	=====	=====	

Reason — The reason the transaction was reversed.

A complete listing of the reversal reasons can be found in the following documents:

- The BASE24-atm Standard Internal Message (STM) documentation contained in the ***BASE24-atm Transaction Processing Manual***.
- The BASE24-pos Standard Internal Message (PSTM) documentation contained in the ***BASE24-pos Transaction Processing Manual***.
- The message reason codes documentation contained in the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***. This document contains the reversal reasons that can be displayed on the report for BASE24 Remote Banking.
- The Teller Transaction Log File (TTLF) documentation contained in the ***BASE24-teller Files Maintenance Manual***.

Reversal Code — The code carried in the Transaction Log File (TLF), POS Transaction Log File (PTLF), ITS Transaction Log File (ITLF), or Teller Transaction Log File (TTLF) that identifies the reason for the reversal.

For BASE24-atm, the reversal code is determined by the BASE24-atm Authorization process and placed in the RQST.RVSL-CDE field in the BASE24-atm Standard Internal Message (STM).

For BASE24-pos, the reversal code is determined by the BASE24-pos Authorization module and placed in the RVSL-CDE field in the BASE24-pos Standard Internal Message (PSTM).

For BASE24 Remote Banking, the reversal code is determined by the BASE24-telebanking Integrated Authorization Server process and placed in the ITD-RSN-CDE-MSG field in the Internal Transaction Data (ITD).

For BASE24-teller, the reversal code is determined by the BASE24-teller Authorization process and placed in the RVSL-CDE field in the Financial Token in the BASE24-teller Standard Internal Message (TSTM).

For a complete listing of reversal codes and their associated reasons, refer to the documentation listed above for the Reason field.

Local Count — The total number of transactions initiated at a terminal or device connected to the BASE24 system that were reversed for the specified reason.

Foreign Count — The total number of reversal transactions initiated at a host, switch, or third party acquiring interface that were reversed for the specified reason.

BASE24-bp — The total number of reversal transactions initiated by a BASE24-billpay process that were reversed for the specified reason. This field is displayed on the report for BASE24 Remote Banking only.

<totals line> — The grand total number of local, foreign, and BASE24-billpay (BASE24 Remote Banking only) reversal transactions that occurred for the specified date.

Station Reports

These reports provide information regarding the activity at terminals within the BASE24 system for BASE24-atm, BASE24-pos, and BASE24-teller. For BASE24 Remote Banking, transaction originator reports are generated instead of station reports. These reports are described later in this section.

Data is displayed for each terminal defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF). The utility sequentially loads terminal names from the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or TTDF and checks the terminal ID logged in the TLF, PTLF, or TTLF for each transaction to determine whether the transaction occurred on a terminal that is defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or TTDF. If so, terminal statistics including the number of transactions performed, total amount of withdrawals, total amount of deposits, and the beginning cash amount in the terminal are recorded. Statistics also show the number of transactions denied due to specific reason codes and the number of transactions reversed due to specific reversal codes.

Stations Sorted by Name, Detail Report

This report contains one line of data for each terminal listed. The first report is an example of a BASE24-atm Station Sorted by Name, Detail Report.

Note: The Stations Sorted by Name, Detail Report is similar for BASE24-atm, BASE24-pos, and BASE24-teller. However, in BASE24-pos, the Total Withdrawn, Total Deposited, and Beginning Cash headers shown on BASE24-atm reports are listed as Normal Purchase, Pre-Auth Purchase, and Cash Advance headers. In BASE24-teller the Beginning Cash header is not present. For both BASE24-pos and BASE24-teller, reversal amounts are applied to the figures. Examples of the Stations Sorted by Name, Detail Report for BASE24-atm, BASE24-pos, and BASE24-teller are shown on the following pages. Field descriptions are provided for all three reports following the examples.

Page 50	System \ACI	A T M	S T A T I O N R E P O R T				Logical network LLLL MM/DD/YY												
(Stations sorted by station name, detail)																			
Station Name	Trans	Percent of avg	Total withdrawn	Total deposited	Beginning cash	50	53	54	55	56	57	62	69	70-74	1	3	8	10	20
<default statn>	0																		
S1AAT206	128	145%	3,790.00	7,680.20			1	2			1		16		1				
S1AAT207	139	158%	5,926.28	6,483.04	20,945		2						1						
S1AAT208	250	284%	6,365.00	4,280.40	40,050		9	1					1		1				1
S1AAT209	95	108%	2,390.00	1,849.02	35,625			3							1				
S1AAT210	38	43%	820.00	1,371.00	3,260		1					1	3						
S1AAT212	206	234%	7,616.72	13,587.36		1	5	2					11						1
S1AAT215	87	99%	2,965.00	4,390.96	8,840		2					2	2						1
S1AAT216	50	57%	1,990.00	2,817.91	11,130		1												
S1AAT229	100	114%	19,253.88	4,513.58	47,795		2	1				1	1				1		
S1AAT230	243	276%	6,225.00	10,968.67			3	6					3						
S1AAT231	107	122%	5,080.00	10,191.47	24,345		3	1					2						
S1AAT232	60	68%	1,410.00	559.60	508,665											1			
S1AAT233	76	86%	1,980.00	1,242.52	15,070														
S1AAT234	14	16%	10.00	25.00	17,835										1				1
S1AAT235	102	116%	1,985.00	1,529.60	12,775													2	
S1AAT236	124	141%	2,565.00	2,880.82			6	1					7		4			1	
S1AAT250	122	139%	3,335.00	1,721.74	645,840		2	1											
S1AAT251	0				10,635														
S1AAT252	62	70%	1,750.00	2,925.44			2						1						
S1AAT254	153	174%	3,860.00	3,992.88	44,395			3					1						
S1AAT256	96	109%	2,325.00	468.35	7,500		3						3						
S1AAT257	117	133%	3,200.00	2,357.16	20,205		2	7					4		1				
S1AAT258	85	97%	2,140.00	997.49			3			1			3					1	
S1AAT259	68	77%	2,070.00	885.84									2						
S1AAT273	54	61%	1,090.00	1,490.00	9,085		1	1		1			2						1
S1AAT275	131	149%	3,405.00	3,214.69	17,650		2						1						
S1AAT276	77	88%	1,695.00	1,463.15	25,605		3	1		1			2						
S1AAT277	128	145%	4,065.45	4,876.95			2	1		1									
S1AAT278	72	82%	1,270.00	764.69									1						
S1AAT279	158	180%	3,667.71	2,117.06									4		1				
S1AAT283	170	193%	4,570.00	4,944.29			7	4	5				1						1
S1AAT294	137	156%	4,175.00	2,279.59			2	1	7				1		1				
S1AAT295	77	88%	2,720.00	855.00	12,730		1				1								
S1AAT297	24	27%	655.00	256.74	24,175														
S1AAT298	169	192%	5,785.92	13,175.94	25,965		9	1		1					2				
S1AAT299	215	244%	7,180.00	3,291.18			3					1	2		2				
S1AAT301	147	167%	4,899.90	3,761.10	32,190	1	3	2											
S1AAT302	143	163%	4,315.00	2,691.78	14,530	1	2	4	1			1	4		1			1	1
S1AAT303	107	122%	4,467.72	1,388.91	9,895														
S1AAT316	284	323%	6,940.00	8,029.12	645,840		2												
S1AAT317	86	98%	2,220.00	1,576.12	32,175		3	1	3										
S1AAT318	148	168%	5,225.00	12,499.27	25,020		1	5	1	1			4		1				1

Page	50	System \ACI	P O S	S T A T I O N R E P O R T										Logical network LLLL MM/DD/YY									
(Stations sorted by station name, detail)																							
Station Name	Trans	Percent of avg	Normal Purchase	Pre-Auth Purchase	Cash Advance	50	52	53	54	56	61	62	69	71	1	3	8	10	20				
<default statn>	0																						
S1APS206	128	145%	1,790.00	1,680.20			1	2		1			16			1							
S1APS207	139	158%	1,926.28	1,483.04	945		2						1										
S1APS208	250	284%	2,365.00	1,280.40	1,050		9	1					1			1			1				
S1APS209	95	108%	990.00	1,849.02	2,625			3								1							
S1APS210	38	43%	820.00	1,371.00	260		1					1	3										
S1APS212	206	234%	2,616.72	3,587.36		1	5	2					11						1				
S1APS215	87	99%	965.00	2,390.96	840							2	2						1				
S1APS216	50	57%	1,990.00	817.91	730		1																
S1APS229	100	114%	1,253.88	2,513.58	1,795		2	1				1	1						1				
S1APS230	243	276%	2,225.00	968.67			3	6					3				1						
S1APS231	107	122%	2,080.00	1,191.47	945		3	1					2										
S1APS232	60	68%	1,410.00	559.60	265																		
S1APS233	76	86%	980.00	1,242.52	1,070								4			1			1				
S1APS234	14	16%	10.00	25.00	1,835																		
S1APS235	102	116%	985.00	1,529.60	775		6	1					7			4			2				
S1APS236	124	141%	565.00	880.82			2	1											1				
S1APS250	122	139%	335.00	1,721.74	840																		
S1APS251	0			0																			
S1APS252	62	70%	1,750.00	2,925.44			2						1										
S1APS254	153	174%	760.00	1,992.88	995			3					1										
S1APS256	96	109%	725.00	468.35	1,500		3						3										
S1APS257	117	133%	600.00	1,357.16	905		2	7					4			1							
S1APS258	85	97%	140.00	697.49			3		3				3						1				
S1APS259	68	77%	770.00	385.84									2										
S1APS273	54	61%	190.00	490.00	1,085		1	1					2										
S1APS275	131	149%	205.00	214.69	750		2						1										
S1APS276	77	88%	495.00	463.15	605		3	1				1	2										
S1APS277	128	145%	565.45	876.95			2	1					1										
S1APS278	72	82%	270.00	64.69									1										
S1APS279	158	180%	667.71	117.06			7	4					4			1							
S1APS283	170	193%	570.00	944.29				1					1			1							
S1APS294	137	156%	1,175.00	279.59			2	1				1											
S1APS295	77	88%	2,720.00	855.00	12,730		1	1					2										
S1APS297	24	27%	655.00	256.74																			
S1APS298	169	192%	785.92	1,175.94	175																		
S1APS299	215	244%	180.00	291.18	1,965		9	1		1						2			1				
S1APS301	147	167%	899.90	761.10		1	3	2					2			2							
S1APS302	143	163%	315.00	691.78	2,190		2	4				1	4			1			1				
S1APS303	107	122%	467.72	388.91	1,530																		
S1APS316	284	323%	940.00	129.12	1,895		2																
S1APS317	86	98%	220.00	576.12	840		3	1															
S1APS318	148	168%	225.00	499.27	2,175		1	5		1			4			1			1				

Page	50	System \ACI	TELLER	S T A T I O N R E P O R T										Logical network LLLL				MM/DD/YY	
		(Stations sorted by station name, detail)																	
Codes	Station Name	Trans	Percent	Total	deposited	F01	F07	F08	F0F	F0I	F0Z	F10	F11	F12	1	3	8	10	
20			of avg	withdrawn															
<default statn>				0															
	S1ATT206	128	145%	3,790.00	7,680.20	1	2	2	1		16				1				
	S1ATT207	139	158%	5,926.28	6,483.04								1						
	S1ATT208	250	284%	6,365.00	4,280.40		9	1				1	1					1	
	S1ATT209	95	108%	2,390.00	1,849.02			3						1					
	S1ATT210	38	43%	820.00	1,371.00		1				1	3							
	S1ATT212	206	234%	7,616.72	13,587.36	1	5	2				11						1	
	S1ATT215	87	99%	2,965.00	4,390.96			2				2	2						
1																			
	S1ATT216	50	57%	1,990.00	2,817.91		1												
	S1ATT229	100	114%	19,253.88	4,513.58		2	1			1	1				1			
	S1ATT230	243	276%	6,225.00	10,968.67			3	6				3			1			
	S1ATT231	107	122%	5,080.00	10,191.47	3	1				2								
	S1ATT232	60	68%	1,410.00	559.60			1								1			
	S1ATT233	76	86%	1,980.00	1,242.52							4		1				1	
	S1ATT234	14	16%	10.00	25.00											2			
	S1ATT235	102	116%	1,985.00	1,529.60														
	S1ATT236	124	141%	2,565.00	2,880.82			6	1					7			1		
4																			
	S1ATT250	122	139%	3,335.00	1,721.74		2	1											
	S1ATT251	0																	
	S1ATT252	62	70%	1,750.00	2,925.44				2							1			
	S1ATT254	153	174%	3,860.00	3,992.88			3				1							
	S1ATT256	96	109%	2,325.00	468.35			3								3			
	S1ATT257	117	133%	3,200.00	2,357.16		2	7	1			4		1					
	S1ATT258	85	97%	2,140.00	997.49				3	3	1			3					
1																			
	S1ATT259	68	77%	2,070.00	885.84							2							
	S1ATT273	54	61%	1,090.00	1,490.00		1	1	2	1		2					1		
	S1ATT275	131	149%	3,405.00	3,214.69			2					1						
	S1ATT276	77	88%	1,695.00	1,463.15		3	1	2	1			2						
	S1ATT277	128	145%	4,065.45	4,876.95				2	1	1								
	S1ATT278	72	82%	1,270.00	764.69								1						
	S1ATT279	158	180%	3,667.71	2,117.06				7	4	5					4		1	
	S1ATT283	170	193%	4,570.00	4,944.29				1	7				1	1	1			
	S1ATT294	137	156%	4,175.00	2,279.59			2	1			1		2					
	S1ATT295	77	88%	2,720.00	855.00		1	1											
	S1ATT297	24	27%	655.00	256.74														
	S1ATT298	169	192%	5,785.92	13,175.94		9	1		1					2				
1																			
	S1ATT299	215	244%	7,180.00	3,291.18		1	3				1		2		2			
	S1ATT301	147	167%	4,899.90	3,761.10		1	3	2										
	S1ATT302	143	163%	4,315.00	2,691.78		2	4	1			1	4		1		1	1	
	S1ATT303	107	122%	4,467.72	1,388.91			1											
	S1ATT317	86	98%	2,220.00	1,576.12		3	1	3										

Station Name — The name of the terminal as it is defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF). This is the terminal at which a transaction originated.

Trans — The number of transactions performed at the named device. This includes approvals, denials, and reversals, but not administrative transactions. Administrative transactions include transactions performed by employees at the financial institution or force cutover records generated by the BASE24 system. For example, removing deposits from a terminal is an administrative transaction.

Percent of avg — A percentage indicating the number of transactions performed at a specific device as compared to the average number of transactions expected per device. The average number of transactions expected per device is calculated by the utility and is given a percentage value of 100 percent. The average number of transactions per device is calculated by adding the number of transactions processed for all active devices and dividing this total by the number of active devices.

For example, if the average number of transactions expected per device is 67 per processing day, when 67 transactions have been performed at a specific device, that device has reached 100 percent of transaction activity expected for the processing day. A number less than 67 indicates that the device is below the average transaction activity level expected for the device. A number greater than 67 indicates that the device is above the average transaction activity level expected for the device. Therefore, a value of 200 percent indicates that the device produced twice as many transactions as the average device, and a value of 50 percent indicates that only half of the average was produced.

Total withdrawn — The total amount of approved withdrawal transactions initiated at the specified terminal. This field is displayed only on the BASE24-atm report.

Total deposited — The total amount of approved deposit transactions initiated at the specified terminal. This field is displayed only on the BASE24-atm report.

Beginning cash — The beginning cash or cash value amount from the first balance record found in the TLF. When a balance transaction occurs, a record is written to the TLF containing the hopper totals from the BASE24-atm Terminal Data files. If you do not maintain correct cash or cash value totals in the BASE24-atm Terminal Data files, this figure is not useful. A *hopper* is the canister that stores funds or cash value items within a terminal. This field is displayed only on the BASE24-atm report.

Normal Purchase — The amount of approved normal purchase transactions initiated at the specified terminal. This field is displayed only on the BASE24-pos reports.

Pre-Auth Purchase — The amount of approved preauthorization purchase transactions initiated at the specified terminal. This field is displayed only on the BASE24-pos report.

Cash Advance — The amount of approved cash advance transactions initiated at the specified terminal. This field is displayed only on the BASE24-pos reports.

Common Denial Reasons — The number of transactions that were denied for a reason associated with the denial code listed in this column. Denial codes listed in this column for BASE24-atm include the following:

- 50 = Unauthorized usage
- 53 = Invalid PIN
- 54 = Database problem
- 55 = Ineligible transaction
- 56 = Ineligible account
- 57 = Transaction not supported
- 62 = PIN tries exceeded
- 69 = No sharing between the card issuer and terminal owner
- 70 = System error
- 71 = Contact card issuer
- 72 = Destination not available
- 73 = Routing lookup problem
- 74 = Message edit error

For more information on denial codes and reasons for BASE24-atm transactions, refer to the ***BASE24-atm Transaction Processing Manual***.

Denial codes listed in this column for BASE24-pos include the following:

- 50 = Declined
- 52 = PIN tries exceeded
- 53 = Sharing is not allowed
- 54 = No Atalla box
- 56 = No support
- 61 = PBF record not found
- 62 = PBF update error
- 69 = Bad message edit
- 71 = Invalid routing to Authorization

For more information on denial codes and reasons for BASE24-pos transactions, refer to the ***BASE24-pos Transaction Processing Manual***.

Denial codes listed in this column for BASE24-teller include the following:

F0F = Account owner not in logical network
F0I = System error
F0Z = Override needed
F01 = Message edit error
F07 = Host is down
F08 = Database problem
F10 = Ineligible transaction
F11 = Invalid Teller level
F12 = Transaction not supported by issuer

For information on denial codes and reasons for BASE24-teller transactions, refer to the ***BASE24-teller Transaction Processing Manual***.

Common Reversal Codes — The number of transaction reversals that occurred from the device with the reason listed in this column. Reversal codes listed in this column include the following:

1 = Timeout
3 = Destination not available
8 = Transaction canceled
10 = Hardware error
20 = Suspect transaction

For more information on reversal codes, refer to the BASE24-atm Standard Internal Message (STM) documentation in the ***BASE24-atm Transaction Processing Manual***, the BASE24-pos Standard Internal Message (PSTM) documentation in the ***BASE24-pos Transaction Processing Manual***, or the BASE24-teller Standard Internal Message Header (TSTMH) documentation in the ***BASE24-teller Transaction Processing Manual***.

Stations Sorted by Activity Report

This report is a condensed version of the Stations Sorted by Name, Detail Report and does not exist in the Transaction Analysis Utility Transaction History File in this format. The Stations Sorted by Activity Report shows the same terminal name as the Stations Sorted by Name, Detail Report. However, it reports only on the number of transactions for that terminal and the percent of average generated by the terminal. This allows the header columns to be listed across the page three times in this report. The data is read left to right and top to bottom, and the station with the highest activity is sorted first. This way, the terminal with the most activity is listed first and terminals with less activity appear in descending order.

An example of the Stations Sorted by Activity Report is shown on the following page for BASE24-atm. The format of the report is identical for BASE24-pos and BASE24-teller. Field descriptions are provided following the example.

Page	63	System \ACI	A T M	S T A T I O N		R E P O R T		Logical network LLLL		MM/DD/YY
(Stations sorted by activity)										
		Station Name	Count, Percent	Station Name	Count, Percent	Station Name	Count, Percent	Station Name	Count, Percent	
		S1BAT864	88	100%	S1AAT215	87	99%	S1AAT914	87	99%
		S1BAT821	86	98%	S1AAT409	86	98%	S1AAT317	86	98%
		S1AAT258	85	97%	S1AAT362	85	97%	S1BAT815	84	95%
		S1AAT361	83	94%	S1BTRFD1	83	94%	S1AAT428	82	93%
		S1BAT471	81	92%	S1BAT451	79	90%	S1AAT562	79	90%
		S1AAT411	79	90%	S1AAT824	78	89%	S1AAT295	77	88%
		S1AAT276	77	88%	S1BAT547	76	86%	S1AAT233	76	86%
		S1AAT841	75	85%	S1BAT541	75	85%	S1BAT453	74	84%
		S1BAT790	73	83%	S1AAT278	72	82%	S1BAT565	71	81%
		S1BAT498	69	78%	S1AAT259	68	77%	S1BAT493	68	77%
		S1AAT425	68	77%	S1BAT786	68	77%	S1BAT812	68	77%
		S1BAT898	68	77%	S1AAT809	68	77%	S1BAT502	67	76%
		S1BAT899	67	76%	S1AAT820	67	76%	S1BAT537	65	74%
		S1BAT566	64	73%	S1AAT364	64	73%	S1AAT369	63	72%
		S1AAT826	63	72%	S1BAT822	63	72%	S1BAT452	62	70%
		S1AAT252	62	70%	S1BAT874	62	70%	S1BAT630	61	69%
		S1BAT691	61	69%	S1AAT232	60	68%	S1BAT764	59	67%
		S1BAT834	59	67%	S1BAT862	57	65%	S1AAT864	58	66%
		S1BAT696	58	66%	S1BAT699	57	65%	S1BAT840	57	65%
		S1AAT345	57	65%	S1BAT539	56	64%	S1AAT370	54	61%
		S1AAT912	54	61%	S1BAT835	54	61%	S1AAT273	54	61%
		S1BAT909	54	61%	S1AAT342	54	61%	S1BAT481	53	60%
		S1BAT695	53	60%	S1AAT647	53	60%	S1BAT760	52	59%
		S1BAT844	52	59%	S1BAT908	52	59%	S1AAT915	51	58%
		S1AAT216	50	57%	S1AAT668	50	57%	S1BAT567	49	56%
		S1AAT655	49	56%	S1AAT832	48	55%	S1AAT857	46	52%
		S1BAT801	46	52%	S1AAT424	44	50%	S1BAT690	44	50%
		S1AAT919	44	50%	S1AAT327	44	50%	S1BAT500	43	49%
		S1BAT778	43	49%	S1BAT847	43	49%	S1BAT525	42	48%
		S1BAT782	41	47%	S1AAT346	41	47%	S1BAT694	41	47%
		S1BAT492	40	45%	S1AAT406	40	45%	S1AAT918	40	45%
		S1BAT789	40	45%	S1BAT845	40	45%	S1AAT977	40	45%
		S1BAT693	38	43%	S1AAT210	38	43%	S1AAT810	38	43%
		S1BAT718	37	42%	S1BAT470	37	42%	S1BAT818	37	42%
		S1AAT435	37	42%	S1AAT980	37	42%	S1BAT844	34	39%
		S1AAT978	34	39%	S1AAT674	34	39%	S1AAT850	33	38%
		S1BAT591	33	38%	S1AAT916	33	38%	S1AAT911	32	36%
		S1AAT401	31	35%	S1BAT849	3	35%	S1AAT812	30	34%
		S1BAT449	29	33%	S1BAT816	29	33%	S1AAT975	29	33%
		S1BAT716	28	32%	S1BAT613	28	32%	S1AAT904	27	31%
		S1AAT972	27	31%	S1BAT712	26	30%	S1AAT430	26	30%
		S1AAT976	26	30%	S1AAT672	26	30%	S1AAT813	26	30%
		S1AAT851	25	28%	S1BAT819	25	28%	S1BAT865	25	28%

Station Name — The name of the terminal as it is defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF). This is the terminal at which a transaction originated.

Count — The number of transactions that occurred at the terminal for the specified processing day.

Percent — A percentage indicating the number of transactions performed at a specific device as compared to the average number of transactions expected per device. The average number of transactions expected per device is calculated by the utility and is given a percentage value of 100 percent. The average number of transactions per device is calculated by adding the number of transactions processed for all active devices and dividing this total by the number of active devices.

For example, if the average number of transactions expected per device is 67 per processing day, when 67 transactions have been performed at a specific device, that device has reached 100 percent of transaction activity expected for the processing day. A number less than 67 indicates that the device is below the average transaction activity level expected for the device. A number greater than 67 indicates that the device is above the average transaction activity level expected for the device. Therefore, a value of 200 percent indicates that the device produced twice as many transactions as the average device, and a value of 50 percent indicates that only half of the average was produced.

Stations Sorted by Line Report

This report sorts the same data as in the Stations Sorted by Name, Detail Report by grouping terminal names within the names of lines to which they are attached, and providing total activity for each line.

This report is useful when terminals are directly attached, polled, and controlled by the BASE24 system. Terminals that dial up or use virtual circuits for transaction authorization do not always use the same line into the BASE24 system, so these situations void the data in this report. A virtual circuit means that more than one circuit exists over the same line. In this case, the BASE24 or HP NonStop line name is not logged in the TLF, PTLF, or TTLF.

The utility produces the report by matching the terminal name from the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or TTDF to the station name configured in the Network Environment File (NEF) for each XPNET node. The XPNET process uses the NEF to control the operation of communications devices within each XPNET node.

In addition to the re-sorted station activity data, the utility also totals the transaction activity within each line. A totals line lists the line name and the total number of transactions for the reporting period for the line. This data can be used to identify lines that are reaching capacity and to balance activity among lines.

This report is optional. It is produced when the NEF parameter is provided in the RUN command. For more information about the NEF runtime parameter, refer to the “Reporting Options” discussion provided earlier in this section.

An example of the Stations Sorted by Line Report is shown on the following page for BASE24-atm. The format of the report is identical for BASE24-pos and BASE24-teller. Field descriptions are provided following the example.

Page 69	System ACI	A T M	S T A T I O N R E P O R T				Logical network L1LL				MM/DD/YY
(Stations sorted by line)											
Station Name		Count, Percent	Station Name		Count, Percent	Station Name		Count, Percent			
S1AAT863		485 480%	S1AAT320		399 395%	S1AAT436		377 373%			
S1AAT802		234 232%	S1AAT298		228 226%	S1BAT871		227 225%			
S1AAT671		351 348%	S1BAT901		331 328%						
L1A^LINE1, DEVICE ID \$SDLC8, SDLC MULTI-POINT SUPERVISOR											
S1BAT765		274 271%	S1BAT870		272 260%	S1BAT903		200 198%	2632, 5%		
S1BAT810		195 193%	S1BAT861		194 199%	S1BAT827		270 267%			
S1AAT363		257 254%	S1AAT316		253 250%	S1AAT825		250 248%			
S1AAT360		269 266%	S1AAT230		268 265%	S1BAT856		268 265%			
S1BAT523		262 259%	S1AAT801		261 258%	S1AAT319		259 256%			
L1A^LINE2, DEVICE ID \$SDLC3, SDLC MULTI-POINT SUPERVISOR											
S1BAT860		247 245%	S1BAT813		241 239%	S1AAT800		234 232%			
S1BAT825		224 222%									
L1A^LINE3, DEVICE ID \$SDLC1, SDLC MULTI-POINT SUPERVISOR											
									946, 2%		

Station Name — The name of the terminal as it is defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF). This is the terminal at which a transaction originated.

Count — The number of transactions that occurred at the terminal for the specified processing day.

Percent — A percentage indicating the number of transactions performed at a specific device as compared to the average number of transactions expected per device. The average number of transactions expected per device is calculated by the utility and is given a percentage value of 100 percent. The average number of transactions per device is calculated by adding the number of transactions processed for all active devices and dividing this total by the number of active devices.

For example, if the average number of transactions expected per device is 67 per processing day, when 67 transactions have been performed at a specific device, that device has reached 100 percent of transaction activity expected for the processing day. A number less than 67 indicates that the device is below the average transaction activity level expected for the device. A number greater than 67 indicates that the device is above the average transaction activity level expected for the device. Therefore, a value of 200 percent indicates that the device produced twice as many transactions as the average device, and a value of 50 percent indicates that only half of the average was produced.

<information provided for the line> — The symbolic name, device ID, and protocol specified for the line to which the terminals are connected. The line information is taken from the Network Environment File (NEF). This area also lists the total number of transactions performed at the terminals listed above for the specified line, and the percent of this transaction activity in relationship to the total transaction activity for all terminals in the system.

Station Activity Summary Report

This report summarizes station activity according to the number of terminals that fall within specified ranges of transaction activity. A graphical representation of the percent of transaction activity during each transaction activity range is displayed to the right of each range. This graph provides an overall picture of transaction activity distribution.

The lower portion of the report lists the total number of transactions originating at stations controlled by the BASE24 system, the average number of transactions originated per station, the number of active stations, the number of stations performing stand-in processing, and the total number of transactions for which stations stood in.

An example of the Station Activity Summary Report is shown on the following page for BASE24-atm. The format of the report is identical for BASE24-pos and BASE24-teller. Field descriptions are provided following the example.

Page 66	System \ACI	A T M	S T A T I O N	R E P O R T	Logical network LLLL	MM/DD/YY
Station Activity Summary						
Transaction Activity Range	Number of Stations Within Activity Range / Graph					
-----	-----					
601...	0					
551..600	0					
501..550	0					
451..500	1					
401..450	0					
351..400	0					
301..350	4					
251..300	6 *					
201..250	19 ***					
151..200	34 *****					
101..150	55 *****					
51..100	94 *****					
1.. 50	132 *****					
No Activity	191 *****					

	536 total					
	=====					
Total station activity: 30504						
Average activity per station: 88.42						
Number of active stations: 345						
Number of stations performing stand-in activity: 0						
Total transactions for which stations stood in: 0						

Transaction Activity Range — A range indicating the number of transactions performed at a particular terminal on the specified date. The various categories listed are fixed in value, from a high of over 600, in increments of 50, down to the range of 1 to 50, and those terminals that had no transaction activity for the specified processing date.

Number of Stations Within Activity Range — The number of terminals with transaction activity in the range shown in the Transaction Activity Range field for the specified processing date.

Graph — A graphical representation of the number of terminals in each range is shown on the right-hand side of the report. These ranges include a field for terminals that had no activity during the specified processing date. From this graph, you can quickly determine which transaction range had the most activity on a particular processing date.

Total station activity — The total number of transactions originated at devices controlled by the BASE24 system.

Average activity per station — The average number of transactions originated from active stations. This number is calculated by dividing the value in the Total station activity field by the value in the Number of active stations field.

Number of active stations — The number of stations, as defined in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF) that originated at least one transaction during the reporting period.

The next two lines provide information used when offline authorization is performed by terminals. These transactions are sent to the BASE24 system as force post transactions. Force post transactions are transactions that have taken place and need to be logged, even though they were not authorized by the BASE24 system.

Number of stations performing stand-in activity — The number of stations that performed stand-in processing (offline authorization). Stand-in processing can occur when a break in the communications lines exists. When a break occurs, the terminal can still authorize transactions. When communications resumes, the terminal sends a list of transactions that were authorized while the

lines were unavailable to the BASE24 system. This function is optional. In most cases when a break in communications exists, the institution shuts down the terminal so that transactions cannot take place.

Total transactions for which stations stood-in — The total number of transactions that were authorized by terminals while communications lines were unavailable.

Device Type Report

The Device Type Report provides statistics regarding transaction activity according to the model of terminal. It reports only on terminals directly attached and controlled by BASE24-atm, BASE24-pos, and BASE24-teller. For BASE24 Remote Banking, the Transaction Originator Type Report is generated instead of this report. This report is described later in this section.

The statistics on the Device Type Report include the names of the devices used, the number of transactions performed at a specific device type, the percent of the total number of transactions performed for a specific device type, the number of devices associated with a specific device type, and the percent of terminals of the same device type. In addition, totals are shown for the total number of transactions performed for the processing day and the number of different device types used to perform the transactions. The report headers are repeated on the right side of the page in order to condense the data.

An example of the Device Type Report is shown on the following page for BASE24-atm. The format of the report is identical for BASE24-pos and BASE24-teller. Field descriptions are provided following the example.

Page	67	System \ACI	A T M		D E V I C E		T Y P E		R E P O R T		Logical network LLLL		MM/DD/YY	
			Device Name	Trans, Percent	Number, Percent	Device Name	Trans, Percent	Number, Percent	Trans, Percent	Number, Percent	Trans, Percent	Number, Percent	Trans, Percent	Number, Percent
			Docutel 5100	2	0	Diebold 910	12987	43%	106	20%				
			Docutel 2300	264	1%	Diebold 911 Outside	11113	36%	134	25%				
			Diebold 911 Lobby	2825	9%	IBM 3624 Ver 7 DC	2308	8%	61	11%				
			Diebold 912	818	3%	IBM 4730 D Console	73		1					
Total transactions: 30390														
Device types used: 8														

Device Name — The model of terminal used to perform transactions. The terminal name is derived from a code for the terminal type as logged in the Transaction Log File (TLF), the POS Transaction Log File (PTLF), or the Teller Transaction Log File (TTLF). This code is converted to a name by the program and placed in the Device Name field.

For BASE24-atm, the device types are as follows:

- 01 = Diebold 910
- 02 = IBM 3624 Version 8—dual cartridge
- 04 = Burroughs RT650/750
- 07 = Fujitsu 910 mode
- 08 = Fujitsu 911 mode
- 09 = Diebold 911 Outside
- 10 = Diebold 911 Lobby
- 11 = Docutel 5100
- 14 = IBM 3624 Version 7—single cartridge
- 15 = IBM 3624 Version 7—dual cartridge
- 18 = IBM 3624 Version 8—single cartridge
- 20 = Diebold 912
- 22 = NCR 5XXX
- 26 = IBM 4730 single console
- 27 = IBM 4730 dual console
- 30 = Diebold 10XX/478X
- 32 = Olivetti 6000
- 48 = IBM 3624 Version 8—single cartridge D01
- 49 = IBM 3624 Version 8—dual cartridge D02
- B3 = Triton Mini ATM
- B4 = Tidel Anycard ATM
- B5 = Fujitsu 7000
- B6 = Diebold CashSource Plus
- B7 = NCR Cash ATM

For BASE24-pos, the device types are as follows:

- 41 = ACI Standard POS Device Handler
- 43 = NCR 2123 and NCR 2126
- 44 = NCR 2127
- 45 = NCR 7000
- 70 = Visa I, APACS 30
- 71 = Visa II, APACS 40
- 78 = Merchant Host Interface
- D0 = Hypercom TDS
- D1 = Custom POS device 1
- D2 = Custom POS device 2
- D3 = Custom POS device 3
- D4 = Custom POS device 4
- D5 = Custom POS device 5
- D6 = BASE24 SPDH (fuel)
- D7 = BASE24 SPDH (electronic cash register)
- D8 = BASE24 SPDH (script dispenser)

For BASE24-teller, the device types are taken from the SYS.RESPONDER.STA-NAME field in the TTLF.

Trans — The number of transactions processed for the specified terminal type.

Percent — The percent of the total number of transactions processed for the specified terminal type.

Number — The number of terminals for this device type.

Percent — A percentage of the total number of terminals with this device type.

Total transactions — The total number of transactions that occurred for the processing day by terminals controlled by the BASE24 system.

Device types used — The total number of different terminal types used in the BASE24 system for the specified processing day.

Device Handler Report

This report provides information regarding the number of transactions processed by BASE24 Device Handler processes or modules. The Device Handler process or module controls a terminal and converts messages from the terminal into BASE24 message format. The utility matches the terminal name logged in the Transaction Log File (TLF), POS Transaction Log File (PTLF), or Teller Transaction Log File (TTLF) with the corresponding name in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF). The BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, and TTDF are sequentially read and loaded into memory during program initialization.

For BASE24 Remote Banking, the Transaction Originator Type Report is generated instead of the Device Handler Report. The Transaction Originator Type Report is described later in this section.

Statistics shown on this report include the symbolic name of the Device Handler process or Device Handler/Router/Authorization process (for BASE24-pos only), the number of active terminals that sent transactions to the Device Handler process or module, the number of transactions associated with the Device Handler process or module, and the percentage of transactions sent to a specific Device Handler process or module in relationship to the total number of transactions performed during the processing day.

Note: For BASE24-pos, the Device Handler name in the BASE24-pos Terminal Data files is not enforced to be accurate. In other words, a BASE24-pos Device Handler/Router/Authorization process does not verify that the name specified in the BASE24-pos Terminal Data files matches its own symbolic name. This is because BASE24-pos terminals often dial up or use virtual circuits, which results in transaction requests entering the system on different communications lines. For this reason, this report may not be useful for BASE24-pos, and an inaccurate representation of transaction activity can arise. However, the Authorization Report provides a subset of this information. This is due to the fact that the BASE24-pos Device Handler and Authorization modules run in a single process. An example of the Device Handler Report is shown on the next page, followed by descriptions of its fields.

Page	68	System \ACI	A T M	D E V I C E	H A N D L E R	R E P O R T	Logical network LLLL	MM/DD/YY
Device Handler Name				Active Terminals		Transactions	Percent	
-----				-----		-----	-----	
P1A^S910^2				46		5679	19%	
P1A^I3624V7^1				14		1178	4%	
P1A^S910^1				38		4619	15%	
P1B^S910^1				23		2296	8%	
P1A^C910^1				39		1969	6%	
P1B^I3624V7^1				30		1130	4%	
P1B^S910^2				22		1631	5%	
P1B^C910				32		2927	10%	
P1B^C910^1				44		5121	17%	
P1A^I3614V6				7		507	2%	
P1B^I3614V6				13		425	1%	
P1A^C910				27		2685	9%	
P1A^2300				8		245	1%	
P1B^I3624				1		73		
P1B^2300				1		19		
Total transactions:				30504				
Number of device handlers used:				15				

Device Handler Name — The symbolic name of the Device Handler process or Device Handler/Router/Authorization process (for BASE24-pos only) as specified in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF) and the Network Environment File (NEF).

Active Terminals — The number of terminals that sent transactions to the named Device Handler process or Device Handler/Router/Authorization process.

Transactions — The number of transactions sent to the named Device Handler process or Device Handler/Router/Authorization process.

Percent — The percentage of transactions sent to the named Device Handler process or Device Handler/Router/Authorization process in relationship to the total number of transactions performed during the processing day.

Total transactions — The total number of transactions performed during the processing day.

Number of device handlers used — The number of different Device Handler processes or Device Handler/Router/Authorization processes that received transactions during the processing day.

Transaction Originator Reports

These reports provide information regarding the activity at remote banking devices within the BASE24 Remote Banking system. For BASE24-atm, BASE24-pos, and BASE24-teller, station reports are generated instead of transaction originator reports.

Data is displayed for each interface instance defined in the ITS Interface Configuration File (IFCF). The utility sequentially loads interface instance names from the IFCF and checks the BUS-MLF.ASCII.TXN-ORIGINATOR.FLG and BUS-MLF.BCD.ACQ-OBJ-ID fields in the ITLF for each transaction to determine whether the transaction occurred on an interface instance that is defined in the IFCF. If so, interface instance statistics including the number of transactions performed, total amount of transfers, and total amount of payments are recorded. Statistics also show the number of transactions denied due to specific reason codes and the number of transactions reversed due to specific reversal codes.

An interface instance is not a process or a station. Each interface instance provides a means of dividing up interface processing according to specific business relationships and purposes. Up to 90 remote banking endpoint stations can be associated with each interface instance. All CSR terminals for the ACI-provided Customer Service Interface are defined by a single interface instance.

Sorted by Interface Instance Name

The Transaction Originator Report (Sorted by Interface Instance Name) contains one line of data for each interface instance listed. A sample of this report is shown on the next page. Field descriptions are provided following the report sample.

Page	12	System \ACI	R	B	TRANSACTION ORIGINATOR REPORT												Logical network	LLLL	MM/DD/YY
(Transaction Originators sorted by Interface Instance Name)																			
Intf Instance	Trans	Percent of avg	Total Transfers	Total Payments	114	118	191	902	906	908	912	941	942	943	Common Reversal Codes				
BANK1INTF1	12	54%		2.00							1				4001	4002	4003	4004	4013
BANK1INTF2	33	150%		160.00	1			11											
BANK1INTF3	0																		
TCP1	0																		
VRU1	0																		
VRU3	0																		
VRU4	0																		

Intf Instance — The name of the interface instance as it is defined in the ITS Interface Configuration File (IFCF).

Trans — The number of transactions performed at the interface instance. This includes approvals, denials, and reversals.

Percent of avg — A percentage indicating the number of transactions performed at a specific interface instance as compared to the average number of transactions expected per interface instance. The average number of transactions expected per interface instance is calculated by the utility and is given a percentage value of 100 percent. The average number of transactions per interface instance is calculated by adding the number of transactions processed for all active interface instances and dividing this total by the number of active interface instances.

For example, if the average number of transactions expected per interface instance is 67 per processing day, when 67 transactions have been performed at a specific interface instance, that interface instance has reached 100 percent of transaction activity expected for the processing day. A number less than 67 indicates that the interface instance is below the average transaction activity level expected for the interface instance. A number greater than 67 indicates that the interface instance is above the average transaction activity level expected for the interface instance. Therefore, a value of 200 percent indicates that the interface instance produced twice as many transactions as the average interface instance, and a value of 50 percent indicates that only half of the average was produced.

Total Transfers — The total amount of approved transfer transactions initiated at the specified interface instance.

Total Payments — The total amount of approved payment transactions initiated at the specified interface instance.

Common Denial Reasons — The number of transactions that were denied for a reason associated with the denial code listed in this column. Denial codes listed in this column for BASE24 Remote Banking include the following:

114	=	No account of type
118	=	Card/ID not found
191	=	Payment date invalid
902	=	Transaction invalid
906	=	Cutover in progress
908	=	Route destination not found
912	=	Issuer unavailable

- 941 = Invalid currency code
- 942 = Amount format error
- 943 = Customer vendor format error

For more information on denial codes and reasons for BASE24 Remote Banking transactions, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.

Common Reversal Codes — The number of transaction reversals that occurred from the interface instance with the reason listed in this column. Reversal codes listed in this column include the following:

- 4001 = Unspecified, no action taken
- 4002 = Suspected malfunction
- 4003 = Format error, no action taken
- 4004 = Completed partially
- 4013 = Unable to deliver message to point of service

For more information on reversal codes, refer to the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***.

Sorted by Activity

The Transaction Originator Report (Sorted by Activity) is a condensed version of the Transaction Originator Report (Sorted by Interface Instance Name) and it does not exist in the Transaction Analysis Utility Transaction History File in this format. The Transaction Originator Report (Sorted by Activity) shows the same interface instance names as the Transaction Originator Report (Sorted by Interface Instance Name); however, it reports only on the number of transactions for that interface instance and the percent of average generated by the interface instance. This allows the header columns to be listed across the page three times in this report. The data is read left to right and top to bottom, and the interface instance with the highest activity is sorted first. This way, the interface instance with the most activity is listed first and interface instances with less activity appear in descending order.

A sample of this report is shown on the next page. Field descriptions are provided following the report sample.

Page 13	System \ACI	R B	TRANSACTION ORIGINATOR REPORT				Logical network LLLL MM/DD/YY			
(Interface Instances sorted by activity)										
Intf Instance			Count, Percent	Intf Instance	Count, Percent	Intf Instance	Count, Percent			
BANK1INTF2			33 150%	BANK1INTF1	12 54%	TCP1	0			
VRU1			0	BANK1INTF3	0	VRU3	0			
VRU4			0							

Intf Instance — The name of the interface instance as it is defined in the ITS Interface Configuration File (IFCF).

Count — The number of transactions performed at the interface instance for the specified processing day. This includes approvals, denials, and reversals.

Percent — A percentage indicating the number of transactions performed at a specific interface instance as compared to the average number of transactions expected per interface instance. The average number of transactions expected per interface instance is calculated by the utility and is given a percentage value of 100 percent. The average number of transactions per interface instance is calculated by adding the number of transactions processed for all active interface instances and dividing this total by the number of active interface instances.

For example, if the average number of transactions expected per interface instance is 67 per processing day, when 67 transactions have been performed at a specific interface instance, that interface instance has reached 100 percent of transaction activity expected for the processing day. A number less than 67 indicates that the interface instance is below the average transaction activity level expected for the interface instance. A number greater than 67 indicates that the interface instance is above the average transaction activity level expected for the interface instance. Therefore, a value of 200 percent indicates that the interface instance produced twice as many transactions as the average interface instance, and a value of 50 percent indicates that only half of the average was produced.

Activity Summary Report

The Transaction Originator Report (Activity Summary) summarizes interface instance activity according to the number of interface instances that fall within specified ranges of transaction activity. A graphical representation of the percent of transaction activity during each transaction activity range is displayed to the right of each range. This graph provides an overall picture of transaction activity distribution.

The lower portion of the report lists the total number of transactions originating at interface instances controlled by the BASE24 system.

An example of the Transaction Originator Report (Activity Summary) is shown on the following page. Field descriptions are provided following the example.

Page 14	System \ACI	R B	TRANSACTION ORIGINATOR REPORT		Logical network LLLL	MM/DD/YY
Activity Summary						
Transaction Activity Range		Number of Intf Instances Within Activity Range / Graph				
-----		-----				
601**		0				
551**	600	0				
501**	550	0				
451**	500	0				
401**	450	0				
351**	400	0				
301**	350	0				
251**	300	0				
201**	250	0				
151**	200	0				
101**	150	0				
51**	100	0				
1**	50	2				
No Activity		5				

		7				
		total				
		=====				

Transaction Activity Range — A range indicating the number of transactions performed at a particular interface instance on the specified date. The various categories listed are fixed in value, from a high of over 600, in increments of 50, down to the range of 1 to 50, and those interface instances that had no transaction activity for the specified processing date.

Number of Intf Instances Within Activity Range — The number of interface instances with transaction activity in the range shown in the Transaction Activity Range field for the specified processing date.

Graph — A graphical representation of the number of interface instances in each range is shown on the right-hand side of the report. These ranges include a field for interface instances that had no activity during the specified processing date. From this graph, you can quickly determine which transaction range had the most activity on a particular processing date.

Total number of transactions — The total number of transactions originated at interface instances controlled by the BASE24 system.

Transaction Originator Type Report

The Transaction Originator Type Report provides statistics regarding transaction activity according to the type of transaction originator. Statistics for all interface instances of the same type are calculated together. The report includes statistics for transactions originated from BASE24-billpay Scheduled Transaction Initiator processes, Billpay Server processes, and the host, as well as from all configured interface instances.

The statistics on this report include the types of transaction originators used, the number of transactions performed at a specific transaction originator type, the percent of the total number of transactions performed for a specific transaction originator type, the number of interface instances associated with a specific transaction originator type, and the percent of interface instances of the same transaction originator type. Transaction originator type statistics for the Remote Banking Standard Interface are further broken down by the following source codes: audio device (AD), personal computer (PC), screenphone, and, optionally, any user-defined source codes.

Totals are shown for the total number of transactions performed for the processing day and the number of different transaction originator types used to perform the transactions. A sample of this report is shown on the next page. Field descriptions are provided following the report sample.

Page 15	System \ACI	R B	T R A N S A C T I O N	O R I G I N A T O R	T Y P E	R E P O R T	Logical network	LLLL	MM/DD/YY
Trans Originator Type									

Trans Originator Type				Trans	Percent	Number	Percent	-----	
VRU Total				0	-----	3	50%	-----	
AD				0	-----	0		-----	
PC				0	-----	0		-----	
Screenphone				0	-----	0		-----	
CSI				20	45%	0		-----	
Host				0	-----	*****	****	-----	
STI				24	54%	*****	****	-----	
XPBP				0	-----	*****	****	-----	
Total Trans:				44					
Trans Originator Types Used:				2					

Trans Originator Type — The type of transaction originator derived from the acquirer object ID, transaction originator, and source code logged for the transaction in the ITS Transaction Log File (ITLF). Valid transaction originator types are as follows:

VRU = Remote Banking Standard Interface
CSI = Customer Service Interface
Host = Host
STI = Scheduled Transaction Initiator process
XPBP = Billpay Server process

Totals for the Remote Banking Standard Interface are further broken down by source code. Values for reserved source codes are as follows. User-defined source codes may also be displayed on the report.

AD = Audio device (for example, an interactive voice response system)
PC = Personal computer
Screenphone = Screen phone

Trans — The number of transactions processed for the specified transaction originator.

Percent — The percent of the total number of transactions processed for the specified type of transaction originator.

Number — The number of interface instances for this type of transaction originator. Asterisks are displayed in this field for the Host, STI, and XPBP transaction originator types because interface instances do not apply to these types.

Percent — A percentage of the total number of interface instances with this transaction originator type.

Total Trans — The total number of transactions that occurred for the processing day by transaction originators controlled by the BASE24 system.

Trans Originator Types Used — The total number of different transaction originator types used in the BASE24 system for the specified processing day.

Authorization Report

This report provides statistics regarding the number of transactions processed by BASE24 Authorization processes or modules. Unlike the Device Handler Report, this includes all transactions processed, regardless of their origin.

For BASE24-atm, BASE24-pos, and BASE24-teller, statistics provided on this report include the following:

- The process pair directory (PPD) name of the Authorization process or Device Handler/Router/Authorization process (for BASE24-pos only)
- The number of transactions handled by a specific Authorization process or Device Handler/Router/Authorization process
- The percentage of transactions handled by a specific Authorization process or Device Handler/Router/Authorization process in relationship to the total number of transactions performed during the processing day

For BASE24 Remote Banking, statistics provided on this report include the following:

- An identifier for the authorizer, which is either IAS for BASE24-telebanking Integrated Authorization Server processes or BPSRV for BASE24-billpay Billpay Server processes
- The number of transactions handled by BASE24-telebanking or BASE24-billpay
- The percentage of transactions handled by BASE24-telebanking or BASE24-billpay in relationship to the total number of transactions performed during the processing day

Examples of the Authorization Report are shown on the following pages. Field descriptions are provided following the report samples.

Page	69	System \ACI	ATM	AUTHORIZATION	REPORT	Logical network	LLLL	MM/DD/YY
				Authorization Name	Count	Percent		
				-----	----	-----		
				PIAA	9892	32%		
				PIAC	5419	18%		
				PIAZ	6530	21%		
				PIBA	1198	4%		
				PIBC	2665	9%		
				PIBZ	4219	14%		
				PICA	27			
				PICC	16			
				PICZ	386	1%		
				PIC9	95			
				PIC8	57			
				Total transactions: 30504				
				Authorization processes used: 11				

Page 16	System \ACI	R B	A U T H O R I Z A T I O N	R E P O R T	Logical network	LLLL	MM/DD/YY
			Authorization Name	Count	Percent		
			-----	-----	-----		
			BPSRV	10	22%		
			TIAS	35	77%		
			Total transactions:				
			Authorization processes used:				
				45			
				2			

Authorization Name — The BASE24 process that authorizes transactions. The Authorization process name is logged in the Transaction Log File (TLF), POS Transaction Log File (PTLF), or Teller Transaction Log File (TTLF). The name is not the symbolic BASE24 name, but rather the PPD name, which is recognized by the HP NonStop operating system.

For BASE24 Remote Banking, this field displays IAS for transactions authorized by any Integrated Authorization Server process or BPSRV for transactions authorized by any Billpay Server process.

Count — The number of transactions handled by the specified authorizer.

Percent — The percentage of the total number of transactions processed by this specified authorizer.

Total transactions — The total number of transactions authorized during a specified processing day.

Authorization processes used — The total number of active Authorization processes used by terminals during the processing day. For BASE24 Remote Banking, all Integrated Authorization Server processes are counted as one and all Billpay Server processes are counted as one.

Institution Reports

The Institution Reports detail the number of transactions processed on behalf of the financial institutions (FIIDs) defined to the system. During initialization, the utility reads the Institution Definition File (IDF) and Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) to build a list of FIID names (the Remote Banking transaction analysis utility currently reads the IDF only). The utility creates and initializes a table in memory from the transaction log file (i.e., TLF, PTLF, ITLF, or TTLF) that contains transaction counts for these FIIDs. If data appears in this report for the FIID called "<DF>" (default), it is because more FIIDs were encountered when building the table than were defined in the ICF or ICFE and IDF, causing the size of the table to be exceeded.

For each transaction logged to the TLF, PTLF, or TTLF, two FIIDs exist. For BASE24-atm and BASE24-teller, these are the FIIDs for the card issuer and the terminal owner (acquirer). For BASE24-pos, these are the FIIDs for the card issuer and the retailer (acquirer). The card issuer FIID identifies the financial institution that issued the card. The other retailer or terminal owner FIID identifies the source of the transaction request received from an interchange.

For each transaction logged to the ITLF, only the customer ID issuer FIID is logged. The customer ID issuer identifies the financial institution that issued the customer ID.

For BASE24-atm, BASE24-pos, and BASE24-teller, two parts are generated for this report. The formats of both parts of the report are the same. Both parts of the report list all transactions that occurred during the reporting period. One part is subtitled "by Card Owner FI-ID" and the other is subtitled "by Terminal Owner FI-ID". The first part shows the number of transactions for each card-issuing FIID and the second part shows the number of transactions originating at each terminal-owning FIID. For BASE24 Remote Banking, only one part, subtitled "by Issuer FI-ID", is generated for this report. This part shows the number of transactions for each customer ID-issuing FIID.

For each FIID, the number of transactions approved or denied by a host, the BASE24 system, or an interchange, are listed. Totals for all authorizers are also provided by FIID. For BASE24 Remote Banking, the BASE24 statistics are further divided between BASE24-telebanking and BASE24-billpay.

Note: The transactions on this report are broken down within each of three categories by approvals, denials, and totals. Percentages are provided for all numbers in various manners. The number of approvals and denials within a

category and FIID is one ratio of 100 percent. The totals among all categories within an FIID are another ratio of 100 percent. Finally, the total transactions for each FIID is expressed as a percentage of the logical network transactions.

By Card Owner

Samples of the Institution Report By Card Owner are provided on the following pages. The first sample shows the format for a BASE24-atm or BASE24-pos report. The second sample shows the format for a BASE24-teller report. A description of each report field is provided following the report samples.

Page 70	System \ACI	ATM	I N S T I T U T I O N R E P O R T				Logical network LLLL	MM/DD/YY
by Card Owner FI-ID								
Host								
FI-ID	Approved	Denied	Total					
-----	-----	-----	-----					
***** No Activity *****								
BASE24								
FI-ID	Approved	Denied	Total					
-----	-----	-----	-----					
BNK1	0,	9, 100%	9, 100%					
BNK2	0,	81, 100%	81, 100%					
Total	0,	90, 100%	90, 77%					
=====	=====	=====	=====					
Switch(not on us)								
FI-ID	Approved	Denied	Total					
-----	-----	-----	-----					
BNK2	27, 100%	0,	27, 25%					
Total	27, 100%	0, 0%	27, 23%					
=====	=====	=====	=====					
Total								
FI-ID	Approved	Denied	Total					
-----	-----	-----	-----					
BNK1	0,	9, 100%	9, 100%					
BNK2	27, 100%	81, 100%	108, 100%					
Total	27, 97%	90, 77%	117, 100%					
=====	=====	=====	=====					

Page 70	System \ACI	TELLER	I N S T I T U T I O N R E P O R T			Logical network	LLLL	MM/DD/YY
			by Card Owner FI-ID					
			Host					
FI-ID	Approved	Denied	Ovrrd	Total				
-----	-----	-----	-----	-----				
	*****	No Activity	*****					
	BASE24							
FI-ID	Approved	Denied	Ovrrd	Total				
-----	-----	-----	-----	-----				
TBK1	0	14	0	14	100%			
-----	-----	-----	-----	-----	-----			
Total	0	14	0	14	100%			
=====	=====	=====	=====	=====	=====			
	Switch(Not on us)							
FI-ID	Approved	Denied	Ovrrd	Total				
-----	-----	-----	-----	-----				
	*****	No Activity	*****					
	Total							
FI-ID	Approved	Denied	Ovrrd	Total				
-----	-----	-----	-----	-----				
TBK1	0	14	0	14	100%			
-----	-----	-----	-----	-----	-----			
Total	0	14	0	14	100%			
=====	=====	=====	=====	=====	=====			

The Institution Report by Card Owner is divided into the following categories. If no transaction activity occurred for a particular category, the phrase “***** No Activity *****” is printed on the report for that category.

Host

The entries in the Host section contain statistics about transactions authorized by a host for the listed FIIDs. The host statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

BASE24

The entries in the BASE24 section contain statistics about transactions authorized by the BASE24 system for the listed FIIDs. The BASE24 statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

Switch (Not On Us)

The entries in the Switch section contain statistics about transactions authorized by an interchange for the listed FIIDs. The interchange statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

Total

This section shows the combined grand total and percentage of all approved, denied, and overridden transactions processed by a host, the BASE24 system, or interchanges for a particular FIID during the specified processing day.

Each of these headings contains the following fields:

FI-ID — The financial institution of the card owner for which transactions were processed.

Approved — The number and percentage of approved transactions for this financial institution.

Denied — The number and percentage of denied transactions for this financial institution.

Ovrrd — The number and percentage of overridden transactions for this financial institution. This field appears on the report for BASE24-teller only.

Total — The total number of transactions for this financial institution. This figure is followed by the percentage of transactions for this financial institution in the logical network.

By Issuer

A sample of the Institution Report By Issuer is provided on the next page. This report is generated by the Remote Banking Transaction Analysis utility only. A description of each report field is also provided following the report sample.

Page 17	System \ACI	R B	I N S T I T U T I O N R E P O R T			Logical network LLLL	MM/DD/YY
			by Issuer FI-ID				
			Host				
FI-ID			Approved	Denied	Total		
-----			-----	-----	-----		
			***** No Activity *****				
			BASE24-tb				
FI-ID			Approved	Denied	Total		
-----			-----	-----	-----		
BK40			22, 62%	13, 37%	35, 77%		
-----			-----	-----	-----		
Total			22, 62%	13, 37%	35, 77%		
=====			=====	=====	=====		
			BASE24-bp				
FI-ID			Approved	Denied	Total		
-----			-----	-----	-----		
BK40			10, 100%	0,	10, 22%		
-----			-----	-----	-----		
Total			10, 100%	0,	10, 22%		
=====			=====	=====	=====		
			Switch(Not on us)				
FI-ID			Approved	Denied	Total		
-----			-----	-----	-----		
			***** No Activity *****				
			Total				
FI-ID			Approved	Denied	Total		
-----			-----	-----	-----		
BK40			32, 71%	13, 28%	45, 100%		
-----			-----	-----	-----		
Total			32, 71%	13, 28%	45, 100%		
=====			=====	=====	=====		

The Institution Report by Issuer is divided into the following categories. If no transaction activity occurred for a particular category, the phrase “***** No Activity *****” is printed on the report for that category.

Host

The entries in the Host section contain statistics about transactions authorized by a host for the listed FIIDs. The host statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The total number and percentage of transactions approved and denied during the reporting period for the particular institution on the specified processing day

BASE24-tb

The entries in the BASE24-tb section contain statistics about transactions authorized by BASE24-telebanking for the listed FIIDs. The BASE24-telebanking statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The total number and percentage of transactions approved and denied during the reporting period for the particular institution on the specified processing day

BASE24-bp

The entries in the BASE24-bp section contain statistics about transactions authorized by BASE24-billpay for the listed FIIDs. The BASE24-billpay statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The total number and percentage of transactions approved and denied during the reporting period for the particular institution on the specified processing day

Switch (Not On Us)

The entries in the Switch section contain statistics about transactions authorized by an interchange for the listed FIIDs. The interchange statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The total number and percentage of transactions approved and denied during the reporting period for the particular institution on the specified processing day

Total

This section shows the combined grand total and percentage of all approved and denied transactions processed by a host, BASE24-telebanking, BASE24-billpay, or interchanges for a particular FIID during the specified processing day.

Each of these headings contains the following fields:

FI-ID — The financial institution of the customer ID owner for which transactions were processed.

Approved — The number and percentage of approved transactions for this financial institution.

Denied — The number and percentage of denied transactions for this financial institution.

Total — The total number of transactions for this financial institution. This figure is followed by the percentage of transactions for this financial institution in the logical network.

By Terminal Owner FIID

Samples of the Institution Report By Terminal Owner are provided on the following pages. The first sample shows the format for a BASE24-atm or BASE24-pos report. The second sample shows the format for a BASE24-teller report. A description of each report field is also provided following the report samples.

Page 70 System \ACI A T M I N S T I T U T I O N R E P O R T Logical network LLLL MM/DD/YY

by Terminal Owner FI-ID

FI-ID	Host	
---	Approved	Denied
---	-----	-----
---		Total
---	-----	-----

***** No Activity *****

FI-ID	BASE24	
---	Approved	Denied
---	-----	-----
---		Total
---	-----	-----
BNK1	0,	9, 100%
BNK2	0,	81, 100%
---	-----	-----
Total	0,	90, 77%
---	=====	=====

Switch(not on us)

FI-ID	Switch(not on us)	
---	Approved	Denied
---	-----	-----
---		Total
---	-----	-----
BNK2	27, 100%	0,
---	-----	-----
Total	27, 100%	0, 0%
---	=====	=====

FI-ID	Total	
---	Approved	Denied
---	-----	-----
---		Total
---	-----	-----
BNK1	0,	9, 100%
BNK2	27, 100%	81, 100%
---	-----	-----
Total	27, 97%	90, 77%
---	=====	=====

Page 70	System \ACI	TELLER	I N S T I T U T I O N R E P O R T				Logical network	LLLL	MM/DD/YY
			by Terminal Owner FI-ID						
			Host						
FI-ID	Approved	Denied	Ovrrd	Total					
----	-----	-----	-----	-----					
***** No Activity *****									
BASE24									
FI-ID	Approved	Denied	Ovrrd	Total					
----	-----	-----	-----	-----					
TBK1	0	14	0	14 100%					
Total	0	14	0	14 100%					
=====	=====	=====	=====	=====					
Switch(Not on us)									
FI-ID	Approved	Denied	Ovrrd	Total					
----	-----	-----	-----	-----					
***** No Activity *****									
Total									
FI-ID	Approved	Denied	Ovrrd	Total					
----	-----	-----	-----	-----					
TBK1	0	14	0	14 100%					
Total	0	14	0	14 100%					
=====	=====	=====	=====	=====					

The Institution Report by Terminal Owner is divided into the following categories. If no transaction activity occurred for a particular category, the phrase “***** No Activity *****” is printed on the report for that category.

Host

The entries in the Host section contain statistics about transactions authorized by a host for the listed FIIDs. The host statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

BASE24

The entries in the BASE24 section contain statistics about transactions authorized by the BASE24 system for the listed FIIDs. The BASE24 statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

Switch (Not On Us)

The entries in the Switch section contain statistics about transactions authorized by an interchange for the listed FIIDs. The interchange statistics shown for each FIID include the following:

- The number and percentage of transactions approved during the entire reporting period
- The number and percentage of transactions denied during the entire reporting period
- The number and percentage of transactions overridden during the entire reporting period (for BASE24-teller only)
- The total number and percentage of transactions approved, denied, and overridden during the reporting period for the particular institution on the specified processing day

Total

This section shows the combined grand total and percentage of all approved, denied, and overridden transactions processed by a host, the BASE24 system, or interchanges for a particular FIID during the specified processing day.

Each of these headings contains the following fields:

FI-ID — The financial institution of the terminal owner for which transactions were processed.

Approved — The number and percentage of approved transactions for this financial institution.

Denied — The number and percentage of denied transactions for this financial institution.

Ovrrd — The number and percentage of overridden transactions for this financial institution. This field appears on the report for BASE24-teller only.

Total — The total number of transactions for this financial institution. This figure is followed by the percentage of transactions for this financial institution in the logical network.

Category Report

This report is optional and can only be produced for BASE24-pos. Furthermore, this report cannot be printed in hard copy format by the POS Transaction Analysis utility.

The report provides a further breakdown of transaction activity by source, destination, and card or account type for each of the intervals for which the utility was run. In this manner, activity can be tracked by source, destination, and card or account type by time of day.

For each interval, a record is written to the Transaction Analysis Utility Transaction History File itemizing the number of transactions for the following unique keys:

- PTLF.HEAD.CRD.FIID — Identifies the institution that issued the card for that transaction.
- PTLF.HEAD.RETL.KY.RDFKEY.FIID — Identifies the retailer who owns the terminal that originated the transaction.
- PTLF.AUTH.ORIG — Identifies the device handler that accepted the transaction.
- PTLF.AUTH.DEST — Identifies the database that authorized the transaction.
- PTLF.AUTH.CRD-TYP — Identifies the type of card or service used. For example, a bank or credit card.
- PTLF.AUTH.TRAN-CDE.AA — Identifies the type of account the funds are withdrawn from. For example, a checking or savings account.

Warning: This report can produce a large amount of data for systems that have many different transaction paths. Therefore, the amount of data generated from this report may restrict disk space.

ACI Worldwide, Inc.

3: BLOCKR2 Utility

This section describes the BLOCKR2 utility. It explains how you can use the BLOCKR2 utility to determine the sizing requirements of a structured Enscribe file. This makes it possible to maximize the use of file space. This section includes the following information:

- An overview of the BLOCKR2 utility
- Procedures for installing and running the BLOCKR2 utility
- Procedures for setting up the input files used with the BLOCKR2 utility
- A description of the output produced by the BLOCKR2 utility

Overview

The BLOCKR2 utility is an analysis utility used for Enscribe file sizing. You can use the utility for HP NonStop disk drive versions DP1 and DP2 (Disk Process 1 and 2). The BLOCKR2 utility computes the optimum block size and number of extents to allow for BASE24 structured Enscribe data files. The BLOCKR2 utility also displays the amount of file space that is currently unused or cannot be used. This is considered wasted file space. Wasted file space includes bytes unavailable for use within allocated data and index blocks, and data space lost in the last allocated block and extent. The BLOCKR2 utility calculates the block size and file extent size providing the least amount of wasted file space.

Generally, the BLOCKR2 utility is run on all BASE24 structured Enscribe data files when a BASE24 system is installed for use in a production environment. However, you can also use the BLOCKR2 utility when you want to do one or more of the following:

- Increase the number of records in one or more Enscribe files
- Delete records from one or more Enscribe files
- Increase transaction volume
- Modify data in one or more Enscribe files

BLOCKR2 Utility Installation Procedures

Before you can use the BLOCKR2 utility, it must be properly installed. To install the BLOCKR2 utility, restore the BA xx BLKR subvolume (where xx represents the current release number) from tape to the volume where the BASE24 performance tools reside. The BA xx BLKR subvolume contains all of the files necessary to run the BLOCKR2 utility.

After this subvolume is restored, the following steps are required to complete installation. For detailed information on MAKE, refer to the *ACI Make User Guide*.

1. Modify the BLOCKRFM and BLOCKRMM files as needed for your site. The BLOCKRFM file contains configuration options and the BLOCKRMM file contains file defines.
2. Run the BLOCKRM MAKE file to compile the BLOCKR2 executable file.

Running the BLOCKR2 Utility

Once the BLOCKR2 utility has been installed, it can be executed at any time from a TACL prompt. The BLOCKR2 utility can be run in *interactive* or *noninteractive* mode. In interactive mode, the utility can be run as needed to size individual files through a series of online prompts displayed at the home terminal.

In noninteractive mode, the BLOCKR2 utility is run as a background process (i.e., as a continuously running or temporarily executing process). In this case, the BLOCKR2 utility uses read files you create to perform the appropriate file sizing functions instead of prompting for information at the home terminal. The IN file contains the same file sizing data for which the operator is prompted when the BLOCKR2 utility is run in interactive mode. Running the BLOCKR2 utility in noninteractive mode is particularly useful if you have several files with the same sizing requirements.

An output location can be specified in the RUN command for the BLOCKR2 utility, regardless of whether the utility is run in interactive or noninteractive mode. The output location can be a terminal, the spooler, or any type of Enscribe file.

Note: ACI recommends that data files use a maximum block size of 4096 and a maximum number of extents of 16.

Command Entry for Running the BLOCKR2 Utility

The command syntax for running the BLOCKR2 utility is described below, followed by an explanation of each variable that can be entered. This subsection also includes examples of the options available when running the utility. To run the BLOCKR2 utility, move to the volume where the BASE24 performance tools reside and enter the following command.

Syntax

RUN *program* [*/[in-file][, [out-file]] /*]

program — The executable file name of the BLOCKR2 utility (e.g., BLOCKR2).

in-file — The name of a file that contains input data to be used by the BLOCKR2 utility. This field is optional. If input data files are used, the BLOCKR2 utility is run in noninteractive mode. If no input data file is specified in the command syntax, the program defaults to interactive mode.

out-file — The name of the file where the output that results from running the BLOCKR2 utility is sent. The output file can also be a spooler location. If an out file or spooler location is specified in the command syntax, the BLOCKR2 utility output is displayed at the home terminal. The valid output file options are as follows:

File name	The name of the file where the output that results from running the BLOCKR2 utility is sent. The output file does not have to be created prior to the execution of the BLOCKR2 utility. If the output file does not currently exist, the program creates it.
Spooler location	A spooler location where the output that results from running the BLOCKR2 utility is sent. This allows output from the utility to be printed in hard copy form.
Default	If no out file or spooler location is specified in the command, the output is displayed at the home terminal.

Command Examples

The command for running the BLOCKR2 utility can be modified to allow entry of input data either interactively at the home terminal or noninteractively through use of an input file. In addition, the command syntax allows you to send the output that results from running the BLOCKR2 utility to a specific location such as a file, a printer, or the home terminal. The following examples describe the options available when entering the BLOCKR2 utility RUN command. The examples are followed by a table that summarizes each option available when running the BLOCKR2 utility.

Using the Home Terminal Default

You can run the BLOCKR2 utility interactively and send both input and output to the home terminal. In this case, no input or output file is specified in the BLOCKR2 utility RUN command. Once the BLOCKR2 utility RUN command is entered, the utility receives information interactively through a series of prompts and subsequent operator entries from the home terminal. Thus, the input data used by the BLOCKR2 utility is entered manually by the operator. The BLOCKR2 utility output is displayed at the home terminal after the operator has entered all of the input data. The BLOCKR2 utility RUN command that allows both the input and output to be sent to the home terminal is as follows:

```
RUN BLOCKR2
```

Using the Home Terminal and Specified Output Locations

In some cases, you may want to send the output data calculated by the BLOCKR2 utility to a spooler location, a file, or a printer location instead of having the data displayed at the home terminal. By modifying the RUN command, it is possible to use the BLOCKR2 utility interactively and specify a particular output location.

In this case, no input file is specified in the BLOCKR2 utility RUN command. Once the BLOCKR2 utility RUN command is entered, the utility receives information interactively through a series of prompts and subsequent operator entries from the home terminal. Thus, the input data used by the BLOCKR2 utility is entered manually by the operator. In the example, the output is sent to a spooler location. You can use the following command to run the BLOCKR2 utility interactively and send the output to a spooler location.

```
RUN BLOCKR2/ OUT $S.#BLOCKR2/
```


Using Input Files Without Specifying Output Locations

It is possible to run the BLOCKR2 utility noninteractively from a TACL prompt through the use of input files. Input files can be created and then specified in the BLOCKR2 utility RUN command.

When an input file is specified in the BLOCKR2 utility RUN command, the utility uses the information in the file and runs noninteractively from a TACL prompt. The following example shows the command syntax for running the BLOCKR2 utility using an input file without specifying an output location.

```
RUN BLOCKR2/ IN INBLOCK /
```

Using Specified Input and Output Files

Input and output files can be specified in the command for running the BLOCKR2 utility. The BLOCKR2 utility uses the information in the file and runs noninteractively from a TACL prompt. The information resulting from running the BLOCKR2 utility is written to the output file or printer location. The following example shows the command syntax for running the BLOCKR2 utility using an input and an output file.

```
RUN BLOCKR2/ IN INBLOCK, OUT OUTBLOCK/
```

It is not necessary to create the output file prior to running the BLOCKR2 utility. If the output file specified in the command syntax does not exist, the BLOCKR2 utility creates it. In the example, the output location is the file OUTBLOCK. If the file OUTBLOCK does not already exist, the BLOCKR2 utility automatically creates it upon entry of this command.

BLOCKR2 Utility Options Summary

The following table summarizes the input and output options described in the previous examples. It shows the variation on the BLOCKR2 utility RUN command, whether an input file is included in the RUN command syntax, whether an output file is included in the RUN command syntax, the resulting location where the output from running the BLOCKR2 utility is sent, and whether the BLOCKR2 utility is running interactively or noninteractively.

RUN Command Entry Variation	In File	Out File	Output Location	Mode
Using the home terminal default	No	No	Home terminal	Interactive
Using the home terminal and specified output locations	No	Yes	Spooler location or out file	Interactive
Using input files without specifying output locations	Yes	No	Home terminal	Noninteractive
Using specified input and output files	Yes	Yes	Spooler location or out file	Noninteractive

Exiting the BLOCKR2 Utility

As described, the BLOCKR2 utility can be run from a TACL prompt interactively through a series of prompts or noninteractively through the use of input files. The BLOCKR2 utility continues to execute as long as input data is received. If the BLOCKR2 utility is run interactively, you can exit the program by pressing the **Enter** key at one of the following prompts.

```
Enter Enscribe DP version (1, 2)
Enter file type
```

If the BLOCKR2 utility is run noninteractively, it stops at the end of the input file. If the BLOCKR2 utility needs to be stopped before the input file has completed, press the **Control-Break** keys to terminate the program.

Running the BLOCKR2 Utility Interactively

The following steps can be performed to run the BLOCKR2 utility interactively.

1. Gather information about the file that requires sizing by using the HP NonStop File Utility Program (FUP) and performing a FUP INFO *file-name*, DETAIL command. Entering this command provides you with information you can use later when running the BLOCKR2 utility.

The FUP INFO *file-name*, DETAIL command provides the file type, the key length, the index block size, the record size, and the block size. Refer to the appropriate HP NonStop manual for more information about FUP.

2. Move to the subvolume where the BASE24 performance tools reside by entering the following command from a TACL prompt.

```
VOLUME volume
```

3. Enter one of the following commands to run the BLOCKR2 utility interactively.

```
RUN BLOCKR2  
RUN BLOCKR2/ OUT $S.#BLOCKR2/
```

Entering the first of these command causes the output resulting from running the BLOCKR2 utility to be displayed at the home terminal. Entering the second command cause the output to be sent to a spooler location.

4. Provide the following information by responding to the prompts described below.
 - a. Enter the HP NonStop disk drive format version.

Prompt: Enter the Enscribe DP Version (1,2)
Entry: 1 or 2

You can also exit the BLOCKR2 utility from this prompt by pressing the **Enter** key.

- b. Enter the type of file for which the BLOCKR2 utility is being run. Valid file types are as follows:

K = Key-sequenced
E = Entry-sequenced
R = Relative

Prompt: Enter the file type (E, K, R)
Entry: E, K, or R

You can also exit the BLOCKR2 utility from this prompt by pressing the **Enter** key.

- c. Enter the file key length. This prompt is only displayed if the file type selected at the previous prompt was K (key-sequenced). The key length can be any value from 1 to 255.

Prompt: Enter key length

Entry: 1–255

- d. Enter the index block size of the file. This prompt is displayed only if the file type is K (key-sequenced) and the DP version is 1. If these conditions are met, you can optionally enter an index block size.

The index block size must be a multiple of sector size 512 and cannot be greater than 4096. If you press the **Enter** key without specifying an index block, the report generated by the BLOCKR2 utility lists all the possible index block sizes.

Prompt: Enter index block size (optional - otherwise shows all possible)

Entry: *index-block-size*

Note: With DP version 2, the index block size defaults to the data block size specified at the upcoming “Enter block size” prompt.

- e. Enter the record size. The record size is based on the type of file being sized. The maximum record size for entry sequenced and relative files is 4072 bytes. The maximum record size for key sequenced files is 4062 bytes.

Prompt: Enter record size

Entry: *record-size*

- f. Enter the data block size. The block size must be a multiple of sector size 512 and cannot be greater than 4096. Responding to this prompt is optional. If you press the **Enter** key without specifying a block size, the report that is generated lists all the possible block sizes.

Prompt: Enter block size (optional - otherwise shows all possible)

Entry: *data-block-size*

- g. Define the number of records you want to be contained in the file. Any number equal to or greater than 1 is a valid value.

Prompt: Enter number of records

Entry: *number-records*

- h. Enter the number of extents to use for sizing the file. This entry defines the number of contiguous blocks of storage, starting on a sector boundary and containing a multiple of 2048 bytes. This field is set based on the DP version. The maximum number of extents for DP1 is 16 and the maximum number of extents for DP2 is 978.

Prompt: Enter number of extents to use

Entry: *number-extents*

Note: ACI recommends that data files use a maximum block size of 4096 and a maximum number of extents of 16.

- 5. Once all of the input information is entered, press the **Enter** key to generate the BLOCKR2 utility output.

Running the BLOCKR2 Utility Noninteractively

The BLOCKR2 utility can be run in noninteractive mode. In this case, the BLOCKR2 utility RUN command is entered from a TACL prompt as described in the “Running the BLOCKR2 Utility” subsection. Once the BLOCKR2 utility RUN command is entered, the utility gets the information it requires from the input file. The BLOCKR2 utility can be run noninteractively by entering one of the following commands.

```
RUN BLOCKR2/ IN in-file /  
RUN BLOCKR2/ IN in-file, OUT output-location/
```

In the first command example, no output location is specified. In this case, the output that results from running the BLOCKR2 utility is displayed at the home terminal when the utility has completed execution. If the second command example is entered, an output file is specified and the information that results from running the BLOCKR2 utility is written to an output file or spooler location.

Setting Up Input Data Files

When you run the BLOCKR2 utility in noninteractive mode, input data files must be set up prior to execution. The following table defines the fields used when creating an input file to run the BLOCKR2 utility. The table contains the input field name, a description of the field, and the valid values that can be entered in the field.

Input Field Name	Input Field Description	Valid Values
DP version	The HP NonStop format disk drive process version. This field specifies either DP1 or DP2.	1 or 2
File type	The type of file that is being sized.	E = Entry-sequenced K = Key-sequenced R = Relative
Key length	The length of the key to the file. This field is only used with key-sequenced files.	1–255
Index block size	The index block size (in multiples of 512). The index block size is only used with DP version 1 key-sequenced files.	0–4096
Record size	The record size of the file being sized.	0–4072 = Entry-sequenced 0–4072 = Relative 0–4062 = Key-sequenced
Block size	The block size (in multiples of 512) of the file being sized. The block size must be a multiple of 512 and cannot be greater than 4096.	0–4096
Number of records	The number of records contained in the file being sized.	Any whole number equal to or greater than 1.
Number of extents	The number of contiguous blocks of storage to be used.	DP1 = 1–16 DP2 = 1–978

Note: ACI recommends that data files use a maximum block size of 4096 and a maximum number of extents of 16.

Input Data File Templates

Several template files are provided to customers. You can use these template files as starting points for running the BLOCKR2 utility in noninteractive mode. You can modify the template files as required. They are located on the subvolume where the BASE24 performance tools reside.

Two templates exist for key-sequenced files; one for DP1 and another for DP2. The only difference in the template files is that the template file for DP1 contains the Index Block Size field. One template file also exists for entry-sequenced and relative files. The content of each template file and the order in which the data should be entered in the input file is shown below.

Key-Sequenced Template

The following is an example of a template file for a key-sequenced file (DP version 1). The example includes all of the fields contained in an input file for a key-sequenced file using an HP NonStop disk drive format version of DP1.

```
1          !DP Version (1)
K          !File type (key-sequenced)
10         !Key length
4096       !Index block size
400        !Record size
4096       !Block size
250        !Number of records
16         !Number of extents
<EOF>
```

The only difference between the template file for a key-sequenced file using DP1 and a key-sequenced file using DP2 is that DP1 requires an Index Block Size field and DP2 does not.

Entry-Sequenced and Relative Template

The following is an example of a template file for a relative file. The example includes all of the fields contained in an input file for a relative file. The template file for an entry-sequenced file is exactly like the example shown below, except the value in the file type field is E (entry-sequenced) instead of R.

```
2          !DP Version (2)
E          !File type (entry-sequenced)
430        !Record size
1024       !Block size
500        !Number of records
20         !Number of extents
R          !File type (relative), change to E if entry-sequenced
210        !Record size
4096       !Block size
420        !Number of records
100        !Number of extents
<EOF>
```

BLOCKR2 Utility Output

This subsection contains examples of the output generated by the BLOCKR2 utility. Although the same report header is produced for all output generated by the utility, the manner in which data is displayed on the report varies according to the type of file being sized. The examples in this subsection include the report header and the output generated by the BLOCKR2 utility for the following structured file types:

- Key-sequenced
- Entry-sequenced
- Relative

The following report header is produced for all output generated by the BLOCKR2 utility. The report header example is followed by descriptions of its fields.

```

-----
Rec   Number of   Blk   Recs   Number of   File size   Amount   Pct   Num   Ext
size   records    size  /blk   blocks      (EOF)    wasted  wasted  ext  size
-----
```

Rec size — The number of bytes in each record in the file.

Number of records — The number of records in the file.

Blk size — The data block size.

Recs/blk — The number of records contained in a single data block. This number is calculated by dividing the block size by the record size.

Number of blocks — The number of data blocks required. This number is calculated by dividing the number of records by the number of records per block.

File size (EOF) — The size of the file in bytes to the end-of-file (EOF). This number is calculated by multiplying the block size and the number of blocks.

Amount wasted — The amount of unused or unavailable space in the file.

Pct wasted — The percentage of unused space and space that cannot be used contained in the file.

Num ext — The number of extents allocated for the file.

Ext size — The number of pages contained in each extent. This number is calculated by dividing the number of pages needed by the number of extents to be allocated for the file.

Key-Sequenced DP2 File

The following is an example of the output produced by the BLOCKR2 utility for a key-sequenced file running on the HP NonStop format disk drive version 2 (DP2). If the disk drive version is DP1, an index block size field is displayed on the report.

The file sizing input data used to produce the example report is as follows:

Input Data	
Field	Value
DP version	2
File type	K
Key length	22
Record size	118
Block size	Press Enter key (list all possibilities)
Number of records	500
Number of extents	15

Key-sequenced DP2 files consist of 3 different block types: a bit map block, a data block, and index blocks. In the output generated by the BLOCKR2 utility, each block size possibility is grouped into several lines of data. The first line of data within each group contains data block information. The second line through the bit map block line contains the index block data. The line following the index block information contains the bit map block. The last line in each group contains the totals for the data block, bit map block, and index blocks.

In the example, block size 1024 has the least percentage of wasted file space (9.5 percent) and, therefore, provides optimum use of file space. The minimum extent size to use for this file is 3. Any modifications to the file being sized can be done through FUP. Refer to the appropriate HP NonStop manual for more information about FUP.

Rec size	Number of records	Blk size	Recs /blk	Number of blocks	File size (EOF)	Amount wasted	Pct wasted	Num ext	Ext size
118	500	512	4	125	64,000	1,536	2.3%		DATA
25	125	512	18	7					IDX LVL 1
25	7	512	18	1					IDX LVL 2
		512		1					BITMAP
			** TOTAL **		68,000	23,852	25.9%	15	3
118	500	1024	8	63	64,512	3,512	5.4%		DATA
25	63	1024	36	2					IDX LVL 1
25	2	1024	36	1					IDX LVL 2
		1024		1					BITMAP
			** TOTAL **		68,608	27,193	9.5%	15	3
118	500	2048	16	32	65,536	4,488	6.8%		DATA
25	32	2048	74	1					IDX LVL 1
		2048		1					BITMAP
			** TOTAL **		69,632	28,090	30.5%	15	3
118	500	4096	33	16	65,536	4,968	7.6%		DATA
25	16	4096	150	1					IDX LVL 1
		4096		1					BITMAP
			** TOTAL **		73,728	26,770	29.0%	15	3

Entry-Sequenced DP2 Example

The following is an example of the output produced by the BLOCKR2 utility for an entry-sequenced file running on the HP NonStop format disk drive version 2 (DP2). If the disk drive version is DP1, an index block size field is displayed on the report.

The file sizing input data used to produce the example report is as follows:

Input Data	
Field	Value
DP version	2
File type	E
Record size	1016
Block size	Press Enter key (list all possibilities)
Number of records	500
Number of extents	100

In an entry-sequenced file, all blocks are data blocks. Each line in the following example represents a different data block possibility for an entry-sequenced file given the input data described earlier.

Rec size	Number of records	Blk size	Recs /blk	Number of blocks	File size (EOF)	Amount wasted	Pct wasted	Num ext	Ext size
1016	500	2048	1	500	1,024,000	504,000	49.2%	100	5
1016	500	4096	4	125	512,000	102,650	16.7%	100	3

Relative DP2 Example

The following is an example of the output produced by the BLOCKR2 utility for a relative file running on the HP NonStop format disk drive version 2 (DP2). If the disk drive version is DP1, an Index block size field is displayed on the report.

The file sizing input data used to produce the example report is as follows:

Input Data	
Field	Value
DP version	2
File type	R
Record size	110
Block size	Press Enter key (list all possibilities)
Number of records	500
Number of extents	10

In a relative file, the first block is a bit map block that indicates which data blocks contain at least one record. The second block is always the first data block. In the output generated by the BLOCKR2 utility, the entry for each block size consists of up to three lines of information. The first line contains data block information. The second line contains the bit map block size. The third line contains totals for the data block and the bit map block.

Rec size	Number of records	Blk size	Recs /blk	Number of blocks	File size (EOF)	Amount wasted	Pct wasted	Num ext	Ext size
110	500	512	4	125	64,000	6,786	10.4%		DATA
		512		1					BITMAP
			** TOTAL **		64,512	22,658	27.7%	10	4
110	500	1024	8	63	64,512	8,142	12.4%		DATA
		1024		1					BITMAP
			** TOTAL **		65,536	23,502	27.7%	10	4
110	500	2048	18	28	57,344	720	1.3%		DATA
		2048		1					BITMAP
			** TOTAL **		59,392	2,768	4.5%	10	3
110	500	4096	36	14	57,344	1,028	1.8%		DATA
		4096		1					BITMAP

ACI Worldwide, Inc.

4: MACKIE Draw Utilities

This section describes the MACKIE Draw utilities. Two separate MACKIE Draw utilities are discussed: MACPUP and MACPRT. These two utilities work together to provide a system configuration diagram that includes CPU and device configurations. You can use the configuration information produced by these programs to monitor a specific system and to assist in deciding where changes can be made to the configuration to enhance performance. This section includes the following information:

- An overview of the MACKIE Draw utilities
- Procedures for installing and running the MACKIE Draw utilities
- An overview of the output produced by the MACKIE Draw utilities

Note: The MACKIE Draw utilities are only relevant to Himalaya systems (i.e., K1000/K2000/K10000/K20000) or older systems. These utilities will not work for newer systems that use the Servernet architecture (i.e., S-series systems).

Overview

The MACKIE Draw utilities are system analysis utilities used to produce a physical diagram of an HP NonStop system configuration. There are two MACKIE Draw utilities. The MACPUP utility gathers system configuration data and writes it to an output file. The MACPRT utility uses this file to create a physical system configuration diagram, which can then be printed out.

The MACPUP utility gathers system configuration data from two main HP NonStop sources: the Peripheral Utility Program (PUP) and the PEEK utility. For more information about these utilities, refer to the appropriate HP NonStop manual.

The MACPUP utility also displays a table of statistics at the home terminal that shows how the information it gathered was sorted and merged into a single output file for use by the MACPRT utility. The HP NonStop SORT product is responsible for performing SORT and MERGE functions. For more information about these functions, refer to the appropriate HP NonStop manual.

The MACPUP and MACPRT utilities provide data that can otherwise be difficult to gather and store in one place. You can use the system configuration diagram for problem resolution and analysis, or simply for informational purposes.

MACKIE Draw Installation Procedures

Before you can use the MACKIE Draw utilities, they must be properly installed. To install these utilities, restore the BA xx MAC subvolume (where xx represents the current release number) from tape to the volume where the BASE24 performance tools reside. The BA xx MAC subvolume contains all of the files necessary to run the MACKIE Draw utilities.

After this subvolume is restored, the following steps are required to complete installation. For detailed information on MAKE, refer to the *ACI Make Release 5 User Guide*.

1. Modify the MACFM and MACMM files as needed for your site. The MACFM file contains configuration options and the MACMM file contains file defines.
2. Run the MACM MAKE file to compile MACPUP and MACPRT executable files.

Running the MACKIE Draw Utilities

Once the MACKIE Draw utilities have been installed, you can run them at any time from a TACL prompt. The MACPUP utility must be run first since it generates an output file that is used by the MACPRT utility. The input for running the MACPUP utility is acquired by the program from the system using the HP NonStop Peripheral Utility Program (PUP) and PEEK utility. For more information concerning PUP and the PEEK utility, refer to the appropriate HP NonStop manual. The remainder of this subsection describes the command syntax for running the MACPUP and MACPRT utilities.

Running the MACPUP Utility

The command syntax for running the MACPUP utility is described below, followed by an explanation of each field that you can enter. This subsection also includes examples of the options available when running the MACPUP utility.

To run the MACPUP utility, move to the volume where the BASE24 performance tools reside and enter the following command.

Syntax

RUN *program* [/ *priority-number*, OUT *out-file* /] [;T]

program — The object file name of the MACPUP utility (e.g., MACPUP).

priority-number — The execution priority at which the utility is to be run.

out-file — The name of the file to contain the MACPUP utility output data. If the output file specified does not exist, the MACPUP utility creates it. If the output file exists prior to running the MACPUP utility, the content of the output file is purged before the new data generated by the MACPUP utility is written to the file.

If no output file is specified in the command syntax, the output that results from running the MACPUP utility is sent to an output file named MACWORK. The MACWORK output file is created by the MACPUP utility.

T — Specifies whether to include TACL processes in the CPU section of the system configuration diagram produced by the MACPRT utility. If this parameter is not specified in the RUN command syntax, TACL processes are not included in the diagram.

MACPUP Utility RUN Command Examples

You can modify the command for running the MACPUP utility to achieve certain results. The following examples describe the options available when entering the MACPUP utility RUN command.

Running the MACPUP Utility Without a Specified Output File

Entering the following command causes the MACPUP utility to be executed. In this case, the MACPUP utility also creates an output file named MACWORK since no output file is specified in the command syntax. The output generated by the MACPUP utility is then written to the MACWORK file.

```
RUN MACPUP
```

Running the MACPUP Utility With an Output File Specified

Entering this command causes the MACPUP utility to be executed. In this case, the MACPUP utility is run at a priority of 10 and a file named OUTFILE is specified in the command as the output location. The addition of the T option at the end of the RUN command instructs the MACPUP utility to provide a list of all TACL processes currently running in the system. This list of TACL processes is included in the CPU section of the system configuration diagram produced by the MACPRT utility.

```
RUN MACPUP / PRI 10, OUT OUTFILE /;T
```

Output Produced by the MACPUP Utility

Although the output file created by the MACPUP utility is used as input for the MACPRT utility, the MACPUP utility does generate its own output for display at the home terminal. The MACPUP utility generates a table of sort and merge statistics. These statistics show how the information gathered by the MACPUP utility was sorted and merged into a single output file for use by the MACPRT utility. Sorting is the process of arranging records in a specific order. Merging is the process of combining two or more sets of records into a single output. Records must be sorted before they can be merged. The HP NonStop SORT product is responsible for performing SORT and MERGE functions. For more information concerning the SORT product, refer to the appropriate HP NonStop manual.

The following example shows the sort and merge statistics generated by the MACPUP utility and displayed at the home terminal. The example is followed by a description of each field shown in the example.

```
-----SORT/MERGE STATISTICS:-----
RECORD SIZE                =          132
BUFFER PAGES                =          166
RECORDS                     =          423
SORT ELAPSED TIME           =         15.03
SORT I/O WAIT TIME          =          9.32
SORT CPU TIME                =          5.71
COMPARES                    =         3607
SCRATCH SEEKS                =           0
SCRATCH DISK SIZE (PAGES)   =           0
INITIAL RUNS                 =           0
FIRST MERGE ORDER           =           0
MERGE ORDER                  =           0
INTERMEDIATE PASSES         =           0
NUMBER OF DUPLICATES DROPPED =           0
```

RECORD SIZE — The number of bytes per record contained in the sorted output file.

BUFFER PAGES — The number of 1024-word pages of memory used by the HP NonStop SORT program as sort area.

RECORDS — The number of records contained in the sorted output file.

SORT ELAPSED TIME — The amount of time, in seconds and hundredths of seconds (ss.hh), spent sorting.

SORT I/O WAIT TIME — The amount of time, in seconds and hundredths of seconds (ss.hh), spent in READ, WRITE, and AWAITIO calls to the HP NonStop SORT program.

SORT CPU TIME — The amount of CPU time, in seconds and hundredths of seconds (ss.hh), used during the sort.

COMPARES — The number of times two records were compared during the sort to determine if duplicate records existed.

SCRATCH SEEKS — The number of READ and WRITE operations performed on the scratch file during the sort. The scratch file is where all input files are stored to be sorted and merged.

SCRATCH DISK SIZE (PAGES) — The size of the scratch file in pages. The scratch file is where all input files are stored to be sorted and merged.

INITIAL RUNS — The number of runs generated by the first pass of the HP NonStop SORT program. The input files received by the SORT program are sorted into runs and stored in the scratch file. This field is used to indicate the number of runs it took to sort the input files. The input files can be merged once they have been sorted.

FIRST MERGE ORDER — The number of sort runs merged in the first intermediate pass.

MERGE ORDER — The maximum number of sort runs that were merged.

INTERMEDIATE PASSES — The number of merge cycles between the initial run formation and final merge.

NUMBER OF DUPLICATES DROPPED — The number of duplicate records that were removed during the merge.

Running the MACPRT Utility

The command syntax for running the MACPRT utility is described below, followed by an explanation of each field that you can enter. This subsection also includes examples of the options available when running the MACPRT utility.

To run the MACPRT utility, move to the volume where the BASE24 performance tools reside and enter the following command.

Syntax

RUN *program* [/ IN *in-file*, OUT *spooler-loc* /] [COLUmns *number*,]
[NOPRINT *device-number device-number*]

program — The object name of the MACPRT utility (e.g., MACPRT).

in-file — The name of the file containing the input data. This is the output file generated by running the MACPUP utility.

spooler-loc — The spooler location where the system diagrams created by the MACPRT utility are sent. The output must be sent to the spooler so that the MACPRT diagrams can be printed.

number — The line width, in columns, of the diagram. The number of columns per line must be between 33 and 132. The COLUMN parameter is optional. If no line width is specified in the RUN command, the default width of the diagram is 132 columns.

device-number — Specifies the numbers of the device types to exclude from the configuration diagram generated by the MACPRT utility. These device types are separated by a single space after the NOPRINT parameter. The NOPRINT parameter is optional and up to 21 device types can be listed. For a list of the valid device type numbers, refer to the HP NonStop **Peripheral Utility Program (PUP) Reference Manual**.

MACPRT Utility RUN Command Example

The command syntax for running the MACPRT utility is shown in the following example. In this example, the input file created by the MACPUP utility is MACWORK and the spooler location where the system diagram generated by the MACPRT utility is sent is \$S.#MACRPT. The number of columns on the report is 100 and information for device types 1 and 62 are not included on the report. In addition, the width of the spooler report is 80 columns.

```
RUN MACPRT/ IN MACWORK, OUT $S.#MACRPT/COLUMNS 100, NOPRINT 1 62,  
80
```

Output Produced by the MACPRT Utility

The MACPRT utility produces a system diagram based on the information contained in the output file created by the MACPUP utility. The system diagram, called a Mackie diagram, contains the following:

- System information
- CPU configuration information
- Device configuration information

The benefits of providing this information in diagram form is two-fold. It provides a printout of what can be a complex configuration, saving you the time and effort of having to gather configuration information to create your own diagram. This allows you to see the configuration of a specific system. The diagram also shows important configuration relationships vital in determining where changes can be made in your configuration. It is also helpful in problem resolution and analysis. For example, the diagram provides a summary of memory use, lists the processes running in each CPU and their execution priorities, and shows the relationships between the devices and controllers in a specific system.

System Information

The system diagram produced by the MACPRT utility is preceded by a report header that contains the following system information:

- System day, month, and year
- System time (hh:mm:ss:hh)
- System name
- System number
- HP NonStop Kernel operating system (Guardian) version number

This gives you basic system information to identify the system illustrated.

CPU Configuration Information

The system diagram produced by the MACPRT utility contains the following information about the CPU configuration of a specific system:

- CPU number
- CPU type
- Number of pages in memory
- Size of memory pages
- Number of free pages available in memory
- Number of Process Control Blocks (PCBS) configured for the CPU
- Number of Link Control Blocks (LCBS) configured for the CPU
- Number of Time List Elements (TLES) configured for the CPU
- Configured sizes of the system storage pools (SYSPPOOL and MAPPOOL) for the CPU

In addition, a list of the processes running in each CPU is provided. The first column lists the execution priorities of the processes running in the CPU. Processes are displayed on the diagram according to their priority. Higher priority processes are listed first followed by lower priority processes.

The second column in the diagram under the heading “PROCESSES” lists the process pair directory (PPD) name of the process. A -P next to the PPD name in the list indicates that the process is the primary process. A -B next to the PPD name in the list indicates that the process is a backup process.

The third column in the diagram under the heading “PROCESSES” lists the process names.

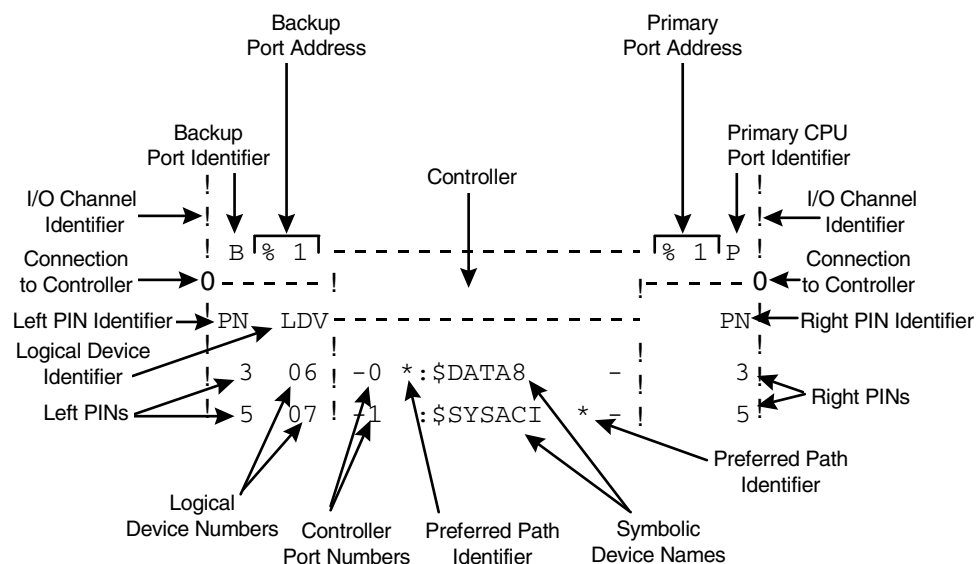
Device Configuration Information

The device configuration information provided in the system diagram created by the MACPRT utility shows the relationships of devices to controllers. The device configuration information is created on the diagram through the use of several special characters. This type of diagram is called a *Mackie diagram*. An example of a Mackie diagram is provided later in this section. The Mackie diagram shows the following relationships between the devices and the controllers:

- The input and output channels between devices, controllers, and CPUs
- The left and right Process ID Numbers (PINs) for each controller
- The logical device number
- The port number for each controller
- The preferred path between the controller and the devices
- An indicator identifying whether the device is primary, backup, or mirror

Example Report Generated by the MACPRT Utility

This subsection provides an example of the system configuration diagram produced by the MACPRT utility. The following illustration shows an example of the symbols used by the MACPRT utility to produce a system configuration diagram. Each symbol is labeled according to what it represents. The illustration is followed by a table that explains the symbols used on the system configuration diagram. The example of the system configuration diagram follows the table.



MACPRT Diagram Symbols

The following table explains the symbols used on the system configuration diagram produced by the MACPRT utility.

Symbol	Description
!	An exclamation mark (!) running vertically down the page represents input/output (I/O) channels between devices, controllers, and CPUs.
<i>number</i>	A number printed in the vertical column in the diagram represents the channel of the CPU to which the controller is connected.
PN	The letters PN in the diagram identify the PIN (Process ID Number) of the current primary or backup process of a device. If PN is displayed to the left of the controller in the diagram, PN represents the left PIN. If PN is displayed to the right of the controller in the diagram, PN represents the right PIN. In the example diagram, the left and right PINs are both 3 for \$DATA8 and both 5 for \$SYSACI. The left PIN is 6 for \$TAPE1 and 7 for \$TAPE2. The right PIN is 10 for \$TAPE1 and 11 for \$TAPE2.
LDV	The letters LDV in the diagram identify the logical device number associated with each device. The logical device is the process used to communicate with a physical device. Each device in the system is associated with a pair of system processes. These two processes reside in separate processor modules. A user process communicates with a logical device, but never directly with a physical device. A logical device number and a device name are associated with every logical device in a system. With the exception of demountable disks, all device names are assigned during system generation. Disks configured as demountable take the device name from the disk volume label when they are initiated. In the example, the logical device number associated with \$DATA8 is 06 and with \$SYSACI is 07. The logical device number associated with \$TAPE1 is 38 and with \$TAPE2 is 39.

Symbol	Description
<i>-number</i>	A minus sign (-) followed by a number to the immediate left of the symbolic device name identifies the controller port number associated with the device. In the example, the controller port numbers are 0 and 1.
*	An asterisk (*) displayed on the diagram to the left or right of the symbolic device name represents the preferred path. This is displayed on the diagram when dual disk controllers are used because more than one path is possible. The devices shown in the example are dual-ported. An access path is the route taken to a physical device from either port of a dual-ported controller by a processor module to which that port is connected. There are two access paths to each physical device (except for dual-ported disks, which have four). These paths are defined during system generation. When the system is cold loaded or reloaded, all I/O processes establish the primary access to each physical device based on physical accessibility and order of processor loading. The access path is rerouted by the operating system if a hardware failure occurs and can be altered by using commands entered through the HP NonStop Peripheral Utility Program (PUP). For more information concerning PUP commands, refer to the appropriate HP NonStop manuals. In addition, of the two paths to a device, one is the primary access path (e.g., the primary I/O process) and the other is the backup. For single-ported devices, the backup path always becomes the active path in the event of any failure to access the device using the primary path.

Symbol	Description
* <i>continued</i>	Disks on HP NonStop systems can be dual ported (i.e., attached to two controllers). This provides four access paths to a disk device (i.e., a primary and backup access path through each controller). Because two possible access paths exist from each processor module, one path is defined as the preferred path (i.e., preferred controller). Both primary and backup access paths for each controller must be from the same processor module pair. In the event of a failure to access the device by the preferred primary path, an attempt is made to access the device by the backup path. In the event of a failure to access the device by the primary processor (e.g., due to the failure of the channel, both controller ports, or a processor) access is by the backup processor.
-P	A minus sign (-) followed by the letter P displayed to the right of the process name in the PROCESSES section of the diagram indicates the process is the primary device of a mirrored disk or is the write device of a synchronous controller.
-M	A minus sign (-) followed by the letter M displayed to the right of the process name in the PROCESSES section of the diagram indicates the process is the mirror device.
-B	A minus sign (-) followed by the letter B displayed to the right of the process name in the PROCESSES section of the diagram indicates the process is the backup controller path to the primary device of a mirrored disk or is the read device of a synchronous controller.
-MB	A minus sign (-) followed by the letters MB displayed to the right of the process name in the PROCESSES section of the diagram indicates the process is the backup controller path to the mirror device.
% <i>number</i>	A percent sign (%) followed by a number on the diagram identifies the primary or backup port address.
P	The letter P displayed to the right or left of the port address in the diagram indicates this is the primary port.

Symbol	Description
B	The letter B displayed to the right or left of the port address in the diagram indicates this is the backup port.
<i>\$symbolic-name</i>	The symbolic name of the device.

Example System Configuration Diagram

The illustration on the following page shows an example of the complete system configuration diagram generated by the MACPRT utility. The example contains three controllers between CPU 0 and CPU 1. The first two controllers are dual disk controllers for devices \$DATA8 and \$SYSACI. The third controller is a tape controller. The 0 in the vertical column indicates that the controllers are connected to channel 0 on both CPUs. The devices associated with the tape controller are \$TAPE1 and \$TAPE2. The exclamation mark (!) to right of the symbolic device names represents an alternative connection.

Copyright (C) 1985, 1990 by Applied Communications, Inc.
All rights reserved.

MONDAY NOVEMBER 05, YYYY 01:11:13.27 PM

SYSTEM NAME: \ACI2 SYSTEM NUMBER: 2
GUARDIAN VERSION: M20

```

-----
! CPU NUMBER: 0 ! CPU NUMBER: 1 !
! CPU TYPE: VLX ! CPU TYPE: VLX !
! MEMORY: 0 PAGES ! MEMORY: 0 PAGES !
! PAGE SIZE: 2 KB ! PAGE SIZE: 2 KB !
! FREE PAGES: 0 ! FREE PAGES: 0 !
! PCBS: 0 ! PCBS: 0 !
! LCBS: 0 ! LCBS: 0 !
! TLES: 0 ! TLES: 0 !
! SYSPPOOL: 0 ! SYSPPOOL: 0 !
! MAPPOOL: 0 ! MAPPOOL: 0 !
!
! PROCESSES: ! PROCESSES: !
!
! 199 $HLP00 CMONHLP ! 199 $HLP01 CMONHLP !
! 175 $XMM -P MEASMON ! 175 $XMM MEASMON !
! 151 $DM00 DMON ! 151 $DM01 DMON !
! 150 $AA07 TACL ! 150 $TB04 TACL !
! 150 $TB17 TACL ! 150 $TC14 TACL !
!
!
-----
!
!
! B % 1-----% 1 P !
0-----! !-----0
!PN LDV-----PN!
! ! !
! 3 06!-0 *: $DATA8 -! 3!
! 5 07!-1 : $SYSACI *-! 5!
! ! !
! ! !
! ! !
! P %11-----%11 B !
0-----! !-----0
! -----!
! P % 1-----% 1 B !
0-----! !-----0
! -----!
! ! !
! 6 38!-0 : $TAPE1 10!
! 7 39!-1 : $TAPE2 11!

```

ACI Worldwide, Inc.

5: The DISCRPT Utility

This section describes the DISCRPT utility. You can use the DISCRPT utility to produce reports that show the configuration of each disk volume in your system and provide statistics concerning disk space usage in the system. This section includes the following information:

- An overview of the DISCRPT utility
- Procedures for installing the DISCRPT utility
- Configuration requirements for the DISCRPT utility
- Procedures for setting up the DISCRPT utility and the files used with the utility
- Procedures for running the DISCRPT utility
- A description of the output produced by the DISCRPT utility

Overview

The DISCRPT utility is a system analysis utility used to produce reports containing statistical information about disk usage and configuration. The DISCRPT utility gathers input data provided to it using three HP NonStop utilities to produce reports about disk configuration, cache memory configuration, disk space usage, and file usage. It then consolidates and organizes this data and produces reports containing information organized according to disk volume.

The DISCRPT utility uses data from the following HP NonStop utilities:

- Peripheral Utility Program (PUP)
- File Utility Program (FUP)
- Disk Space Analysis Program (DSAP)

You can use the DISCRPT utility reports to analyze disk space allocation and cache memory configurations. You can also use them to monitor how full each disk volume in the system is becoming and the files currently used in the system.

Note: Because PUP will no longer be supported beginning with the HP NonStop G-series NonStop Kernel operating system, this utility will only work on systems using an earlier release of the HP NonStop Kernel operating system.

DISCRPT Utility Installation Procedures

Before you can use the DISCRPT utility, it must be properly installed. To install these utilities, restore the BAxxDISC subvolume (where xx represents the current release number) from tape to the volume where the BASE24 performance tools reside. The BAxxDISC subvolume contains all of the files necessary to run the DISCRPT utility.

After this subvolume is restored, the following steps are required to complete installation. For detailed information on MAKE, refer to the *ACI Make Release 5 User Guide* manual.

1. Modify the DSCRPTFM and DSCRPTMM files as needed for your site. The DSCRPTFM file contains configuration options and the DSCRPTMM file contains file defines.
2. Run the DSCRPTM MAKE file to compile the DSCRPT executable file.

Once the DISCRPT utility is installed, some preparation is required before running it. In order for the DISCRPT utility to produce an accurate report of disk cache memory configuration, the disk cache counters for each disk must be reset. The disk cache counters should be reset before the beginning of the daily peak period to achieve accurate reporting. Reset the disk cache counters by entering the following PUP command at a TACL prompt, where *disk-name* is the name of the disk for which cache counters are to be reset. A super group ID is required to perform this command.

```
PUP LISTCACHE $disk-name, INIT
```

The DISCRPT utility should be run immediately after the peak period and before any batch jobs such as reports, extracts, backups, etc.

Once the DISCRPT utility has been installed correctly and the disk cache counters have been reset, the DISCRPT utility can be used.

Setting Up the DISCRPT Utility

The DISCRPT utility writes the information it collects to an output file. You can then use the output file to run the two reports the DISCRPT utility generates. The output file must be created prior to the execution of the DISCRPT utility and the name of the file is required in the DISCRPT RUN command. You can create an output file by entering the following File Utility Program (FUP) command at a TACL prompt. Once the output file is created, you can enter the file name in the RUN command for the DISCRPT utility.

```
FUP CREATE file-name, TYPE E, REC 132, BLOCK 4096, EXT (10, 10),  
BUFFERED
```

In the example, the FUP command is used to create a buffered output file that is entry-sequenced with a record length of 132 bytes, a block size of 4096 bytes, and maximum primary and secondary extent sizes of 10. The BUFFERED parameter should be used only if the file resides on a DP2 volume. For more information concerning FUP, refer to the appropriate HP NonStop manual.

Specifying File Locations

The DISCRPT utility uses information provided by PUP, FUP, and DSAP to create its output. Before running the DISCRPT utility, the locations of the PUP, FUP, and DSAP object files must be specified using TACL ASSIGN statements. The DISCRPT utility then uses these object files to gather the information it requires to produce its output. ASSIGN statements can be specified at a TACL prompt by entering the following commands.

Syntax

CLEAR ALL

ASSIGN *assign-name*, *file-name*

ASSIGN *assign-name*, *file-name*

ASSIGN *assign-name*, *file-name*

assign-name — Specifies a name for the assign.

file-name — The fully qualified names (i.e., \$volume.subvolume.filename) of the object files for PUP, FUP, and DSAP. Three separate ASSIGN commands must be entered at the TACL prompt to specify the file locations.

An example of setting the file assigns for PUP, FUP, and DSAP follows.

Example

CLEAR ALL

ASSIGN PUP^NAME, \$SYSTEM.SYSTEM.PUP

ASSIGN FUP^NAME, \$SYSTEM.SYSTEM.FUP

ASSIGN DSAP^NAME, \$SYSTEM.SYSTEM.DSAP

Note: File locations for PUP, FUP, and DSAP must be specified using ASSIGN statements. If the file locations are not specified, the DISCRPT utility cannot produce its output.

Running the DISCRPT Utility

The DISCRPT utility can be executed from a TACL prompt. The command syntax for running the DISCRPT utility is described below, followed by an explanation of each variable in the command. This subsection also includes an example of the options available when running the DISCRPT utility.

To run the DISCRPT utility, move to the volume where the BASE24 performance tools reside and enter the following command:

Syntax

RUN *program* [/ PRI *priority*, OUT *out-file* /]

program — The object file name of the DISCRPT utility (e.g., DISCRPT).

priority — The execution priority at which the DISCRPT utility is to run. The default priority is 10.

out-file — The name of the file that is to contain output data generated by the DISCRPT utility. If no out file is specified in this field, the DISCRPT utility creates a default file named DISCROUT.

The following example shows the command used to execute the DISCRPT utility.

Example

RUN DISCRPT/ PRI 10, OUT DISCROUT/

The utility in this example runs at a priority of 10 and the output resulting from running the DISCRPT utility is written to an output file named DISCROUT.

Output Produced by the DISCRPT Utility

The output file created by the DISCRPT utility contains two reports: the Disc Report Detail and the Disc Report Summary. These two reports contain information retrieved from PUP, FUP, and DSAP for each disk volume in the system and consolidate the data in one central location.

Disc Report Detail

The Disc Report Detail is a collection of information obtained for each disk volume in the system from calls to the following HP NonStop utilities:

- Peripheral Utility Program (PUP)
- File Utility Program (FUP)
- Disk Space Analysis Program (DSAP)

The Disc Report Detail is divided into five sections. The first section is the report header, which contains the report title, system ID, current date, and current time.

The second section provides an introduction to the report. It contains the disk volume, the number of pages allocated to cache, the percentage of the disk that is full, the disk subtype, the disk capacity, the primary CPU, and the backup CPU.

The third section of the report is called **CACHE MEMORY CONFIGURATIONS**. This section provides statistical data about the cache memory for a particular disk volume in the system. For more information about cache memory, refer to the appropriate HP NonStop manual.

The fourth section of the report is called **NOTES** and it displays data on space allocation and media failure for a particular disk volume. It identifies space allocation anomalies and any unspared defective sectors contained on the primary disk.

The fifth section of the report is called **PERMANENT FILES OPEN** and it contains a list of the files currently open in the system (temporary files are not included) according to subvolume and file name. It also provides the total number of permanent files and temporary files currently opened.

The following is an example of the Disc Report Detail produced by the DISCRPT utility. The example is followed by a description of each field displayed on the report.

```

                                DISC REPORT DETAIL

                                \ACI2
                        Thursday, November 8, YYYY - HH:MM:SS PM

VOLUME: $DATA8                      ALLOCATED 93, 486 PAGES, 77.7% FULL
4114/4115, 264MB                      CPUS 1, 1

CACHE MEMORY CONFIGURATIONS:

COUNTERS RESET : SEP 10 YYYY, 07:50
ELAPSED TIME : 59 DAYS, 05:46
CACHE BLOCK SIZE : 512      1024      2048      4096
BLOCKS REQUESTED :
BLOCKS ALLOCATED : 16      16      16      64
BLOCKS INUSE : 2      15      16      64
BLOCKS DIRTY : 0%      0%      0%      0%
CACHE READS : 74%      57%      100%      95%
CACHE READ HITS : 91%      54%      61%      99%
CACHE READ MISSES : 9%      46%      39%      1%
CACHE WRITES : 26%      43%      0%      5%
CACHE WRITE DIRTYS : 0%      0%      0%      6%
CACHE WRITE CLEANS : 87%      99%      82%      90%
CACHE WRITE MISSES : 13%      1%      18%      4%
CACHE CALLS : 67877      15232967      50207      22931538
CACHE FAULTS : 28      25      149      627
AUDIT FORCES : 0      0      0      0
BYTES ALLOCATED TO CACHE : 312K      WRITES/CONTROL POINT : 0.00

NOTES:

Space Allocation Consistency Analysis:

No space allocation anomalies.

Media Failure Analysis:

Primary disk has no unspared defective sector(s).
Mirror disk is down.

PERMANENT FILES OPEN:

ACIMEAS.M901108      PATHACI.PATHCTL      SPOOLACI.CDATA
SPOOLACI.CTRLSP2      SPOOLACI.CTRLSP3      SPOOLACI.CTRLSP4

TOTAL PERMANENT FILES OPEN: 06
TOTAL TEMPORARY FILES OPEN: 09

```

VOLUME — The volume for which cache configuration statistics were produced. In the example, the volume is \$DATA8.

ALLOCATED — The number of blocks allocated to cache.

FULL — The percentage of cache that is currently full.

4114/4115 — The disk subtype.

264MB — The disk capacity.

CPUS — The primary and backup CPU numbers.

CACHE MEMORY CONFIGURATIONS

This section contains cache configuration statistics for the specified system and volume. The data displayed in this section results when the DISCRPT utility makes a call to run the PUP LISTCACHE *\$disk-name*, STAT command.

COUNTERS RESET — The last date (MMM DD YYYY) and time (hh:mm) the cache counters were reset. If the counters have not been initialized since the system time was first set, this field shows the date and time of the last system cold load. If the counters have been initialized, this field shows the date and time at which the counters were initialized.

ELAPSED TIME — The total amount of elapsed time in days, hours, and minutes (e.g., 59 days, 05:46) since the most recent PUP LISTCACHE, INIT command was performed.

CACHE BLOCK SIZE — Lists the block size of the cache (i.e., 512, 1024, 2048, and 4096).

BLOCKS REQUESTED — The number of blocks requested for the cache configuration. This can be changed using the PUP SETCACHE command.

BLOCKS ALLOCATED — The number of blocks allocated to cache.

BLOCKS INUSE — The number of blocks in cache that contain a valid disk block in memory. This value is usually equal to the value displayed in the BLOCKS ALLOCATED field.

BLOCKS DIRTY — The percentage of allocated blocks that are currently *dirty*. Dirty blocks are blocks in cache that have been changed, but are not yet written to disk.

CACHE READS — The percentage of cache calls displayed in the CACHE CALLS field that were made for a user read request. When added together, the value in this field and the value in the CACHE WRITES field should equal 100 percent.

CACHE READ HITS — The percentage of time the requested block was found in cache memory. When added together, the value in this field and the value in the CACHE READ MISSES field should equal 100 percent. A high percentage of cache read hits indicates high activity by a sequential-access file of the indicated cache block size. A random-access file should not have a high hit rate.

CACHE READ MISSES — The percentage of time the disk process could not find the requested block in cache memory and had to bring the block in from disk.

Note: A user write request does not affect the read counters even if the cache write request causes a disk read. A 40 percent miss rate (equal to a 60 percent hit rate) is acceptable for key-sequenced files accessed randomly, but is too high for files accessed sequentially.

CACHE WRITES — The percentage of cache calls displayed in the CACHE CALLS field that the disk process made for user write requests resulting in cache writes. The percentage displayed in the CACHE READS field should equal 100 percent minus the percentage displayed in this field.

CACHE WRITE DIRTYS — The percentage of cache writes displayed in the CACHE WRITES field for which the block found in cache memory was already dirty (changed). When the disk process finds a dirty block, it is not required to perform a disk read, but it is required to perform a disk write.

CACHE WRITE CLEANS — The percentage of cache writes displayed in the CACHE WRITES field that resulted in the block being found in cache memory, but not being dirty (changed). The disk process is not required to perform a disk read, but is required to perform a disk write. This counter increases each time the disk process writes to a file.

CACHE WRITE MISSES — The percentage of cache writes displayed in the CACHE WRITES field that resulted in the disk process not finding a block in cache memory and having to read the block from the disk. For cache write misses, the disk process is required to perform both a disk read and a disk write.

CACHE CALLS — The total number of cache reads and cache writes performed since the specified date. If a user request bypasses cache (e.g., by specifying direct I/O), none of the cache counters are affected by the call.

CACHE FAULTS — The number of times a cache call was made to find a block in cache memory, but because of memory limitations, the block could not be found in cache. This number should be very small or zero. If it is not, you can use the PUP SETCACHE command to add more memory or you can reconfigure the cache.

AUDIT FORCES — The number of times a dirty block that has been audited required an audit-trail write to disk to make room for a new block to be read from disk.

BYTES ALLOCATED TO CACHE — The total memory, in bytes, allocated to cache. This number equals the sum of the value displayed in the BLOCKS ALLOCATED field multiplied by the cache block size. This number cannot exceed 6000K.

WRITES/CONTROL POINT — The number of buffers that were required to be written to disk at the most recent control point. This number is derived by dividing the number of writes forced at a control point by the number of control points encountered since the counters were initialized. This number should be very small or zero. If it is not, notify the system manager. For more information about control points, refer to the appropriate HP NonStop manual.

NOTES

This section contains disk analysis data resulting from when the DISCRPT utility makes a call to run the DSAP *\$disk-name* command.

Space Allocation Consistency Analysis — A listing of any space allocation anomalies that exist for the specified disk. If no anomalies exist, a message indicating there are no space anomalies is displayed.

Media Failure Analysis — A listing of any media problems that exist for the specified disk. If no problems exist, a message indicating there are no media problems is displayed.

PERMANENT FILES OPEN

This section contains a list of all permanent files that are currently open in the system according to subvolume and file name (temporary files are not included). The data displayed in this section results when the DISCRPT utility makes a call to run the FUP LISTOPENS *\$disk-name.*.** command on each volume in the system.

TOTAL PERMANENT FILES OPEN — The total number of permanent files open in the system.

TOTAL TEMPORARY FILES OPEN — The total number of temporary files open in the system.

Disc Report Summary

The Disc Report Summary contains a graph indicating the percentage of each disk volume in the system that is full.

This report is divided into two sections. The first section is a report header that contains the report title; the system ID; and the current day, date, and time. The second section of the report contains the disk volume, the disk subtype, the disk capacity, the numerical percentage of the disk that is full, and a graphical representation of the percentage of the disk that is full.

The following is an example of the Disc Report Summary produced by the DISCRPT utility. The example is followed by a description of each field displayed on the report.

DISC REPORT SUMMARY											
\ACI2											
Thursday, November 8, YYYY - HH:MM:SS PM											
VOLUME	PERCENT FULL	0	10	20	30	40	50	60	70	80	90 100
		!	!	!	!	!	!	!	!	!	!
\$DATA8	77.7%	*****!									
4114/4115, 264MB								!		!	
\$SYSACI	83.5%	*****									
4114/4115, 264MB								!		!	
\$SPAN2	79.1%	*****!									
4170, 895MB								!		!	
\$PLUTO	72.4%	*****									
4130, 415MB								!		!	

VOLUME — The volume for which disk statistics were produced.

PERCENT FULL — The percentage of the disk that is currently full.

4114/4115 — The disk subtype.

264MB — The disk capacity.

ACI Worldwide, Inc.

A: Error Messages

This appendix describes the error messages generated by the BASE24 performance tools documented in this manual. These error messages may occur as a result of running one of the BASE24 performance tools. Error messages are logged at the terminal where the BASE24 performance tools are run. Error messages are documented in alphabetical order by performance tool, with all of the transaction analysis utility error messages grouped together.

As you review these error messages, note that some of these messages contain an HP NonStop Kernel operating system error number. An HP NonStop Kernel operating system error indicates that the operating system encountered an error while running a program (for example, a security error occurred while accessing a file or a file name was not properly entered). The number associated with the HP NonStop Kernel operating system error identifies the type of error that occurred and allows you to reference the error. Errors not associated with an HP NonStop Kernel operating system error number are considered to be logic errors. Logic errors generally arise when data has been improperly entered.

For additional information regarding HP NonStop Kernel operating system error messages, refer to the appropriate HP NonStop manual.

Transaction Analysis Utility Error Messages

The following error messages are produced when problems occur while running the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities.

Abort Program; Fatal errors exist

Cause: The Remote Banking Transaction Analysis utility aborted due to fatal errors.

Remedy: Check for associated error messages and correct the errors if possible. If the problem persists, contact HELP24.

Assign NETLIST-PROG not found in LCONF

Cause: The NETLIST-PROG assign could not be found in the Logical Network Configuration File (LCONF). This assign specifies the location of the Netlist utility or Netlist emulation program used to retrieve line information from the Network Environment File (NEF) for the Stations Sorted by Line Report.

Remedy: Verify that the NETLIST-PROG assign exists in the LCONF. If not, add it. If the problem persists, contact HELP24.

AUTH table overflows by # entries; heavy use of default AUTH and HISF possible

Where: # = Number of excess entries

Cause: The number of unique Authorization process names encountered in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF) exceeds those found in the Authorization Report Table. The Authorization Report is incomplete.

Remedy: Verify that the Authorization process destinations in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or TTDF are valid and retry the program. If the problem persists, contact HELP24.

AUTH table overflows by # entries; transactions forced to DEFAULT
AUTH

Where: # = Number of excess entries

Cause: The number of unique FIIDs encountered in the Transaction Log File (TLF) was greater than the number that were found in the Institution Definition File (IDF) and the Interchange Configuration File (ICF). This should not occur if all these files are from the same BASE24 system. The Institution Report produced is incomplete.

Remedy: Verify that the information in the IDF and ICF is correct. If the problem persists, contact HELP24.

BAD PTLF FILE BLOCK LENGTH, BLOCK MUST BE 4096. PTLF: *file-name*

Where: *file-name* = Name of the POS Transaction Log File (PTLF)

Cause: The utility encountered a PTLF with a block length other than 4096.

Remedy: Correct the block length of the PTLF to 4096 and run the utility again. If the problem persists, contact HELP24.

BAD PTLF FILE TYPE, MUST BE ENTRY SEQUENCED. PTLF: *file-name*

Where: *file-name* = Name of the POS Transaction Log File (PTLF)

Cause: The utility encountered a PTLF with a file type that was not entry sequenced.

Remedy: Correct the file type of the PTLF to be entry sequenced and run the utility again. If the problem persists, contact HELP24.

clockdate & tomorrow params can not be used together; use help
param

Cause: The CLOCKDATE and TOMORROW parameters cannot be used in the same program run. The CLOCKDATE option opens and merges the data from two Transaction Log Files (TLFs): the one with the date specified and the next day's. Thus a midnight to midnight report is produced. If data from the previous day is required when using the CLOCKDATE parameter, specify the date in the FILE parameter. The TOMORROW parameter specifies that

the Transaction Analysis utilities produce a report including transactions occurring between midnight and financial institution cutover (i.e., transactions from yesterday's and today's TLF, PTLF, ITLF, or TTLF).

Remedy: Correct the RUN command and run the utility again. The CLOCKDATE and TOMORROW options are discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

CLOCKDATE parameter needs no value assigned: *text*; use help param

Where: *text* = Invalid value entered for the CLOCKDATE parameter

Cause: A value was specified for the CLOCKDATE parameter. This parameter does not require a value.

Remedy: Correct the CLOCKDATE parameter and run the utility again. The CLOCKDATE parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Date parameter must be specified; use HELP param

Cause: A date value specified for the FILE or HISTFILE parameter is invalid or missing.

Remedy: Correct the FILE or HISTFILE parameter and run the utility again. Both parameters are discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

DH table OVERFLOWS by # entries; heavy use of default DH possible

Where: # = Number of excess entries

Cause: The number of unique Device Handler process names encountered in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or Teller Terminal Data File (TTDF) exceeds those found in the Device Handler Report Table. The Device Handler Process Report is incomplete.

Remedy: Verify that the Device Handler routing entries in the BASE24-atm Terminal Data files, BASE24-pos Terminal Data files, or TTDF are valid. If the problem persists, contact HELP24.

Disassemble IFCF Error (*file-name*)

First segment ID not found (*seg-num*)

Reports RBTOD, RBTOA, RBTOS, and RBTOT will not be produced

Where: *file-name* = File name of the ITS Interface Configuration File (IFCF)
 seg-num = Number of the segment on which disassembly error occurred

Cause: The Remote Banking Transaction Analysis utility encountered an error while attempting to disassemble the indicated segment from the IFCF. The Transaction Originator Reports will not be produced.

Remedy: Correct the IFCF and rerun the utility. If the problem persists, contact HELP24.

Dump

Cause: The software is programmed to create a Dump File if it encounters an operating system trap.

Remedy: Send the Dump File and a description of the exact run statement to HELP24.

eof hit on next days TLF before todays TLF; REMAINDER OF TLF
IGNORED

Cause: The ATM Transaction Analysis utility reached the end-of-file on the next day's TLF before it reached the end-of-file on the current day's TLF.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

eof hit on next days TTLF before todays TTLF; REMAINDER OF TTLF
IGNORED

Cause: The Teller Transaction Analysis utility reached the end-of-file on the next day's TTLF before it reached the end-of-file on the current day's TTLF.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

eof hit on next days PTLF before todays PTLF; REMAINDER OF PTLF
IGNORED

Cause: The POS Transaction Analysis utility reached the end-of-file on the next day's PTLF before it reached the end-of-file on the current day's PTLF.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Duplicate NEF names *file-name* - *file-name*

Where: *file-name* = Name of the NEF

Cause: The same Network Environment File (NEF) name was specified more than once for the NEF parameter.

Remedy: Correct the NEF names in the NEF parameter and run the utility again.

Error # examining temporary NEFS file - filename *file-name*

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the temporary Network Environment Source
 File (NEFS) created by the Netlist emulation process

Cause: An error occurred while attempting to read the temporary NEFS.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in creating NETLIST process - error #

Where: # = HP NonStop Kernel operating system error number

Cause: An error occurred while attempting to create a Netlist emulation
 process.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in defining a memory pool, error #

Where: # = HP NonStop Kernel operating system error number

Cause: An error occurred while attempting to define a memory pool in
 extended memory.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in generating unique process name

Cause: An error occurred while attempting to create a process name for the
 Netlist emulation process.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in getpool call

Cause: An error occurred in a GETPOOL call to the operating system to
 obtain a block of storage from a memory pool.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in opening NETLIST process - error #

Where: # = HP NonStop Kernel operating system error number

Cause: An error occurred while attempting to open the Netlist emulation process.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error # opening temporary NEFS file - *file-name*

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the temporary Network Environment Source File (NEFS)

Cause: An error occurred while attempting to open or purge the specified temporary NEFS.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in writing to NETLIST process - error #

Where: # = HP NonStop Kernel operating system error number

Cause: An error occurred while attempting to write to the Netlist emulation process.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Error in putpool call

Cause: An error occurred in a PUTPOOL call to the operating system to return a block of storage to a memory pool.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

```
ERROR ITD TAG (tag-num) ITD Error Value (#);  
--ITD Errors in ITLF (file-name)  
--Possible bad ITLF record
```

Where: *tag-num* = Tag number of the Internal Transaction Data (ITD) field on which the error occurred
 # = Error value
 file-name = Name of the ITS Transaction Log File (ITLF) on which the error occurred

Cause: An error occurred while attempting to retrieve the specified ITD data from the ITLF.

Remedy: Correct the ITLF record. If the problem persists, contact HELP24.

```
ERROR no assigns found in the LCF (file-name); FATAL ERROR
```

Where: *file-name* = Name of the Log Configuration File (LCF)

Cause: The Remote Banking Transaction Analysis utility could not find any assigns for the ITS Transaction Log File (ITLF) in the LCF. Processing stopped and no reports were produced.

Remedy: Correct the LCF and rerun the utility.

```
Error # on NEF IO
```

Where: # = HP NonStop Kernel operating system error number

Cause: The TLFRD program was unable to get the line and station names from the specified Network Environment File (NEF). The line report is incomplete or is not produced.

Remedy: Contact HELP24 if it is a release compatibility problem. The TLFRD program must be compatible with the version of XPNET being used.

Error # on read of NEF. Bytes req #, bytes read #

Where: # = HP NonStop Kernel operating system error number
 # = Number of bytes required
 # = Number of bytes read

Cause: An HP NonStop Kernel operating system sequential Input/Output read error occurred. The TLFRD program was unable to get the line and station names from the specified Network Environment File (NEF). The line report is incomplete or is not produced.

Remedy: Contact HELP24 if it is a release compatibility problem. The TLFRD program must be compatible with the version of XPNET being used.

ERROR # opening *file-name* for output; PROGRAM ABORTED

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the out file which the utility could not open

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the output file named in the run statement.

Remedy: Verify that the output file named in the run statement exists. If it does not, create it and run the utility again. If the output file already exists, correct the output file name in the run statement and run the utility again.

ERROR # opening HISTORY file (*file-name*)

- Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the Transaction Analysis Utility Transaction History File that the utility could not open
- Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the specified Transaction Analysis Utility Transaction History File.
- Remedy: Correct the Transaction Analysis Utility Transaction History File name specified in the RUN command and run the utility again.

ERROR (#) opening ICF (*file-name*); VERY HEAVY USE OF DEFAULTS PROBABLE

- Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the ICF that could not be opened
- Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Interchange Configuration File (ICF). The ICF name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, the institution reports are not produced.
- Remedy: Check the LCONF to determine if the ICF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening IDF (*file-name*); VERY HEAVY USE OF DEFAULTS PROBABLE

- Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the IDF that could not be opened
- Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Institution Definition File (IDF). The IDF name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, the Institution Reports are not produced.
- Remedy: Check the LCONF to determine if the IDF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening IFCF (*file-name*)

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the IFCF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the ITS Interface Configuration File (IFCF). The IFCF name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, the Transaction Originator reports are not produced.

Remedy: Check the LCONF to determine if the IFCF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening LCF (*file-name*); FATAL ERROR

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the LCF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Log Configuration File (LCF). The LCF name is retrieved from the Logical Network Configuration File (LCONF). The utility stops running and no reports are produced.

Remedy: Check the LCONF to determine if the LCF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening LCONF (*file-name*)

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the LCONF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Logical Network Configuration File (LCONF) specified in the run statement.

Remedy: Check the LCONF name and correct if necessary. If the problem persists, contact HELP24.

ERROR (#) opening LCONF (*file-name*); PROGRAM ABORTED

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the LCONF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the Remote Banking Transaction Analysis utility was attempting to open the Logical Network Configuration File (LCONF) specified in the run statement.

Remedy: Check the LCONF name and correct if necessary. If the problem persists, contact HELP24.

ERROR (#) opening ITLF (*file-name*)

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the ITLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open an ITLF. The utility continues to run in spite of this error.

Remedy: Check the ITLF name and correct if necessary. If the problem persists, contact HELP24.

ERROR (#) opening PTDF (*file-name*); VERY HEAVY USE OF DEFAULTS PROBABLE

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the BASE24-pos Terminal Data file that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while attempting to open a BASE24-pos Terminal Data file. The BASE24-pos Terminal Data file name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, reports dealing with station activity are not produced.

Remedy: Check the LCONF to determine if the BASE24-pos Terminal Data file name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening PTLF (*file-name*); Program continues

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the next day's PTLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the next day's POS Transaction Log File (PTLF). This error only occurs if the CLOCKDATE or TOMORROW parameters are used. The utility continues to run in spite of this error. However, station reports are not produced.

Remedy: Check the value contained in the CLOCKDATE or TOMORROW parameter and correct it if it is invalid. If the problem persists, contact HELP24. The CLOCKDATE and TOMORROW parameters are discussed in section 2.

ERROR (#) opening TDF (*file-name*); VERY HEAVY USE OF DEFAULTS
PROBABLE

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the TDF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while attempting to open the Terminal Data File (TDF). The TDF name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, reports dealing with station activity are not produced.

Remedy: Check the LCONF to determine if the TDF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening TLF (*file-name*); Program continues

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the next day's TLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the next day's Transaction Log File (TLF). This error only occurs if the CLOCKDATE or TOMORROW parameters are used. The utility continues to run in spite of this error. However, station reports are not produced.

Remedy: Check the value contained in the CLOCKDATE or TOMORROW parameter and correct it if it is invalid. If the problem persists, contact HELP24. The CLOCKDATE and TOMORROW parameters are discussed in section 2.

ERROR (#) opening TTDF (*file-name*); VERY HEAVY USE OF DEFAULTS PROBABLE

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the TTDF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while attempting to open the Teller Terminal Data File (TTDF). The TTDF name is retrieved from the Logical Network Configuration File (LCONF). The utility continues to run in spite of this error. However, reports dealing with station activity are not produced.

Remedy: Check the LCONF to determine if the TTDF name is correct. If it is not, correct it. If the problem persists, contact HELP24.

ERROR (#) opening TTLF (*file-name*); Program continues

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the next day's TTLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the next day's Transaction Log File (TTLF). This error only occurs if the CLOCKDATE or TOMORROW parameters are used. The utility continues to run in spite of this error. However, station reports are not produced.

Remedy: Check the value contained in the CLOCKDATE or TOMORROW parameter and correct it if it is invalid. If the problem persists, contact HELP24. The CLOCKDATE and TOMORROW parameters are discussed in section 2.

ERROR (#) reading IDF; remainder of file NOT READ; heavy use of default INST and HISF probable

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while reading the Institution Definition File (IDF). The Institution Report is incomplete.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading ICF; remainder of file NOT READ; heavy use of default INST probable

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read the Interchange Configuration File (ICF). The Institution Report produced is incomplete.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading IFCF

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the Remote Banking Transaction Analysis utility was attempting to read the ITS Interface Configuration File (IFCF). The Transaction Originator reports cannot be produced.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading ITLF (*file-name*)
Possible bad ITLF record !

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the ITS Transaction Log File (ITLF)

Cause: An HP NonStop Kernel operating system error occurred while the Remote Banking Transaction Analysis utility was attempting to read the ITS Transaction Log File.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading LCF (*file-name*); FATAL ERROR

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the LCF that could not be read

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read the Log Configuration File (LCF). The utility stops running and no reports are produced.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading LCONF (*file-name*); REQUIRED ASSIGN

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the LCONF that could not be read

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to retrieve a required assign from the LCONF.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading LCONF (*file-name*); REQUIRED PARAM

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the LCONF that could not be read

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to retrieve a required param from the LCONF.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

ERROR (#) reading next day's PTLF; REMAINDER OF PTLF IGNORED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while reading the next day's POS Transaction Log File (PTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading next day's TLF; REMAINDER OF TLF IGNORED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while reading the next day's Transaction Log File (TLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading next day's TTLF; REMAINDER OF TTLF IGNORED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while reading the next day's Teller Transaction Log File (TTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading PTLF; REMAINDER OF PTLF IGNORED.

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred after valid records were found in the POS Transaction Log File (PTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in current day's PTLF; REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the current day's POS Transaction Log File (PTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in current day's TLF; REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the current day's Transaction Log File (TLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in current day's TTLF; REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the current day's Teller Transaction Log File (TTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in next day's PTLF: REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the next day's PTLF. The CLOCKDATE option was used, but no records were available to be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in next day's TLF: REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the next day's TLF. The CLOCKDATE option was used, but no records were available to be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading record in next day's TTLF: REPORT ABORTED

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read a record in the next day's TTLF. The CLOCKDATE option was used, but no records were available to be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading TLF; REMAINDER OF TLF IGNORED.

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred after valid records were found in the Transaction Log File (TLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#) reading TTLF; REMAINDER OF TTLF IGNORED.

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred after valid records were found in the Teller Transaction Log File (TTLF).

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error.

ERROR (#); Unable to allocate extended memory for table: (*rpt-id*)

Where: # = HP NonStop Kernel operating system error number
 rpt-id = The acronym for one of the Transaction Originator reports

Cause: An HP NonStop Kernel operating system error occurred while attempting to allocate extended memory for one of the Transaction Originator reports. The report is not generated.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error and run the Remote Banking Transaction Analysis utility again.

Error (#); Unable to read HISTORY file record header (*hdr*)

Where: # = HP NonStop Kernel operating system error number
 hdr = Transaction Analysis Utility Transaction History File record header value

Cause: An HP NonStop Kernel operating system error occurred while attempting to read a Transaction Analysis Utility Transaction History File record header.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error and run the Remote Banking Transaction Analysis utility again.

Error (#); Unable to read HISTORY file (*file-name*) record (*rpt-id*)

Where: # = HP NonStop Kernel operating system error number
 hdr = Transaction Analysis Utility Transaction History File name
 rpt-id = An acronym identifying the report to which the data in this record pertains

Cause: An HP NonStop Kernel operating system error occurred while attempting to read the specified Transaction Analysis Utility Transaction History File record.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error and run the Remote Banking Transaction Analysis utility again.


```
Error (#) writing header record to the Transaction History File
(file-name)
```

Where: # = HP NonStop Kernel operating system error number
file-name = Transaction Analysis Utility Transaction History File name

Cause: An HP NonStop Kernel operating system error occurred while attempting to write a header record to the specified Transaction Analysis Utility Transaction History File.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error and run the Remote Banking Transaction Analysis utility again.

```
ERROR (#) writing record (rpt-id) to the Transaction History File
(file-name)
```

Where:

- # = HP NonStop Kernel operating system error number
- rpt-id* = An acronym identifying the report to which the data in this record pertains
- file-name* = Transaction Analysis Utility Transaction History File name

Cause: An HP NonStop Kernel operating system error occurred while attempting to write a record to the specified Transaction Analysis Utility Transaction History File.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the meaning of the error and run the Remote Banking Transaction Analysis utility again.

```
ERROR-- No PTLF records meet the criteria set in the run params;  
NO REPORT CREATED. Try different params in the run statement.
```

Cause: No financial transactions were found in the POS Transaction Log Files (PTLFs).

Remedy: No action is required.

ERROR-- No TLF records meet the criteria set in the run params; NO REPORT CREATED. Try different params in the run statement.

Cause: No financial transactions were found in the Transaction Log Files (TLFs).

Remedy: No action is required.

ERROR-- No TTLF records meet the criteria set in the run params; NO REPORT CREATED. Try different params in the run statement.

Cause: No financial transactions were found in the Teller Transaction Log Files (TTLFs).

Remedy: No action is required.

ERROR- No valid records in next day's PTLF; REMAINDER OF NEXT DAY'S PTLF IGNORED

Cause: No financial transactions were found in the next day's POS Transaction Log File (PTLF).

Remedy: No action is required.

ERROR- No valid records in next day's TLF; REMAINDER OF NEXT DAY'S TLF IGNORED

Cause: No financial transactions were found in the next day's Transaction Log File (TLF).

Remedy: No action is required.

ERROR- No valid records in next day's TTLF; REMAINDER OF NEXT DAY'S TTLF IGNORED

Cause: No financial transactions were found in the next day's Teller Transaction Log File (TTLF).

Remedy: No action is required.

ERROR reading PTDF; remainder of file NOT READ; heavy use of defaults probable

Cause: An HP NonStop Kernel operating system error occurred while reading a BASE24-pos Terminal Data file into memory. Reports concerning terminal activity are incomplete.

Remedy: Check the HP NonStop Kernel operating system error and correct the problem.

ERROR reading TDF; remainder of file NOT READ; heavy use of defaults probable

Cause: An HP NonStop Kernel operating system error occurred while reading the TDF into memory. Reports concerning terminal activity are incomplete.

Remedy: Check the HP NonStop Kernel operating system error and correct the problem.

ERROR reading TTDF; remainder of file NOT READ; heavy use of defaults probable

Cause: An HP NonStop Kernel operating system error occurred while reading the TTDF into memory. Reports concerning terminal activity are incomplete.

Remedy: Check the HP NonStop Kernel operating system error and correct the problem.

Error, Unable to access segment-id (#) for table (*tbl-id*). The segment-id (#) is not allocated, or the segment cannot be used by a nonprivileged caller !

Where: # = Extended memory segment identifier
 tbl-id = The acronym for the extended memory table

Cause: The Remote Banking Transaction Analysis utility could not access a segment in the specified extended memory table. Either the segment is not allocated or the segment cannot be used by a nonprivileged caller.

Remedy: Contact HELP24.

ERROR, Unable to create extended memory table (*tbl-id*). The number of bytes required to build the extended memory table exceeds the maximum (127.5MB) Tandem limit. To fix the problem will require reducing the number of records in file (*file-name*). Due to the problem, report (*rpt-id*) will not be produced.

Where: *tbl-id* = The acronym for the Interface extended memory table
 file-name = The name of ITS Interface Configuration File (IFCF)
 rpt-id = The acronym for a Transaction Originator Report

Cause: The Interface extended memory table could not be built because it would exceed the maximum HP NonStop limit. The specified Transaction Originator Reports could not be produced.

Remedy: Reduce the number of records in the IFCF and rerun the Remote Banking Transaction Analysis utility.

ERROR, unknown currency code (###) in LCONF param CURRENCY-CODE

Where: ### = The value for the CURRENCY-CODE param

Cause: The Remote Banking Transaction Analysis utility encountered an unknown CURRENCY-CODE param value.

Remedy: Correct the CURRENCY-CODE param value in the LCONF and run the utility again.

Error writing line # to page # of report

Where: # = Line number on the report where the write error occurred
 # = Page number of the report where the write error occurred

Cause: An error occurred on a write to the output file. The utility continues to run in spite of this error. However, data is missing in the output file.

Remedy: Check the error and correct the problem based on the error.

```
library-proc-name Failure in Proc
???                !
???                information shown below:
ERROR (#)
Field: (field-name)
        (field-name)
Field value: (fld-value)
End of library-proc-name information
```

Where: *library-proc-name* = Name of a library procedure in which an error occurred
 # = HP NonStop Kernel operating system error number
 field-name = Name of the field on which the error occurred
 fld-value = Value of the field on which the error occurred

Cause: An error occurred in the specified library procedure for the Remote Banking Transaction Analysis utility.

Remedy: Send the detailed error message to HELP24 for analysis.

FATAL CONDITION: BASE24-REL param does not match program requirements

Cause: The utility did not recognize the value of the BASE24-REL param in the LCONF.

Remedy: Correct the BASE24-REL param in the LCONF and run the utility again. If the problem persists, contact HELP24.

FATAL ERROR (#) opening PTLF (*file-name*); PROGRAM ABORTED

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the PTLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the POS Transaction Log File (PTLF). The PTLF name is retrieved from the Logical Network Configuration File (LCONF). The utility aborts because of this error.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error. If the problem persists, contact HELP24.

FATAL ERROR (#) opening TLF (*file-name*); PROGRAM ABORTED

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the TLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Transaction Log File (TLF). The TLF name is retrieved from the Logical Network Configuration File (LCONF). The utility aborts because of this error.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error. If the problem persists, contact HELP24.

FATAL ERROR (#) opening TTLF (*file-name*); PROGRAM ABORTED

Where: # = HP NonStop Kernel operating system error number
 file-name = Name of the TTLF that could not be opened

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the Teller Transaction Log File (TTLF). The TTLF name is retrieved from the Logical Network Configuration File (LCONF). The utility aborts because of this error.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error. If the problem persists, contact HELP24.

FATAL no ITLF files were successfully opened

Cause: The Remote Banking Transaction Analysis utility aborted because no ITLFs could be successfully opened.

Remedy: Check for other error messages and correct the problem based on the errors. If the problem persists, contact HELP24.

FATAL no ITLF records to report on

Cause: The Remote Banking Transaction Analysis utility aborted because not ITLFs could be successfully read.

Remedy: Check for other error messages and correct the problem based on the errors. If the problem persists, contact HELP24.

History options selected without providing HISTORY file name.

Cause: A Transaction Analysis Utility History File name was not provided in conjunction with the OPTIONS parameter.

Remedy: Provide a Transaction Analysis Utility History File name to use in conjunction with the OPTIONS parameter and run the utility again.

ICF file empty

Cause: This message is informational only. No records were encountered in the Interchange Configuration File (ICF).

Remedy: Add the appropriate records to the ICF, if ICF records are required.

IDF file empty

Cause: No records were encountered in the Institution Definition File (IDF). The institution reports are incomplete or are not produced.

Remedy: Check the LCONF to verify the correct file name is specified. If the file name is correct, check the file specified in the LCONF to verify records exist. Correct these problems and retry the utility.

Invalid category param value: *text*; use HELP param

Where: *text* = The value specified for the CATEGORY parameter

Cause: The value specified for the CATEGORY parameter is invalid. It must be in the range of 1 to 5000.

Remedy: Correct the CATEGORY parameter and run the utility again. This parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid MAXTPS param value: *text*; use HELP param

Where: *text* = Value specified for the MAXTPS parameter

Cause: An invalid value was entered in the MAXTPS parameter. Valid values are between 0 and 32.

Remedy: Correct the MAXTPS parameter and run the utility again. The syntax for this parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid divideby param value: *text*; use HELP param

Where: *text* = Value specified for the DIVIDEBY parameter

Cause: An invalid value was entered in the DIVIDEBY parameter. Valid values are between 0 and 32768.

Remedy: Correct the DIVIDEBY parameter and run the utility again. The syntax for this parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid entries param value: *text*; use HELP param

Where: *text* = The value specified for the ENTRIES parameter

Cause: The value specified for the ENTRIES parameter is invalid. It must be in the range of 1 through 99.

Remedy: Correct the ENTRIES parameter and run the utility again. This parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid History acronym: *text*; see DDLS

Where: *text* = An acronym specified for the OPTIONS parameter

Cause: An invalid acronym was specified in the OPTIONS parameter. Acronyms indicate which reports are written to the Transaction Analysis Utility History File.

Remedy: Correct the OPTIONS parameter and run the utility again. The OPTIONS parameter is discussed in section 2. The syntax is also produced by the utility if you specify the HELP parameter in the RUN command.

Invalid history file name: *text*; use HELP param

Where: *text* = The value specified for the ENTRIES parameter

Cause: The value specified for the ENTRIES parameter is invalid. It must be in the range of 1 through 99.

Remedy: Correct the ENTRIES parameter and run the utility again. This parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid interval param value: *text*; use HELP param

Where: *text* = The value specified for the INTERVAL parameter

Cause: The value specified for the INTERVAL parameter is invalid. The valid range for this value is 1 through 60 minutes.

Remedy: Correct the value for the INTERVAL parameter and run the utility again. The syntax for this parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid options param: *text*; refer to TOOLS manual

Where: *text* = An invalid value specified for the OPTIONS parameter

Cause: A value specified for the OPTIONS parameter is invalid.

Remedy: Correct the value for the OPTIONS parameter and run the utility again. The valid values for this parameter are discussed in section 2.

Invalid NEF file *file-name*

Where: *file-name* = Name of the NEF specified in the RUN command

Cause: The NEF given in the run statement is not a valid Network Environment File (NEF) produced by the NETGEN program. The utility continues to run in spite of this problem, but the line report is either incomplete or not produced.

Remedy: Correct the NEF name and run the utility again.

Invalid NEF specification: *text*; use HELP param

Where: *text* = Value specified for the NEF parameter

Cause: The NEF parameter does not include a valid Network Environment File (NEF) name.

Remedy: Correct the NEF parameter to contain a valid NEF name. The syntax for this parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Invalid nospool acronym: *text*; see DDLs

Where: *text* = Invalid value entered for the NOSPOOL parameter

Cause: An invalid value was specified for the NOSPOOL parameter.

Remedy: Correct the acronyms used with the NOSPOOL parameter and run the utility again. The valid acronyms for the NOSPOOL parameter are discussed in section 2.

Invalid parameter: *text*; use HELP param

Where: *text* = Invalid parameter value

Cause: An invalid parameter was specified.

Remedy: Correct the invalid parameter specified in the message and run the utility again. The syntax for parameters are discussed in section 2. The syntax for parameters is also produced by the utility if you specify the HELP parameter in the RUN command.

Invalid start/end time param value: *text*; use HELP param

Where: *text* = The value specified for the START or END parameter

Cause: An invalid time value was specified in the START or END parameters. The start and end times must be in the format of HH:MM and between the times of 00:00 and 24:00. This message is generated if either the START or END parameter is invalid.

Remedy: Correct the values in the START and END parameters and run the utility again. The syntax for these parameters are discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

ITLF Identification Record Fields shown below:

CUSTOMER ID: *cust-id*

FIID: *fiid*

ACCOUNT NUM 1: *acct-num-1*

ACCOUNT NUM 2: *acct-num-2*

TXN DATE: *txn-date*

TXN TIME: *txn-time*

TXN AMOUNT: *txn-amt*

SEQUENCE NUM: *txn-seq-num*

End of ITLF Identification Record Fields:

Where: *cust-id* = Customer ID value
 fiid = FIID value
 acct-num-1 = Account 1 number
 acct-num-2 = Account 2 number
 txn-date = Date of this transaction
 txn-time = Time this transaction occurred
 txn-amt = Amount of this transaction
 txn-seq-num = Sequence number assigned to this transaction

Cause: This message is generated by the Remote Banking Transaction Analysis utility to identify an ITS Transaction Log File (ITLF) record on which an error occurred while attempting to read the record.

Remedy: Check the associated error message and correct the problem based on the error.

NEF file *file-name* may be incompatible with this version - WARNING

Where: *file-name* = Name of the NEF that is invalid for the current XPNET version

Cause: The Network Environment File (NEF) is invalid or the release of the utility is not compatible with the release of XPNET that produced the NEF. The utility continues to run in spite of this problem, but the line report is either incomplete or is not produced.

Remedy: Correct the NEF or upgrade the utility to be compatible with the version of XPNET being used and run the utility again.

NEF symbol table format error

Cause: The TLFRD program was unable to get the line and station names from the specified Network Environment File (NEF). The line report is incomplete or is not produced.

Remedy: Contact HELP24 if it is a release compatibility problem. The TLFRD program must be compatible with the version of XPNET being used.

Next day's PTLF file empty; REPORT ABORTED

Cause: The next day's POS Transaction Log File (PTLF) was empty. The CLOCKDATE option was used, but no records existed in the next day's PTLF so none could be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Verify that the next day's PTLF exists and is not empty. Also verify that the CLOCKDATE option is correct, and run the utility again.

Next day's TLF file empty; REPORT ABORTED

Cause: The next day's Transaction Log File (TLF) was empty. The CLOCKDATE option was used, but no records existed in the next day's TLF so none could be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Verify that the next day's TLF exists and is not empty. Also verify that the CLOCKDATE option is correct, and run the utility again.

Next day's TTLF file empty; REPORT ABORTED

Cause: The next day's Teller Transaction Log File (TTLF) was empty. The CLOCKDATE option was used, but no records existed in the next day's TLF so none could be merged. The CLOCKDATE option is discussed in section 2.

Remedy: Verify that the next day's TLF exists and is not empty. Also verify that the CLOCKDATE option is correct, and run the utility again.

NOT FOUND (*item-name*)

Where: *item-name* = The name of a Logical Network Configuration File (LCONF) assign or param

Cause: The specified assign or param was not found in the LCONF.

Remedy: Correct the LCONF and rerun the utility.

Nospool option invalid with history acronym B(BOTH); use HELP param

Cause: The NOSPOOL option cannot be used when the value B is specified with the OPTIONS parameter.

Remedy: Correct the acronyms used with the NOSPOOL parameter or the OPTIONS parameter and run the utility again. Both parameters are discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

No records in the ITLF (*file-name*)

Where: *file-name* = The name of the ITS Transaction Log File (ITLF)

Cause: The Remote Banking Transaction Analysis utility did not find any records in the specified ITLF for the reporting time range.

Remedy: No action is necessary.

One or more required params or assigns are missing

Cause: One or more required assigns or params was not found in the LCONF. The Remote Banking Transaction Analysis utility stops running and no reports are produced.

Remedy: Correct the LCONF and rerun the utility.

Out of range timestamp in field:

(*field-name*)

Timestamp value : (*log-time*)

Where: *field-name* = The name of a field in the ITLF
log-time = The timestamp defining when this record was logged to the ITLF

Cause: The timestamp for an ITLF record was out of range.

Remedy: Correct the ITLF timestamp and run the Remote Banking Transaction Analysis utility again.

PTDF file empty; VERY HEAVY USE OF DEFAULTS PROBABLE

Cause: No records existed in the POS Terminal Data File (PTDF). Reports concerning terminal activity are not produced.

Remedy: Check the PTDF to determine if records exist in it. If they do not, check to determine if the correct PTDF is being accessed. Correct the PTDF and run the utility again.

PTLF file empty; REPORT ABORTED

Cause: No records exist in the POS Transaction Log File (PTLF) for the specified date.

Remedy: No action is required.

Software not licensed by ACI; PROGRAM ABORTED

Cause: The utility was not licensed for this HP NonStop system.

Remedy: Contact HELP24.

STA table OVERFLOWS by # entries; heavy use of default STA, DH,
and AUTH possible

Where: # = Number of excess entries

Cause: The number of unique station names encountered in the Terminal
Data File (TDF), POS Terminal Data File (PTDF), Teller Terminal
Data File (TTDF) exceeds those found in the Station Report Table.
The Station Reports are incomplete.

Remedy: Verify that the station entries in the TDF, PTDF, or TTDF are valid
and retry the utility. If the problem persists, contact HELP24.

Start time must be less than end time; use HELP param

Cause: The utility was run with an end time specified that was earlier than
the start time.

Remedy: Correct the START and END parameters and run the utility again.
The syntax for these parameters is discussed in section 2. The
syntax is also produced by the utility if you include the HELP
parameter in the RUN command.

TDF file empty; VERY HEAVY USE OF DEFAULTS PROBABLE

Cause: No records existed in the Terminal Data File (TDF). Reports
concerning terminal activity are not produced.

Remedy: Check the TDF to determine if records exist in it. If they do not,
check to determine if the correct TDF is being accessed. Correct
the TDF and run the utility again.

The history file read does not match input date

Cause: The utility could not read the Transaction Analysis Utility History
File for the specified date.

Remedy: Correct the input date and run the utility again.

The history param cannot be used when creating reports from a history file

Cause: Both the HISTORY and HISTFILE parameters were included in the RUN command. The HISTORY param cannot be used when you are creating reports from existing Transaction Analysis Utility History Files.

Remedy: Correct the RUN command syntax and run the utility again.

TLF file empty; REPORT ABORTED

Cause: No records exist in the Transaction Log File (TLF) for the specified date.

Remedy: Check the date of the TLF for accuracy. If the problem persists, contact HELP24.

Tomorrow parameter needs no value assigned: *text*; use HELP param

Where: *text* = Invalid value specified for the TOMORROW parameter

Cause: A value was specified for the TOMORROW parameter. No value is required for this parameter.

Remedy: Correct the TOMORROW parameter and run the utility again. The syntax for this parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Too many days specified in HISTFILE parameter; the limit is 10

Cause: The date range value specified for the HISTFILE parameter spans more than 10 Transaction Analysis Utility History Files. If a date range is specified for this parameter it cannot span more than 10 files.

Remedy: Correct the Transaction Analysis utility run statement and run the utility again. The syntax for the HISTFILE parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

Too many NEF files specified; use HELP command

Cause: More than 16 Network Environment Files (NEF) were entered in the ATM, POS, Remote Banking, or Teller Transaction Analysis utilities run statement.

Remedy: Correct the Transaction Analysis utility run statement and run the utility again. The syntax for the NEF parameter is discussed in section 2. The syntax is also produced by the utility if you include the HELP parameter in the RUN command.

TTDF file empty; VERY HEAVY USE OF DEFAULTS PROBABLE

Cause: No records existed in the Teller Terminal Data File (TDF). Reports concerning terminal activity are not produced.

Remedy: Check the TTDF to determine if records exist in it. If they do not, check to determine if the correct TTDF is being accessed. Correct the TTDF and run the utility again.

TTLF file empty; REPORT ABORTED

Cause: No records exist in the Teller Transaction Log File (TTLF) for the specified date.

Remedy: Check the date of the TTLF for accuracy. If the problem persists, contact HELP24.

Unable to allocate extended memory - error #

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error was returned by the operating system after a program call to HP NonStop Kernel operating system ALLOCATESEGMENT. This error causes the utility to unexpectedly halt.

Remedy: Check the reason the error occurred by referring to the appropriate HP NonStop manual. Correct the error and run the utility again. If the problem persists, contact HELP24.

Unable to allocate extended memory - error #

Where: # = HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error was returned by the operating system after a program call to HP NonStop Kernel operating system ALLOCATESEGMENT. This error causes the utility to unexpectedly halt.

Remedy: Check the reason the error occurred by referring to the appropriate HP NonStop manual. Correct the error and run the utility again. If the problem persists, contact HELP24.

Unable to create temporary NEFS file - filename *file-name*

Where: *file-name* = Name of the temporary Network Environment Source File (NEFS) that could not be created.

Cause: The NetList emulation process was unable to create the specified temporary NEFS for a Network Environment File (NEF) specified by the NEF parameter.

Remedy: Rerun the utility. If the problem persists, contact HELP24.

Unable to open NEF *file-name*; Error #

Where: *file-name* = Name of the NEF on which the HP NonStop Kernel operating system error occurred
= HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the specified Network Environment File (NEF).

Remedy: Correct the NEF name specified in the RUN command and run the utility again.

Unable to open HISTORY file *file-name*; Error #

Where: *file-name* = Name of the Transaction Analysis Utility Transaction History File on which the HP NonStop Kernel operating system error occurred
= HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to open the specified Transaction Analysis Utility Transaction History File.

Remedy: Correct the Transaction Analysis Utility Transaction History File name specified in the RUN command and run the utility again.

UNABLE TO READ HISTORY FILE *file-name*; Error #

Where: *file-name* = Name of the Transaction Analysis Utility Transaction History File on which the HP NonStop Kernel operating system error occurred
= HP NonStop Kernel operating system error number

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to read the specified Transaction Analysis Utility Transaction History File.

Remedy: Correct the HP NonStop Kernel operating system error and run the utility again. If the problem persists, contact HELP24.

Unable to read NDF global record on file *file-name*

Where: *file-name* = Name of the NEF

Cause: The Network Environment File (NEF) is invalid or the release of the utility is not compatible with the release of XPNET that produced the NEF. The Network Definition File (NDF) is the header portion of the NEF. If the utility cannot find the NDF, then this error is produced. The utility continues to run in spite of this problem, but the line report is either incomplete or is not produced.

Remedy: Correct the NEF or upgrade the utility to be compatible with the version of XPNET being used, and run the utility again.

Unable to read symbols on file *file-name*

Where: *file-name* = Name of the NEF for which the utility was unable to read line and station names

Cause: The utility was unable to get the line and station names from the specified Network Environment File (NEF). The line report is incomplete or is not produced.

Remedy: Correct the NEF or upgrade the utility to be compatible with the version of XPNET being used and run the utility again. Contact HELP24 if it is not a release compatibility problem.

Unknown BASE24-REL param read from LCONF:##;will default to 40

Where: ## = Value of the BASE24-REL param from the LCONF

Cause: The utility was unable to recognize the release number specified by the BASE24-REL param and assumes the default value of 40.

Remedy: Correct the BASE24-REL param in the LCONF and run the utility again. If the problem persists, contact HELP24.

Unknown protocol encountered *protocol-line*

Where: *protocol-line* = Name of an unknown protocol encountered while extracting an entry from the temporary NEFS to the extended memory list

Cause: The utility was unable to recognize the protocol associated with a communications line while extracting an entry from the temporary NEFS to the extended memory list. The line report is incomplete or is not produced.

Remedy: Correct the NEF or upgrade the utility to be compatible with the version of XPNET being used and run the utility again. Contact HELP24 if it is not a release compatibility problem.

Unknown TLF.AUTH.RESPONDER; # transactions dropped from DEST reporting

Where: # = Number of transactions found in the TLF that contain invalid values in the TLF.AUTH.RESPONDER field

Cause: A number of records were found in the Transaction Log File (TLF) containing invalid values in the TLF.AUTH.RESPONDER field.

Remedy: Identify the records and the processes that are not filling in this field properly and correct them. If an invalid value is entered, the utility cannot determine where the transaction originated and, therefore, cannot include it in the Transaction Log File Report. The valid values for this field are explained in section 2. The utility continues to run in spite of this error. However, all reports produced are incomplete.

WARNING; MAXTPS param not present, Report RBTVR will not be produced

Cause: The MAXTPS param was missing from the RUN command. The Transaction Rate Vs. Response Time Summary Report will not be produced.

Remedy: Add the MAXTPS param to the RUN command and rerun the Remote Banking Transaction Analysis utility.

WARNING Param LOGICAL-NET not found in LCONF

Cause: The LOGICAL-NET param was not found in the Logical Network Configuration File (LCONF). This message is informational only. The only effect that excluding the LOGICAL-NET parameter from the LCONF has on generating reports is that the Logical Network Name is not displayed in the page header.

Remedy: No action is required.

WARNING (#) reading LCONF (*file-name*); OPTIONAL ASSIGN

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the LCONF that could not be read

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to retrieve an optional assign from the LCONF.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

WARNING (#) reading LCONF (*file-name*); OPTIONAL PARAM

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the LCONF that could not be read

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to retrieve an optional param from the LCONF.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

Write error # on file *file-name* - History writing aborted

Where: # = HP NonStop Kernel operating system error number
file-name = Name of the Transaction Analysis Utility History File being written when the HP NonStop Kernel operating system error occurred

Cause: An HP NonStop Kernel operating system error occurred while the utility was attempting to write records to the Transaction Analysis Utility History File. The report is produced; however, no more records are written to the Transaction Analysis Utility History File. In addition, more than one run of reports for the same date to the same Transaction Analysis Utility History File with the same acronyms cannot be accomplished because unique keys to the file are required.

Remedy: Check the HP NonStop Kernel operating system error number and correct the problem based on the error.

BLOCKR2 Error Messages

The following error messages are generated when problems occur while running the BLOCKR2 utility.

BAD DATA IN INPUT FILE : Enscribe DP version !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the input file and run the utility again.

BAD DATA INPUT FILE : Block size !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : File type !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : index !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : Key length !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : Number of extents !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : Number of records !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

BAD DATA INPUT FILE : Record size !!

Cause: Invalid data was specified in the RUN command syntax.

Remedy: Correct the data and run the utility again.

Error on read from IN file #

Where: # = In file name

Cause: An error occurred on a read from the input file.

Remedy: Check the input file, make any corrections, and run the BLOCKR2 utility again. If no input file is specified, input must be manually entered by the operator from the home terminal.

Error on write to OUT file #

Where: # = Out file name

Cause: An error occurred on a write to the output file.

Remedy: Check the output file, make any corrections, and run the BLOCKR2 utility again. If no output file is specified, the output generated by the BLOCKR2 utility is written to the home terminal.

DSCRPT Error Messages

The following error messages are generated when problems occur while running the DSCRPT utility.

ASSIGN NAME MUST BE 8 CHARACTERS OR LESS

Cause: The out file name specified in the RUN command syntax was too long. File names must be a maximum of eight characters.

Remedy: Correct the out file name length and run the utility again.

CAN'T OPEN \$RECEIVE

Cause: An error occurred when attempting to open the \$RECEIVE file.

Remedy: Correct the problem and run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T OPEN DSAP PROCESS.

Cause: The DSCRPT utility was unable to open the HP NonStop Disk Space Analysis Program (DSAP).

REMEDY Correct the problem and run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T OPEN FUP PROCESS.

Cause: The DSCRPT utility was unable to open the HP NonStop File Utility Program (FUP) process.

Remedy: Correct the problem and run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T OPEN OUTPUT FILE.

Cause: The DSCRPT utility was unable to open the output file.

Remedy: Correct the problem and run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T OPEN PUP PROCESS.

Cause: The DSCRPT utility was unable to open the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T READ \$RECEIVE

Cause: An error occurred when attempting to read the \$RECEIVE file.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T START DSAP PROCESS.

Cause: The DSCRPT utility was unable to start the HP NonStop Disk Space Analysis Program (DSAP).

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T START FUP PROCESS.

Cause: The DSCRPT utility was unable to start the HP NonStop File Utility Program (FUP) process.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T START PUP PROCESS.

Cause: The DSCRPT utility was unable to start the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T WRITE TO DSAP PROCESS.

Cause: The DSCRPT utility was unable to write to the HP NonStop Disk Space Analysis Program (DSAP).

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T WRITE TO OUTPUT FILE.

Cause: The DSCRPT utility was unable to write to the output file.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

CAN'T WRITE TO PUP PROCESS.

Cause: The DSCRPT utility was unable to write to the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Run the DSCRPT utility again. If the problem persists, contact HELP24.

ERROR CALLING ACI UTILITY.

Cause: An error occurred while the DSCRPT utility attempted to call either ASCII^INTEGER or INTEGER^ASCII.

Remedy: Verify that all parameters in the RUN command syntax are correct. If the problem persists, contact HELP24.

EXTENDED MEMORY ALLOCATION FAILURE.

Cause: The DSCRPT utility was unable to allocate extended memory.

Remedy: Contact HELP24.

SYSTEM MESSAGE RECEIVED WHILE PROCESSING DSAP INFORMATION

Cause: A system message was received when a message from the HP NonStop Disk Space Analysis Program (DSAP) was expected.

Remedy: Run the DSCRPT utility again. Incomplete reports are generated when this message is received. If the problem persists, contact HELP24.

SYSTEM MESSAGE RECEIVED WHILE PROCESSING FUP LISTOPENS

Cause: A system message was received when a message from the HP NonStop File Utility Program (FUP) was expected.

Remedy: Run the DSCRPT utility again. Incomplete reports are generated when this message is received. If the problem persists, contact HELP24.

TABLE SIZE EXCEEDED - DISK DROPPED

Cause: The maximum number of volumes defined for the DSCRPT utility was 100. However, more than 100 volumes exist on the system.

Remedy: Contact HELP24.

MACPUP Error Messages

The following error messages are generated when problems occur while running the MACPUP utility.

CAN'T CREATE OUTPUT FILE

Cause: The MACPUP utility was unable to create the MACWORK output file specified in the RUN command syntax.

Remedy: Determine the problem with the output file name, correct it, and run the utility again.

CAN'T CREATE TEMP FILE

Cause: The MACPUP utility was unable to create the temporary storage file.

Remedy: Determine the problem with the temporary file name, correct it, and run the utility again.

CAN'T OPEN PEEK.

Cause: The MACPUP utility was unable to open the HP NonStop PEEK utility.

Remedy: Verify that the PEEK utility is on the same volume and subvolume as the TACL object. If the problem persists, contact HELP24.

CAN'T WRITE TO PEEK.

Cause: The MACPUP utility was unable to write to the HP NonStop PEEK utility.

Remedy: Contact HELP24.

ERROR CREATING PUP PROCESS.

Cause: The MACPUP utility was unable to start the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Verify that the PUP utility resides on \$SYSTEM.SYSTEM. If the problem persists, contact HELP24.

ERROR OPENING EXISTING OUTPUT FILE

- Cause:** The output file could not be opened because the file already exists and is currently being used. Because of this, the output file could not be successfully read.
- Remedy:** Correct the output file specified in the RUN command syntax and run the utility again. If the problem persists, contact HELP24.
-

ERROR OPENING OR READING \$RECEIVE #

- Where:** # = HP NonStop Kernel operating system error number
- Cause:** An error occurred when attempting to open or read the \$RECEIVE file.
- Remedy:** Correct the problem and run the MACPUP utility again. If the problem persists, contact HELP24.
-

ERROR OPENING OUTPUT FILE

- Cause:** The MACPUP utility was unable to open the MACWORK output file specified in the RUN command syntax.
- Remedy:** Determine the problem with the output file name, correct it, and run the utility again.
-

ERROR OPENING TEMP FILE

- Cause:** The MACPUP utility was unable to open the MACWORK output template file.
- Remedy:** Determine the problem with the output file, correct it, and run the utility again. If the problem persists, contact HELP24.
-

ERROR WRITING TO OUTPUT FILE

- Cause:** The MACPUP utility was unable to write to the MACWORK output file specified in the RUN command syntax.
- Remedy:** Determine the problem with the output file, correct it, and run the utility again. If the problem persists, contact HELP24.
-

READ/WRITE ERROR ON TEMP FILE

Cause: The MACPUP utility was unable to read or write to the MACWORK output template file.

Remedy: Check subsequent error messages, correct the problem, and run the MACPUP utility again. If no subsequent error messages are displayed, contact HELP24.

SORTMERGERECEIVE - WRONG LENGTH RECORD

Cause: The sort count generated by the HP NonStop SORTMERGERECEIVE process did not match the record length. The SORTMERGERECEIVE process is part of the HP NonStop SORT product. The SORT product is used by the MACPUP utility to produce SORT/MERGE statistics.

Remedy: Contact HELP24.

SORTMERGERECEIVE ERROR

Cause: An error occurred while the MACPUP utility was attempting to call the HP NonStop SORTMERGERECEIVE process. The SORTMERGERECEIVE process is part of the HP NonStop SORT product. The SORT product is used by the MACPUP utility to produce SORT/MERGE statistics.

Remedy: Run the MACPUP utility again. If the problem persists, contact HELP24.

SORTMERGESEND ERROR

Cause: An error occurred while the MACPUP utility was attempting to call the HP NonStop SORTMERGESEND process. The SORTMERGESEND process is part of the HP NonStop SORT product. The SORT product is used by the MACPUP utility to produce SORT/MERGE statistics.

Remedy: Correct the problem based on the HP NonStop SORT error number displayed. If the problem persists, contact HELP24.

SORTMERGESTART ERROR

Cause: An error occurred while the MACPUP utility was attempting to call the HP NonStop SORTMERGESTART process. The SORTMERGESTART process is part of the HP NonStop SORT product. The SORT product is used by the MACPUP utility to produce SORT/MERGE statistics.

Remedy: Correct the problem based on the HP NonStop SORT error number displayed. If the problem persists, contact HELP24.

UNABLE TO OPEN PUP PROCESS.

Cause: The MACPUP utility was unable to open the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Contact HELP24.

UNABLE TO READ FROM CONFIG FILE

Cause: The MACPUP utility was unsuccessful in a read attempt to the \$RECEIVE file.

Remedy: Run the utility again. If the problem persists, contact HELP24.

UNABLE TO WRITE TO PUP PROCESS.

Cause: The MACPUP utility was unable to open the HP NonStop Peripheral Utility Program (PUP) process.

Remedy: Contact HELP24.

WRONG WORK FILE VERSION

Cause: The output file specified in the MACPUP utility RUN command is for an incorrect version of the MACPUP utility.

Remedy: Correct the version of the output file and run the utility again. If the problem persists, purge the output file and specify a different output file in the RUN command syntax and run the utility again.

MACPRT Error Messages

The following error messages are generated when problems occur while running the MACPRT utility.

CAN'T CREATE/OPEN TEMP FILE

Cause: The MACPRT utility was unable to create or open the temporary file used for processing.

Remedy: Determine the problem with the temporary file, correct it, and run the utility again. If the problem persists, contact HELP24.

ERROR ON WORK FILE

Cause: A call to POSITION on the work file was unsuccessful. An open or read on the work file was unsuccessful.

Remedy: Run the MACPRT utility again. If the problem persists, contact HELP24.

ERROR OPENING OR READING \$RECEIVE

Cause: An error occurred when attempting to open or read the \$RECEIVE file.

Remedy: Run the MACPRT utility again. If the problem persists, contact HELP24.

ERROR OPENING THE OUTPUT FILE

Cause: The MACPRT utility was unable to open the specified output file created by the MACPUP utility.

Remedy: Determine the problem with the output file name, correct it, and run the utility again.

INTERNAL ERROR - PICTURE TABLE OVERFLOW

Cause: The internal picture table for the MACPRT utility has become full.

Remedy: Contact HELP24.

INVALID PRINT WIDTH; DEFAULTING TO 80

Cause: An invalid print width was specified in the run statement for the MACPRT utility. Valid widths are between 33 and 132 characters. The print width default is 80 characters.

Remedy: No action is required.

MINIMUM PRINT LINE LENGTH IS 33

Cause: A print width of less than 33 was specified in the run statement for the MACPRT utility.

Remedy: Correct the print line length to be greater than 32, but less than 133 and run the utility again.

WRONG LENGTH CPU RECORD IN WORK FILE

Cause: The number of bytes read from the work file did not match the length of the CPU record.

Remedy: Run the MACPUP and MACPRT utilities again. If the problem persists, contact HELP24.

WRONG LENGTH DEVICE RECORD IN WORK FILE

Cause: The number of bytes read from the work file did not match the length of the device record.

Remedy: Run the MACPUP and MACPRT utilities again. If the problem persists, contact HELP24.

WRONG LENGTH SYSTEM RECORD IN WORK FILE

Cause: The number of bytes read from the work file did not match the length of the system record.

Remedy: Run the MACPUP and MACPRT utilities again. If the problem persists, contact HELP24.

WRONG WORK FILE VERSION

Cause: The output file created by the MACPUP utility is for an incorrect version of the MACPRT utility.

Remedy: Determine the correct version of the MACPUP utility required to generate a new output file compatible with the existing version of the MACPRT utility, or determine the correct version of the MACPRT utility required to process the existing MACWORK output file. If the problem persists, contact HELP24.

Index

A

Administrative transactions, [2-118](#)
Approval Amount Report, [2-90](#)
ATM Transaction Analysis utility
 examples for running, [2-23](#)
 installation procedures, [2-7](#)
 introduction, [1-3](#)
 LCONF assigns and params, [2-9](#)
 running, [2-21](#)
Authorization Report, [2-150](#)

B

BASE24-atm Terminal Data files, [2-118](#)
BASE24-pos Terminal Data files, [2-118](#)
BLOCKR2 utility
 command examples, [3-5](#)
 entry-sequenced file templates, [3-16](#)
 exiting, [3-9](#)
 input data file templates, [3-15](#)
 installation procedures, [3-3](#)
 key-sequenced templates, [3-15](#)
 options summary, [3-8](#)
 output, [3-17](#)
 overview, [3-2](#)
 relative file templates, [3-16](#)
 RUN command syntax, [3-4](#)
 running, [3-4](#)
 running interactively, [3-10](#)
 running noninteractively, [3-13](#)
 setting up input files, [3-14](#)
 using home terminal and specified output locations, [3-6](#)
 using home terminal default, [3-6](#)
 using input files without specifying output locations, [3-7](#)
 using specified input and output files, [3-7](#)

C

CATEGORY parameter, [2-12](#)
Category Report, [2-169](#)
CLOCKDATE parameter, [2-12](#), [2-86](#)

D

Daylight savings time (DST) adjustment, [2-86](#)

Device Handler Report, [2-135](#)
Device Type Report, [2-131](#)
DIVIDEBY parameter, [2-12](#)
DSCRPT utility
 Disc Report Detail, [5-7](#)
 Disc Report Summary, [5-12](#)
 installation procedures, [5-3](#)
 output, [5-7](#)
 overview, [5-2](#)
 running, [5-6](#)
 set up procedures, [5-4](#)
 specifying file locations, [5-5](#)

E

END parameter, [2-13](#)
Enhanced Interchange Configuration File (ICFE), [2-154](#)
ENTRIES parameter, [2-13](#)

F

FILE parameter, [2-14](#)
Financial Institution Identifier (FIID), [2-154](#)

H

HELP parameter, [2-14](#)
HISTFILE parameter, [2-5](#), [2-14](#)
HISTORY parameter, [2-15](#)
HP NonStop Kernel operating system errors, [A-1](#)

I

Institution Definition File (IDF), [2-154](#)
Institution Report
 by card owner, [2-155](#)
 by issuer, [2-160](#)
 by terminal owner, [2-164](#)
 overview, [2-154](#)
Interchange Configuration File (ICF), [2-154](#)
INTERVAL parameter, [2-15](#)
ITS Interface Configuration File (IFCF), [2-138](#)
ITS Transaction Log File (ITLF), [2-2](#)

L

Logic errors, [A-1](#)

Logical Network Configuration File (LCONF) assigns

ATM-TRN-THF, [2-4](#), [2-9](#)

ICF, [2-9](#)

ICFE, [2-9](#)

IDF, [2-9](#), [2-10](#)

IFCF, [2-10](#)

LCF, [2-10](#)

NETLIST-PROG, [2-9](#), [2-10](#)

POS-PTDF, [2-10](#)

POS-PTLF, [2-10](#)

POS-TRN-THF, [2-4](#), [2-10](#)

RB-TRN-THF, [2-4](#), [2-10](#)

TDF, [2-9](#)

TLF, [2-9](#)

TLF-DEFAULT, [2-10](#)

TLF-group, [2-10](#)

TLF-TTDF, [2-10](#)

TLF-TTLF, [2-10](#)

TLR-TRN-THF, [2-4](#), [2-10](#)

Logical Network Configuration File (LCONF) params

BASE24-REL, [2-5](#), [2-9](#)

CURRENCY-CODE, [2-10](#)

DATE-DISPLAY-TYPE, [2-5](#), [2-9](#), [2-10](#)

LOGICAL-NET, [2-9](#), [2-10](#)

M

MACKIE Draw utilities

examples, [4-6](#), [4-10](#)

installation procedures, [4-3](#)

output produced by the MACPRT utility, [4-11](#)

output produced by the MACPUP utility, [4-7](#)

overview, [4-2](#)

report generated by the MACPRT utility, [4-14](#)

running the MACPRT utility, [4-9](#)

running the MACPUP utility, [4-5](#)

running the MACPUP utility with a specified output file, [4-6](#)

running the MACPUP utility with no specified output file, [4-6](#)

MAXTPS parameter, [2-15](#), [2-78](#), [2-82](#)

N

NEF parameter, [2-16](#), [2-124](#)

NOSPOOL parameter, [2-16](#)

O

OPTIONS parameter, [2-5](#), [2-18](#)

P

POS Transaction Analysis utility

examples for running, [2-27](#)

installation procedures, [2-7](#)

introduction, [1-4](#)

LCONF assigns and params, [2-9](#)

running, [2-21](#)

POS Transaction Log File (PTLF), [2-2](#)

Process pair directory (PPD), [2-153](#)

Purchasing options, [1-9](#)

R

Remote Banking Transaction Analysis utility

examples for running, [2-33](#)

installation procedures, [2-7](#)

introduction, [1-5](#)

LCONF assigns and params, [2-10](#)

running, [2-21](#)

Response Time Summary Report, [2-72](#)

Reversal Report, [2-109](#)

S

START parameter, [2-20](#)

Station Activity Summary Report, [2-127](#)

Station reports, [2-114](#)

Stations Sorted by Activity Report, [2-121](#)

Stations Sorted by Line Report, [2-124](#)

Stations Sorted by Name, Detail Report, [2-114](#)

System and file analysis utilities

BLOCKR2 utility, [1-7](#)

DISCRPT utility, [1-8](#)

MACKIE draw utilities, [1-7](#)

T

Teller Terminal Data File (TTDF), [2-114](#), [2-118](#), [2-126](#), [2-135](#)

Teller Transaction Analysis utility

installation procedures, [2-8](#)

introduction, [1-6](#)

LCONF assigns and params, [2-10](#)

running, [2-21](#)

Teller Transaction Log File (TTLF), [2-2](#)

THF

see Transaction Analysis Utility Transaction History File (THF)

see Transaction History File (THF),
BASE24-telebanking

TOMORROW parameter, [2-20](#)

Transaction analysis utilities

ATM command entry, [2-21](#)

installation procedures, [2-6](#)

overview, [2-2](#)

POS command entry, [2-21](#)

Remote Banking command entry, [2-21](#)

reporting options, [2-11](#)

running ATM, [2-21](#)

running POS, [2-21](#)

running Remote Banking, [2-21](#)

running Teller, [2-21](#)

Transaction analysis utilities *continued*

 Teller command entry, [2-21](#)

 Teller examples, [2-37](#)

Transaction Analysis Utility Transaction History File (THF), [2-4](#), [2-42](#)

Transaction Denial Report, [2-101](#)

Transaction History File (THF),
 BASE24-telebanking, [2-4](#)

Transaction Log File (TLF), [2-2](#)

Transaction Log File Report, [2-43](#)

Transaction Log File Report—Summary and
 Totals, [2-54](#)

Transaction log files, [2-3](#)

Transaction Originator Report (Activity
 Summary), [2-144](#)

Transaction Originator Report (Sorted by
 Activity), [2-141](#)

Transaction Originator Report (Sorted by Interface
 Instance Name), [2-138](#)

Transaction Originator reports, [2-138](#)

Transaction Originator Type Report, [2-147](#)

Transaction Rate Vs. Response Time Report, [2-78](#)

Transaction Volume Summary Report, [2-86](#)

V

Virtual circuit, [2-124](#), [2-135](#)

ACI Worldwide, Inc.